Test Item File

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 1: Semiconductor Diodes

1) An intrinsic semiconductor is one that is as pure as present-day technology can make it.

2) Electrons are the minority carriers in an *n*-type material.

3) Holes are the majority carriers in a *p*-type material.

4) The quantum-Volt (qV) is the unit of measurement for electron energy.

5) A free electron has a higher energy state than any that are bound to their nucleus.

6) Si and Ge both have negative temperature coefficients.

7) The amount of energy that is converted to heat at a silicon *p-n* junction can be a significant design consideration.

8) A *normalized* value has a reference magnitude of one.

9) The reverse breakdown voltage of an LED is typically less than 12 V.

10) The amount of photon energy emitted at the *p-n* junction of a silicon diode is negligible.

11) The characteristic of an ideal diode are those of a switch that can conduct current _____.
A) in both directions
B) in one direction only
C) in both directions but in only one direction at a time
D) depends on the circuit it is used in

12) When a diode is doped with either a pentavalent or a trivalent impurity its resistance will _____.
A) increase
B) decrease
C) make the resistance stable against variation due to temperature
D) None of the above

13) To make a *p*-type of semiconductor material you need a doping material that is _____.
A) pentavalent
B) tetravalent
C) trivalent
D) hexavalent

14) The direction of the arrow in the diode symbol points in the direction of _____.
A) positive terminal under forward bias
B) from *n*-type of semiconductor to *p*-type semiconductor material
C) from *p*-type of semiconductor to *n*-type semiconductor material
D) leakage current flow

15) The reverse saturation current of a diode will just about _____ for every 10°C rise in the diode temperature.
A) double
B) half
C) increase proportionately with temperature
D) decrease proportionately with temperature

16) Increasing the temperature of a forward-biased diode _____.
A) causes forward current to increase
B) causes forward current to decrease
C) has no significant effect on the forward current
D) None of these

17) The DC or the static resistance of the diode is given by _____.
A) $R_D = \frac{V_D}{I_D}$
B) $R_D = \frac{VV_D}{VI_D}$
C) $R_D = \frac{V_{D1} - V_{D2}}{I_{D1} - I_{D2}}$
D) All of the above can be used.

18) The piecewise linear model, equivalent circuit of the diode consists of _____.
A) a junction capacitor, a battery, a small resistor, and the ideal diode
B) a battery, a small resistor, and the ideal diode
C) a battery and the ideal diode
D) the ideal diode

19) Some of the modern ohmmeters have a diode test setting. If you do not have one of these ohmmeters then to test the diode you need to check its resistance in the forward and the reverse direction. These resistances should be _____.
A) relatively high in the forward direction and relatively low in the reverse direction
B) relatively low in the forward direction and relatively low in the reverse direction
C) relatively low in the forward direction and relatively high in the reverse direction
D) relatively high in the forward direction and relatively high in the reverse direction

20) In the Zener region the current _____ and the voltage across the diode _____.
A) is almost constant; can increase a lot
B) is almost constant; is almost constant
C) can increase a lot; is almost constant
D) can increase a lot; can increase a lot

21) Suppose that a particular Zener diode has a temperature coefficient of 0.00575. If the temperature of this Zener diode increases by 50° C, what is the change in Vz?
A) $50 \times 0.00575 = 0.2875$
B) $5 \times 0.00575 = 0.02875$
C) $10 \times 0.00575 = 0.0575$
D) Cannot tell without looking at the circuit in which the Zener is used

22) An LED produces visible light when _____.
A) the electrons and the holes combine with each other
B) an electron enters the diffusion region
C) a hole enters the diffusion region
D) the electrons and the holes combine in the diffusion region

23) Light-emitting diodes emit light when the *p-n* junction is _____.
A) forward-biased
B) reverse-biased
C) zero biased
D) operating in the Zener region

24) As semiconductor devices have become _____ one of the primary purposes of the container is simply to provide a means for physical handling.
A) larger
B) widely used
C) miniaturized
D) more powerful

25) An advantage of the miniaturization of electronic devices is that they _____.
A) improve reliability
B) reduce cost
C) increase speed
D) increase availability

26) The characteristics of an ideal diode are those of a switch that can conduct current in _____.
A) both directions
B) only one direction
C) the reverse bias direction
D) None of the above

27) The _____ diode is a short circuit for the region of conduction and it is an open circuit in the region of nonconduction.
A) ideal
B) typical
C) power
D) small-signal

28) The ideal diode symbol has an arrow that points in the direction of _____.
A) the leakage current flow
B) the forward current flow
C) positive terminal under forward bias
D) All of the above

29) The term _____ is applied to any material that supports a generous flow of charge when a voltage source of limited magnitude is applied across its terminals.
A) conductor
B) insulator
C) semiconductor
D) dielectric

30) The term _____ is applied to a material that offers a very low level of conductivity under pressure from an applied voltage.
A) conductor
B) insulator
C) semiconductor
D) ionic

31) The term _____ is applied to a material that has a conductivity level somewhere between the extremes of conductivity.
A) conductor
B) insulator
C) semiconductor
D) ionic

32) Which of the following is not a commonly used semiconductor material"?
A) carbon
B) lead
C) silicon
D) germanium

33) As the device temperature increases, semiconductor materials tend to have _____.
A) an increasing number of free electrons
B) a decreasing number of free electrons
C) lower conduction levels
D) relatively unchanged conduction conduction levels

34) Pentavalent elements have _____ valence electrons.
A) 1
B) 3
C) 4
D) 5

35) Doping is used to _____.
A) decrease the conductivity of an intrinsic semiconductor
B) increase the conductivity of an intrinsic semiconductor
C) stabilize the conductivity of an intrinsic semiconductor
D) increase the insulative quality of an intrinsic semiconductor

36) When pentavalent elements are used in doping, the resulting material is called _____ material and has an excess of _____.
A) *n*-type; valence-band holes
B) *n*-type; conduction-band electrons
C) *p*-type; valence-band holes
D) *p*-type; conduction-band electrons

37) When trivalent elements are used in doping, the resulting material is called _____ material and has an excess of _____.
A) *n*-type; valence-band holes
B) *n*-type; conduction-band electrons
C) *p*-type; valence-band holes
D) *p*-type; conduction-band electrons

38) In an *n*-type material, the majority carriers are _____.
A) conduction-band electrons
B) conduction-band holes
C) valence-band electron
D) valence-band holes

39) The energy required to move an electron in silicon from the valence band to the conduction band is _____.
A) 0.67 eV
B) 10 eV
C) 1.8 eV
D) 1.1 eV

40) When a *p-n* junction's depletion layer is narrowed and the device acts as a nearly perfect conductor, it is _____.
A) forward-biased
B) reverse-biased
C) unbiased
D) None of the above

41) The maximum reverse bias potential that can be applied to a Zener diode before it enters the Zener region is called the _____.
A) threshold voltage
B) PIV
C) barrier voltage
D) depletion voltage

42) When a *p-n* junction is reverse-biased, the depletion layer is _____ and the device acts as a near-perfect _____.
A) narrowed; conductor
B) narrowed; insulator
C) widened; conductor
D) widened; insulator

43) The electrode with *n*-type material of a diode is called the _____.
A) anode
B) cathode
C) depletion region
D) Zener region

44) Silicon diodes have been more significantly developed than germanium because _____.
A) it is cheaper
B) it is easier to produce
C) it is more tolerant of heat
D) it has a lower forward voltage drop

45) In a *p*-type material, the minority carriers are _____.
A) conduction-band electrons
B) conduction-band electrons
C) valence-band electrons
D) valence-band holes

46) Pentavalent atoms are often referred to as _____.
A) donor atoms
B) minority carriers
C) acceptor atoms
D) majority carriers

47) When a *p-n* junction is reverse-biased, its junction resistance is _____.
A) high
B) low
C) determined by the components that are external to the device
D) constantly changing

48) A *p-n* junction is forward biased when _____.
A) the applied potential causes the *n*-type material to be more positive than the *p*-type material
B) the applied potential causes the *n*-type material to be more negative than the *p*-type material
C) both materials are at the same potential
D) None of these

49) A *p-n* junction is reverse biased when _____.
A) the applied potential causes the *n*-type material to be more positive than the *p*-type material
B) the applied potential causes the *n*-type material to be more negative than the *p*-type material
C) the current flow across the junction is based on minority carrier transfer
D) All of the above

50) The isolated atomic energy structure associated with electron orbital shells is called a/an _____.
A) conduction band
B) energy band
C) valence band
D) energy gap

51) The electrode with *p*-type material of a diode is called the _____.
A) anode
B) cathode
C) depletion region
D) Zener region

52) The diffusion capacitance of a diode is a shunt capacitance effect that occurs when the diode _____.
A) is large
B) is small
C) is forward biased
D) is reverse biased

53) The transition capacitance of a diode is a shunt capacitive effect that occurs when the diode _____.
A) is large
B) is small
C) is forward-biased
D) is reverse-biased

54) When tested with an ohmmeter, a diode should have a relatively high resistance for _____ condition.
A) the reverse-biased
B) the forward-biased
C) both reverse and forward-biased
D) zero-biased

55) When tested with an ohmmeter, a diode should have a relatively small resistance for _____ condition.
A) the reverse-biased
B) the forward-biased
C) both reverse- and forward-biased
D) zero-biased

56) The nominal voltage for a 1N961 Fairchild 10-V Zener diode has a temperature coefficient of 0.072. If the temperature increases by 50° C, what is the change in V?
A) 0.54 V
B) 0.36 V
C) 0.72 V
D) 0.108 V

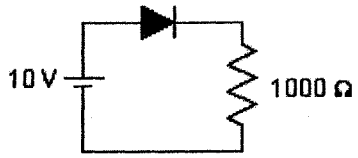
57) The act of giving off light by applying an electrical source of energy is called _____.
A) light power
B) laser
C) photons
D) electroluminescence

ANSWER KEY: Chapter 1: Semiconductor Diodes

1) TRUE
2) FALSE
3) TRUE
4) FALSE
5) TRUE
6) TRUE
7) TRUE
8) TRUE
9) TRUE
10) TRUE
11) B
12) B
13) C
14) C
15) A
16) A
17) A
18) B
19) C
20) C
21) A
22) A
23) A
24) C
25) C
26) D
27) A
28) B
29) A
30) B
31) C
32) B
33) A
34) C
35) B
36) B
37) C
38) A
39) D
40) A
41) B
42) D
43) B
44) A
45) A
46) A
47) A
48) B
49) D
50) B
51) A
52) C
53) D
54) A
55) B
56) B
57) D

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 2: Diode Applications

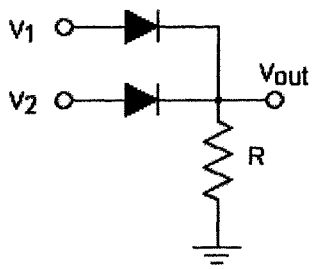
1) For this circuit, determine the load-line intersection with the two axes.



A) $V_D = 10$ V and $I_D = 1$ mA
B) $V_D = 1$ V and $I_D = 1$ mA
C) $V_D = 1$ V and $I_D = 10$ mA
D) $V_D = 10$ V and $I_D = 10$ mA

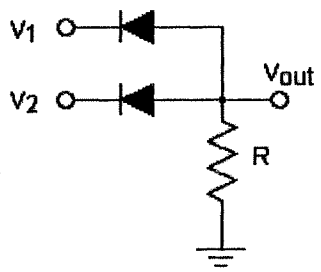
2) If one silicon diode and one germanium diode are connected in series, the voltage drop across the combination of the two diodes will be equal to _____.
A) the forward drop equal to that of the silicon diode
B) the forward drop equal to that of the germanium diode
C) the forward drop equal to that of the sum of the voltage drops across the two diodes
D) the forward drop equal to that of the difference of the voltage drops across the two diodes

3) Name the logic gate that is formed by this circuit.



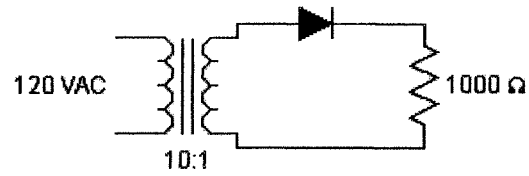
A) positive logic OR gate
B) positive logic AND gate
C) negative logic OR gate
D) negative logic AND gate

4) Name the logic gate that is formed by this circuit.



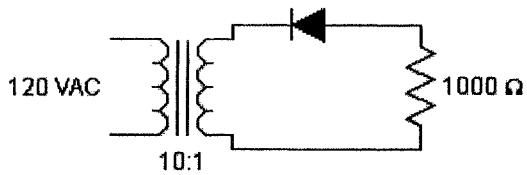
A) positive logic OR gate
B) positive logic AND gate
C) negative logic OR gate
D) negative logic AND gate

5) The current flows through the load resistor in this circuit during the _____.



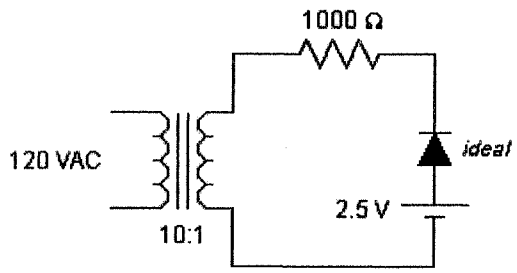
A) positive half cycle of the input waveform
B) negative half cycle of the input waveform
C) entire input waveform
D) The diode will block all current and there will be no current flowing through the load.

6) Calculate the peak current that will flow through this circuit, assuming an ideal diode.



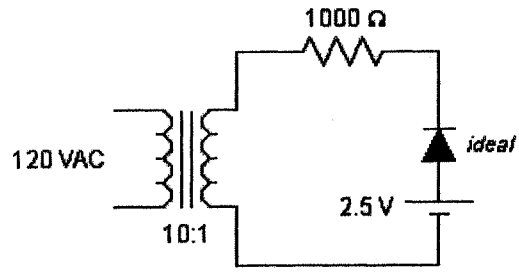
A) 12 mA during the positive half cycle
B) 12 mA during the negative half cycle
C) 16.97 mA during the positive half cycle
D) 16.97 mA during the negative half cycle

7) For this clipping circuit, what will be the maximum output voltage when the diode is conducting?



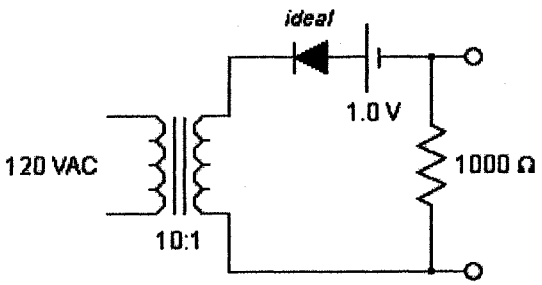
A) + 16.97 Volts
B) - 16.97 Volts
C) + 2.5 Volts
D) + 19.47 Volts

8) For this clipping circuit, what is the maximum output voltage when the diode is not conducting?



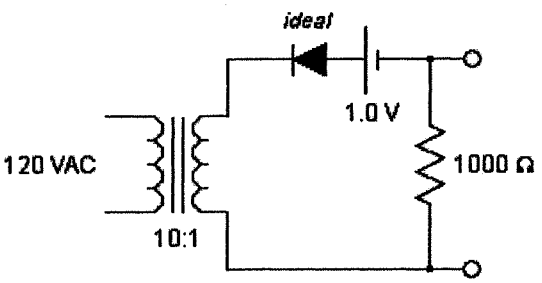
A) + 16.97 V
B) - 16.97 V
C) + 2.5 V
D) + 19.47 V

9) For this clipping circuit, what is the minimum output voltage when the diode is conducting?



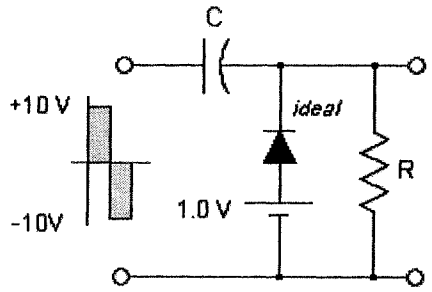
A) - 16.97 V
B) + 16.97 V
C) - 1.0 V
D) - 17.97 V

10) What is the minimum output voltage for this clipping circuit when the diode is not conducting?



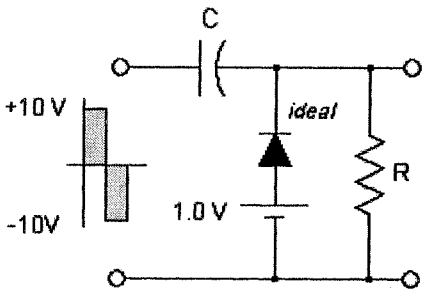
A) - 16.97 V
B) + 16.97 V
C) 0 V
D) - 17.97 V

11) What is the maximum output voltage for this clamping circuit?



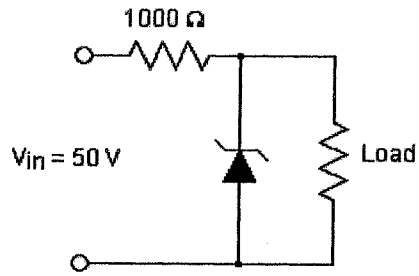
A) + 11 Volts
B) + 21 Volts
C) - 11 Volts
D) - 21 Volts

12) What is the minimum output voltage for this clamping circuit?



A) + 1 V
B) + 21 V
C) - 11 V
D) - 1 V

13) What are the minimum and maximum values of current flowing in the load resistor while the diode is operating in the Zener region?



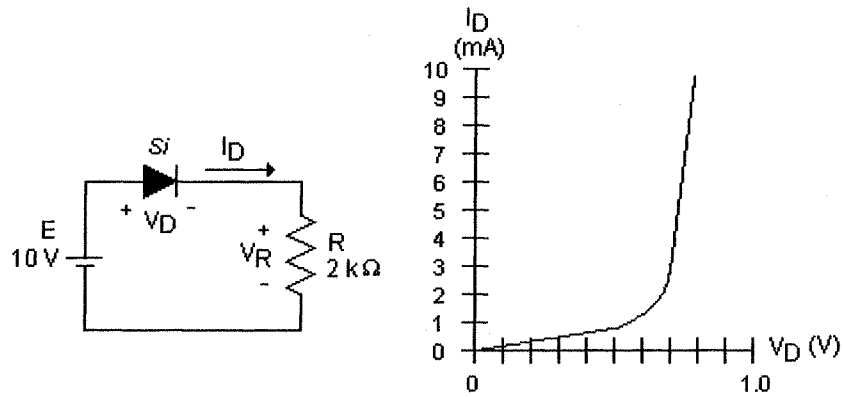
A) 8 mA and 40 mA
B) 8 mA and 35 mA
C) 12.5 mA and 40 mA
D) Need to know the load resistor value in order to determine the values

14) The point of intersection between the characteristic curve of the diode and the resistors loadline is known as the _____.
A) point of operation
B) Q-point
C) quiescent point
D) All of the above

15) Given a series silicon diode circuit with the resistor R = 2 kΩ ohms and an applied voltage of 10 V, what is I_{DQ} ?
A) 4.65 mA
B) 1.0 mA
C) 10 mA
D) 0.5 mA

16) A series silicon diode circuit has a 2 kΩ resistor and a 10 V source. Determine V_{DQ} if I_{DQ} is 4.5 mA.
A) 2 V
B) 0.7 V
C) 11.5 V
D) 1 V

17) For this series diode configuration, use the diode characteristic to estimate the value of V_R.

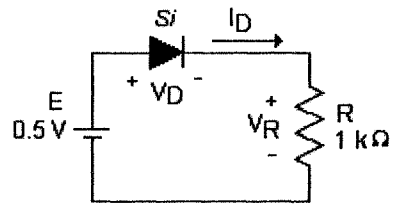


A) 0.92 V
B) 92 mV
C) 9.2 V
D) 10 V

18) Generally a silicon diode is in the _____ state if the current established by the applied voltage source is in the direction of the diode symbol's arrow and V_D is greater than or equal to 0.7 V.
A) off
B) on
C) saturated
D) reverse-biased

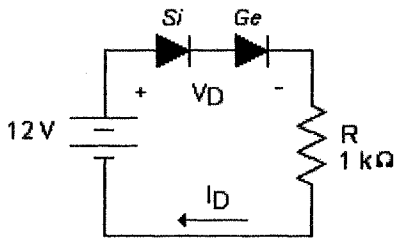
19) Generally a germanium diode is in the _____ state when the current established by the applied voltage source is in the direction of the diode symbol's arrow and V_D is greater than or equal to 0.3 V.
A) off
B) on
C) saturated
D) reverse-biased

20) The practical value of the current I_R in this circuit is _____.



A) 0 A
B) 0.5 mA
C) 0.5 A
D) 5 mA

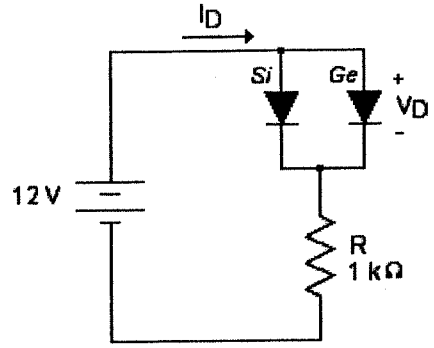
21) The resistor voltage and resistor current in this circuit are _____.



A) 10 V, 5 mA
B) 11 V, 2 mA
C) 11 V, 11 mA
D) 2 V, 11 mA

22) What is the value of the voltage dropped across forward-biased silicon diodes that are connected in parallel with each other?
A) 11.3 V
B) 0.3 5 V
C) 0.7 V
D) 1.4 V

23) The value of V_D in this circuit is _____.



A) 11.3 V
B) 10.6 V
C) 0.7 V
D) 0.3 V

24) When the diode in a half-wave rectifier points toward the load, the output from the rectifier is _____.
A) positive
B) negative
C) either positive or negative, depending on the polarity of the transformer secondary voltage
D) full-wave

25) A half-wave rectifier with the diode arrow pointing away from the load has a DC output voltage of _____ for an AC input voltage of 20 V maximum.
A) 19.3 V
B) 13.65 V
C) 6.14 V
D) 12.49 V

26) A half-wave rectifier is connected to a AC source of 20 V_m. The dc output voltage is _____.
A) 19.3 Vdc
B) 13.65 Vdc
C) 6.14 Vdc
D) None of these

27) Why are bridge rectifiers preferred over full-wave center-tapped rectifiers?
A) They do not require the use of a center-tapped transformer.
B) They provide higher dc output voltages.
C) They require a lower PIV rating.
D) All the above

28) A bridge rectifier has values of $V_m = 177$ V, turns ratio = 5 : 1, and $R_L = 500\ \Omega$. What is the dc output voltage?
A) 3.75 V
B) 9.91 V
C) 19.82 V
D) 6.88 V

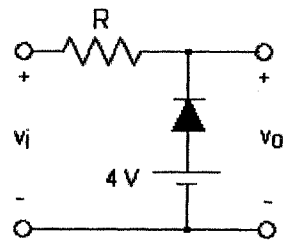
29) A positive full-wave center-tapped rectifier has a secondary voltage of 20 V_m. The peak load voltage for the circuit is _____ if the diode drop is included.
A) 20 V_p
B) 9.3 V_p
C) 19.3 V_p
D) 10 V_p

30) A full-wave center-tapped rectifier has a secondary maximum voltage of 20 V_m and a 4.7 kΩ load resistance. What is the dc load current for the circuit?
A) 1.26 mA
B) 2.61 mA
C) 629.8 mA
D) 1.4 mA

31) Which of the following circuits is used to eliminate a portion of a signal?
A) Clipper
B) Damper
C) Voltage multiplier
D) Voltage divider

32) The two general categories of clippers are _____.
A) dc restorer and dc eliminator
B) half-wave and full-wave
C) series and parallel
D) regenerator and eliminator

33) The circuit shown here is a _____.



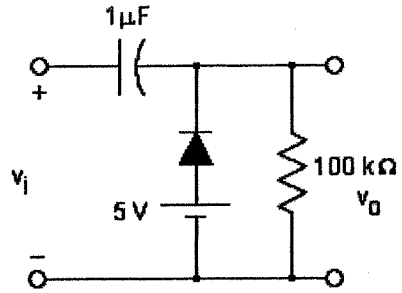
A) series clipper
B) shunt clipper
C) series clamper
D) shunt clamper

34) A(n) _____ is commonly used to provide transient protection.
A) damper
B) multiplier
C) eliminator
D) clipper

35) Which of the following circuits is used to change the dc reference of a signal without changing the shape of the signal?
A) a clipper
B) a damper
C) a voltage multiplier
D) a voltage divider

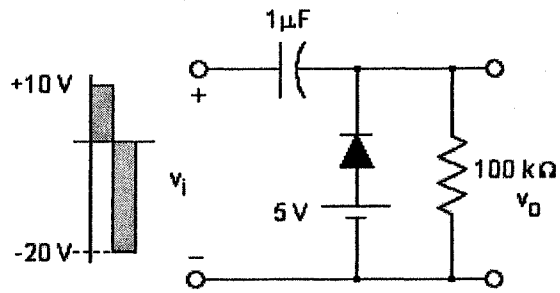
36) A clamper must have a(n) _____ that is large enough to maintain the capacitor's charge during diode conduction.
A) dc restorer
B) RC time constant
C) diode voltage
D) applied voltage

37) This circuit uses a _____.



A) positive clipper
B) negative clipper
C) positive clamper
D) negative clamper

38) Assuming this circuit uses a silicon diode, the output voltage is clamped to _____.



A) 10.7 V
B) 5.7 V
C) 4.3 V
D) 5.3 V

39) The biased damper has a dc reference voltage that is _____.
A) approximately equal to the dc voltage that is applied to the diode
B) approximately equal to zero volts
C) dependent on the peak-to-peak value of the ac input
D) equal to the dc average of the circuits output signal

40) Given that a 1000 Hz signal is applied to a damper with a resistor value of 10 kΩ. What is the minimum value of capacitor needed to maintain safe clamping action?
A) 0.25 pF
B) 10 pF
C) 5 pF
D) 250 pF

41) When the output signal to a clamper circuit is clamped to zero, the total swing of the output is equal to _____.
A) the total diode voltage drop
B) half the total voltage drop
C) the total input voltage swing
D) half the total input voltage swing

42) The Zener diode is on if the applied voltage, V, is _____.
A) $V < V_Z$
B) $V \geq V_Z$
C) $V > 2V_Z$
D) $V < V_Z / 2$

43) When in its "on" state, the voltage across a Zener diode, V_Z _____.
A) gets larger with an increase in applied voltage
B) gets smaller with an increase in applied voltage
C) increases sharply with a decrease in applied voltage
D) None of these

44) The Zener diode must be operated such that _____.
A) $I_Z \times V_Z = P_Z$
B) P_Z is less than the specified P_{Zmax}
C) the applied voltage is greater than V_Z
D) All of these

45) The most frequent application for a _____ is in regulator networks and as a reference voltage.
A) half-wave rectifier
B) full-wave rectifier
C) Zener diode
D) ideal diode

46) A typical Zener diode regulator circuit uses a _____.
A) dropping resistor in series with the load
B) resistor in parallel with the load
C) Zener diode in parallel with the series resistor
D) Zener diode in series with the load

47) When the Zener regulator is used to stabilize the output voltage, given a fixed input voltage and a variable load resistance, a load resistance that is too small results in _____.
A) V_1 being greater than V_Z
B) V_1 being less than V_Z
C) V_1 being equal to V_Z
D) V_Z being equal to V_{in}

48) When a Zener diode circuit is used to stabilize the output voltage given a fixed load resistor and a variable input voltage, the input voltage must be _____.
A) small enough to turn off the Zener diode
B) large enough to turn off the Zener diode
C) small enough to turn on the Zener diode
D) large enough to turn on the Zener diode

49) Two Zener diodes connected _____ can be used as an ac regulator.
A) in parallel with each other
B) in series with the load
C) back-to-back
D) in series with the input voltage

50) A Zener diode is designed to operate in the _____ region of its characteristic curve.
A) forward operating
B) reverse bias
C) reverse breakdown
D) zero voltage

ANSWER KEY: Chapter 2: Diode Applications

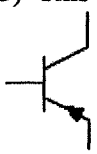
1) D
2) C
3) A
4) A
5) A
6) D
7) C
8) C
9) C
10) C
11) B
12) D
13) A
14) D
15) A
16) D
17) C
18) B
19) B
20) A
21) C
22) C
23) D
24) A
25) C
26) C
27) D
28) C
29) B
30) A
31) A
32) C
33) B
34) D
35) B
36) B
37) D
38) C
39) A
40) D
41) C
42) B
43) D
44) D
45) C
46) A
47) B
48) D
49) C
50) C

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 3: Bipolar Junction Transistors

1) For basic operation of a transistor the base-emitter junction is _____ biased.
A) forward-
B) reverse-
C) not
D) semi-

2) For basic operation of a transistor the collector-base junction is _____ biased.
A) forward-
B) reverse-
C) not
D) semi-

3) This is the symbol for a _____.



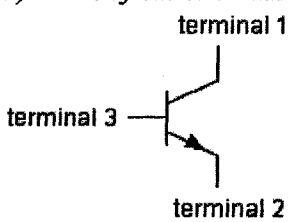
A) *npn*-type BJT
B) *pnp*-type BJT
C) *pnn*-type BJT
D) *npp*-type BJT

4) This is the symbol for a _____.



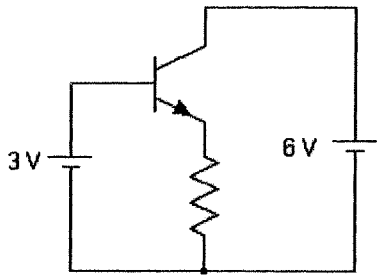
A) *npn*-type BJT
B) *pnp*-type BJT
C) *pnn*-type BJT
D) *ppn*-type BJT

5) Identify the terminals on this BJT.



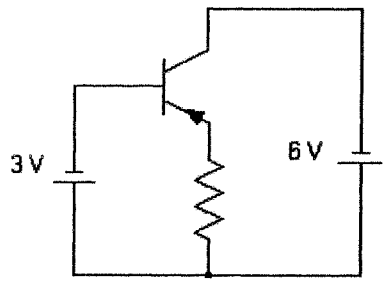
A) 1 = base, 2 = emitter, 3= collector
B) 1 = emitter, 2 = collector, 3 = base
C) 1 = collector, 2 = base, 3 = emitter
D) 1 = collector, 2 = emitter, 3 = base

6) Which of the following is true for this BJT circuit?



A) The base-emitter and collector-base junctions are both forward-biased.
B) The base-emitter junction is forward-biased and the collector-base junction is reversed-biased.
C) The base-emitter junction is reverse-biased and the collector-base junction is forward-biased.
D) The base-emitter and collector-base junctions are both reverse-biased.

7) Which of the following is true for this BJT circuit?



A) The base-emitter and collector-base junctions are both forward-biased.
B) The base-emitter junction is forward-biased and the collector-base junction is reversed-biased.
C) The base-emitter junction is reverse-biased and the collector-base junction is forward-biased.
D) The base-emitter and collector-base junctions are both reverse-biased.

8) 4. The output or the collector characteristics for a common base transistor amplifier shows that as a first approximation the relation between I_E and I_C in the active region is given by _____.
A) $I_E = I_C$
B) $I_E >> I_C$
C) $I_E << I_C$
D) $I_E \approx I_C$

9) In the saturation region, the base-emitter junction _____.
A) and the base-collector junctions are both forward-biased
B) and the base-collector junctions are both reverse-biased
C) is forward-biased while the base-collector junction is reversed-biased
D) is reversed-biased while the base-collector junction is forward-biased

10) In the cut-off region, the base-emitter junction _____.
A) and the base-collector junctions are both forward-biased
B) and the base-collector junctions are both reverse-biased
C) is forward-biased while the base-collector junction is reverse-biased
D) is reversed-biased while the base-collector junction is forward-biased

11) In the active region, the base-emitter junction _____.
A) and the base-collector junctions are both forward-biased
B) and the base-collector junctions are both reverse-biased
C) is forward-biased while the base-collector junction is reversed-biased
D) is reverse-biased while the base-collector junction is forward-biased

12) In a small-signal transistor, the typical range of the parameter α is _____.
A) greater than 1
B) between 0 and 1
C) almost equal to 1 but always less than 1 (0.9 to 1.0)
D) almost equal to 1 but always greater than 1 (1.0 to 1.1)

13) The *common-base, short-circuit, amplification factor* is better known as _____.
A) ac β
B) dc β
C) ac α
D) dc α

14) The *common-emitter, forward-current, amplification factor* is better known as _____.
A) ac β
B) dc β
C) ac α
D) dc α

15) Which of the following expressions is true?
A) $\alpha_{dc} = \frac{I_C}{I_E}$
B) $\alpha_{dc} = \frac{I_C}{I_B}$
C) $\alpha_{dc} = \frac{\Delta I_C}{\Delta I_E}$ where V_{CB} is constant
D) $\alpha_{dc} = \frac{\Delta I_C}{\Delta I_B}$ where V_{CE} is constant

16) Which of the following expressions is true?

A) $\beta_{dc} = \frac{I_C}{I_E}$

B) $\beta_{dc} = \frac{I_C}{I_B}$

C) $\beta_{dc} = \frac{\Delta I_C}{\Delta I_E}$ where V_{CB} is constant

D) $\beta_{dc} = \frac{\Delta I_C}{\Delta I_B}$ where V_{CE} is constant

17) Which of the following expressions is true?

A) $\beta_{ac} = \frac{I_C}{I_E}$

B) $\beta_{ac} = \frac{I_C}{I_B}$

C) $\beta_{ac} = \frac{\Delta I_C}{\Delta I_E}$ where V_{CB} is constant

D) $\beta_{ac} = \frac{\Delta I_C}{\Delta I_B}$ where V_{CE} is constant

18) Which of the following expressions is true?

A) $\alpha_{dc} = \frac{I_C}{I_E}$

B) $\alpha_{dc} = \frac{I_C}{I_B}$

C) $\alpha_{dc} = \frac{\Delta I_C}{\Delta I_E}$ where V_{CB} is constant

D) $\alpha_{dc} = \frac{\Delta I_C}{\Delta I_B}$ where V_{CE} is constant

19) In a small-signal transistor, the typical range of the parameter β is _____.
A) greater than 100
B) between 0 and 100
C) almost equal to 100 but always less than 100 (90 to 100)
D) large and in the range of about 50 to 400

20) A BJT has measured dc current values of $I_B = 0.1$ mA and $I_C = 8.0$ mA. When I_B is varied by 100 μA, I_C changes by 10 mA. What is the value of the β_{ac} for this device?
A) 80
B) 10
C) 100
D) 800

21) A BJT has measured dc current values of I_B = 0.1 mA and I_C = 8.0 mA. When I_B is varied by 100 μA, I_C changes by 10 mA. What is the value of the β_{dc} for this device?
A) 80
B) 10
C) 100
D) 800

22) When a BJT is operating in the saturation region the voltage drop from the collector to the emitter V_{CE} is approximately equal to _____.
A) the collector supply voltage
B) the collector current times the collector resistor
C) zero (about 0.3 Volts)
D) the emitter voltage

23) When a BJT is operating in the active region, the voltage drop from the base to the emitter V_{BE} is approximately equal to the _____.
A) base bias voltage
B) base current times the base resistor
C) diode drop (about 0.7 V)
D) emitter voltage

24) BJTs are commonly used as _____.
A) the primary components in amplifiers
B) series damper circuits
C) the primary components in rectifiers
D) All of the above

25) V_{CE} is measured _____.
A) from the emitter terminal to ground
B) from the collector terminal to the emitter terminal
C) from the collector-emitter junction to ground
D) None of the above

26) Why is the arrow on the BJT schematic symbol important?
A) It identifies the emitter terminal and the type of BJT.
B) It identifies the collector terminal and the type of BJT.
C) It identifies the base terminal and the type of BJT.
D) None of the above

27) In most cases, which two of the three BJT terminal currents are approximately equal in value?
A) collector current and base current
B) collector current and emitter current
C) emitter current and base current
D) All currents are approximately equal.

28) Which of the following biasing combinations is not normally associated with one of the three transistor operating regions?
A) E-B junction = forward, C-B junction = reverse
B) E-B junction = reverse, C-B junction = reverse
C) E-B junction = reverse, C-B junction = forward
D) All of the above

29) The condition where increase in bias current will not cause further increases in collector current is called _____.
A) cutoff
B) saturation
C) active operation
D) All of the above

30) β is the ratio of _____.
A) collector current to emitter current
B) base current to collector current
C) collector current to base current
D) emitter current to collector current

31) A given BJT has an emitter current of 12 mA and a base current of 600 μA. What is the value of β_{dc}?
A) 20
B) 21
C) 19
D) 200

32) A given BJT has an emitter current of 15 mA and a collector current of 14.95 mA. What is the exact value of β?
A) 300
B) 299
C) 1.003
D) 250

33) A given BJT, $\beta = 400$. What is the value of α for the device?
A) 1.0025
B) 0.002
C) 0.9975
D) 1.00

34) A given BJT has an alpha of 0.9985 and a collector current of 15 mA. What is the value of base current?
A) 15.15 mA
B) 14.85 mA
C) 15 mA
D) None of the above

35) Which transistor amplifier configuration is the most commonly used?
A) common-emitter
B) common-collector
C) common-base
D) None of these are used more often than the others.

36) A given transistor has ratings of maximum collector current equal to 200 mA and a beta that varies between 150 and 200. What is the maximum allowable value of base current for the device?
A) 1 mA
B) 4 mA
C) 1.33 mA
D) None of the above

37) A BJT has measured dc current values of I_B = 1 mA and I_C = 80 mA. When I_B is varied by 100 μA, I_C changes by 10 mA. What is the value of β_{dc} for the device?
A) 80
B) 10
C) 100
D) 800

38) A BJT has measured dc current values of I_B = 1 mA and I_C = 80 mA. When I_B is varied by 100 μA, I_C changes by 10 mA. What is the value of β_{ac} for the device?
A) 80
B) 10
C) 100
D) 800

39) When a transistor is in saturation V_{CE} is approximately equal to _____.
A) collector supply voltage
B) collector current times collector resistor
C) 0.3 Volts
D) emitter voltage

40) A transistor has a rating of β = 50 to 450. What value of β should be used for circuit analysis purposes?
A) 50
B) 250
C) 450
D) 150

ANSWER KEY: Chapter 3: Bipolar Junction Transistors

1) A
2) B
3) B
4) A
5) D
6) B
7) B
8) D
9) A
10) B
11) C
12) C
13) C
14) A
15) A
16) B
17) D
18) C
19) D
20) C
21) A
22) C
23) C
24) A
25) B
26) A
27) B
28) C
29) B
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32) B
33) C
34) D
35) A
36) A
37) A
38) C
39) C
40) D

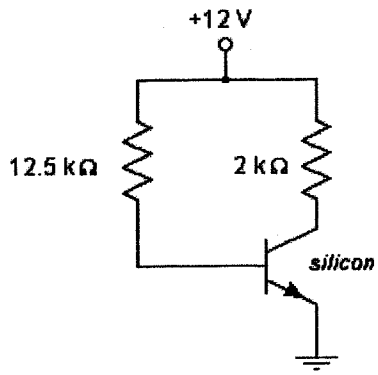
Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 4: DC Biasing-BJTS

1) When a BJT is biased in the active region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.
A) forward; reverse
B) reverse; forward
C) forward; forward
D) reverse; reverse

2) When a BJT is biased in the cut-off region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.
A) forward; reverse
B) reverse; forward
C) forward; forward
D) reverse; reverse

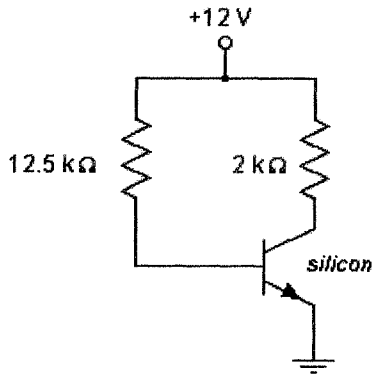
3) When a BJT is biased in the saturation region, its base-emitter junction is _____-biased and its collector-base junction is _____-biased.
A) forward; reverse
B) reverse; forward
C) forward; forward
D) reverse; reverse

4) Calculate the base current for this circuit.



A) 0.904 mA
B) 0.96 mA
C) 0.056 mA
D) 6.0 mA

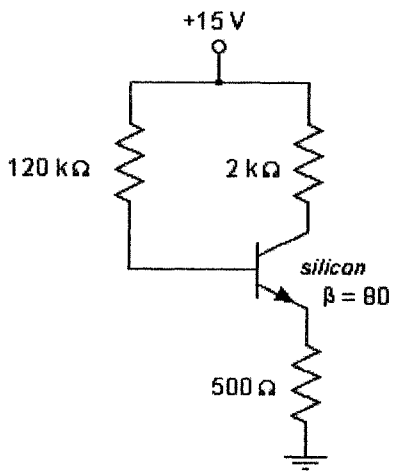
5) Calculate the maximum collector current for this circuit.



A) 0.904 mA
B) 0.96 mA
C) 0.056 mA
D) 6.0 mA

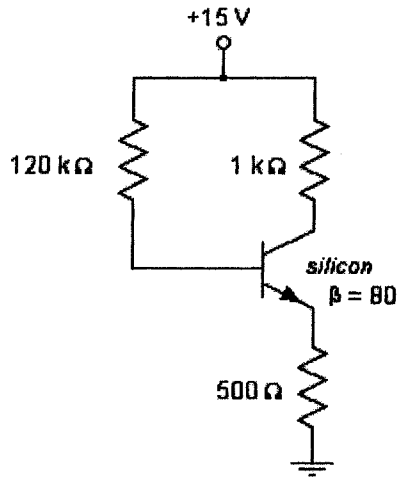
6) When a BJT is biased in the cutoff region the collector-to-emitter voltage is typically equal to _____.
A) the emitter voltage
B) 0.03 V
C) the collector current times the collector resistor
D) the collector supply voltage

7) This emitter-stabilized bias circuit is operating in the _____.



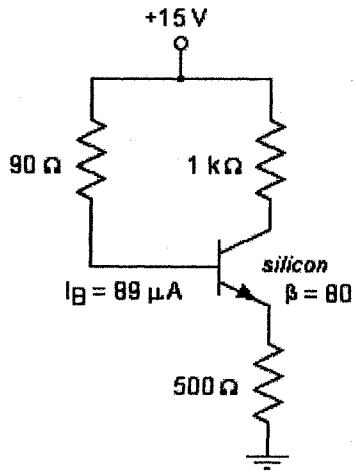
A) saturation region
B) cutoff region
C) active region
D) The transistor is not properly biased.

8) Calculate the base current for this emitter-stabilized bias circuit.



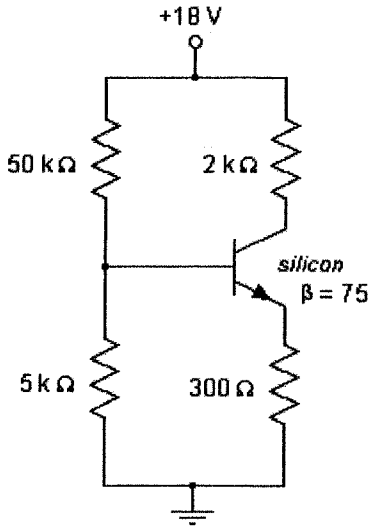
A) 89.0 mA
B) 89.0 μA
C) 0.119 mA
D) None of the above

9) Calculate the collector-emitter voltage for this emitter-stabilized circuit.



A) 4.32 V
B) 10.68 V
C) 0.1335 V
D) 14.24 V

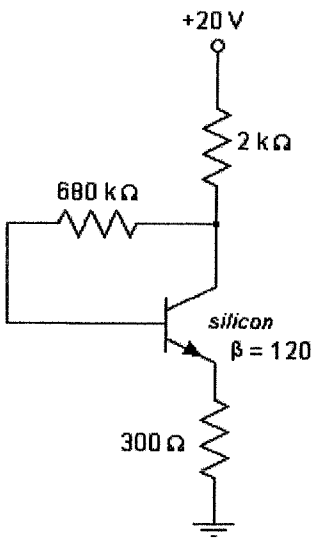
10) Calculate the base current for this voltage-divider bias circuit.



A) 233.78 μA
B) 34.62 μA
C) 596.55 μA
D) 76.8 μA

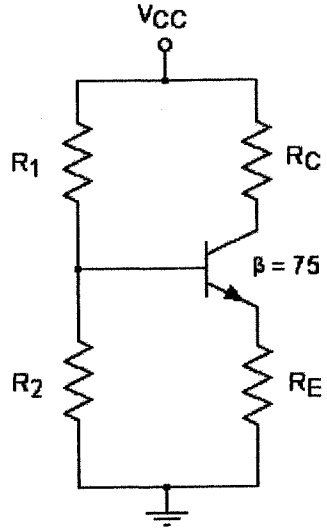
11) When voltage-divider bias is used, it is considered appropriate to use the approximate analysis to determine the bias condition when the resistance R_2 is _____ $(1+\beta)R_E$.
A) greater than
B) less than
C) very much greater than
D) very much less than

12) Calculate the base current for this circuit.



A) 28.4 μA
B) 20.2 μA
C) 28.3 μA
D) Need more information to calculate the base current

13) When designing a current-gain-stabilized voltage-divider bias circuit such as this one, the rule of thumb used for the emitter voltage is _____.



A) $V_E = V_{CC} / 10$
B) $V_{CE} = V_{CC} / 10$
C) $V_B = V_{CC} / 10$
D) $V_C = V_{CC} / 10$

14) When a BJT transistor is used in a switching circuit, it operates in the _____.
A) saturation and active regions
B) active and cutoff regions
C) saturation and cutoff regions
D) active region only

15) The difference between the resulting equations for a network in which an *npn* transistor has been replaced by a *pnp* transistor is _____.
A) the values of the resistors
B) the value of β
C) the sign associates with the particular quantities
D) All of the above

16) When a BJT has its base-emitter junction forward biased and its collector-base junction reverse biased, it is biased in the _____.
A) saturation region
B) active region
C) cutoff region
D) passive region

17) When a BJT has its base-emitter junction reverse biased and its base-collector junction forward biased, it is biased in the _____.
A) saturation region
B) active region
C) cutoff region
D) passive region

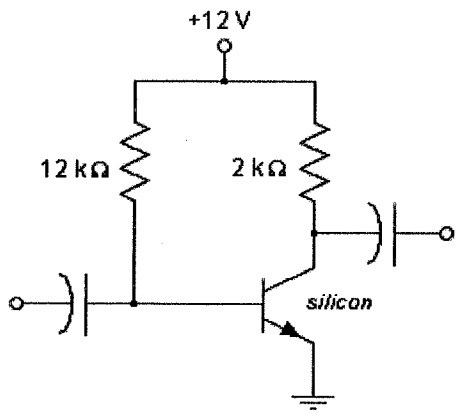
18) When a BJT has its base-emitter junction forward biased and its collector-base junction also forward biased, it is in the _____.
A) saturation region
B) active region
C) cut-off region
D) passive region

19) When a BJT has its base-emitter junction reverse biased and its collector-base junction reverse biased, it is in the _____.
A) saturation region
B) active region
C) cutoff region
D) passive region

20) The term *quiescent* means _____.
A) midpoint-biased
B) at rest
C) active
D) inactive

21) The base current for this circuit is _____.
A) 6 mA
B) 1.37 mA
C) 1.13 mA
D) 12 mA

22) The maximum collector current for this circuit is _____.



A) 1.13 mA
B) 12 mA
C) 6 mA
D) 1.0 mA

23) When a BJT is in cutoff, the collector-to-emitter voltage is typically equal to _____.
A) collector supply voltage
B) collector current times collector resistor
C) 0.3 Volts
D) emitter voltage

24) The change in β and V_{CE} that can occur when the temperature changes is known as _____.
A) midpoint bias
B) midpoint movement
C) output movement
D) Q-point movement

25) A(n) _____ is added to the fixed-bias configuration to improve bias stability.
A) base voltage
B) emitter resistor
C) collector resistor
D) All of the above

26) The input resistance of a stabilized fixed-bias circuit configuration is _____.
A) inversely related to the emitter resistor
B) inversely related to β
C) directly related to the collector resistor
D) directly related to the emitter resistor

27) Two of the factors associated with bias stability are _____.
A) voltage and current
B) the β and the junction temperature
C) age and amount of use
D) None of the above

28) When a transistor is in saturation, the total collector current is limited by _____.
A) collector supply voltage and the total resistance in the collector and emitter circuits
B) collector-to-emitter and collector supply voltage
C) collector supply, collector-to-emitter voltage, and the total collector circuit resistance
D) the transistor

29) Voltage-divider bias stability is _____.
A) dependent on alpha
B) dependent of beta
C) dependent on the collector resistor
D) independent of beta

30) Collector-feedback bias _____.
A) provides a feedback path from collector to base
B) provides an improved level of stability
C) is not totally independent of beta
D) All of the above

31) The collector-feedback bias configuration's input resistance is related to the _____.
A) emitter resistor
B) collector resistor
C) device beta
D) base feedback resistor

32) The emitter-follower configuration has _____.
A) a 180° phase shift
B) an output voltage slightly greater than the input voltage
C) the emitter connected to dc ground potential
D) None of the above

33) A collector-feedback bias circuit is found to be in saturation. Which of the following could cause this condition?
A) The base resistor is open.
B) The collector resistor is open.
C) The transistor is shorted base-to-emitter.
D) A solder bridge across the base resistor.

34) In the design of an emitter-bias stabilized circuit engineering, judgment must be used because the _____.
A) collector resistor is usually unknown
B) emitter resistor is usually unknown
C) relative voltage levels have not been defined
D) All of the above

35) When designing for best bias stability the _____ configuration should be chosen.
A) voltage-divider bias
B) collector-feedback bias
C) fixed-bias
D) emitter-feedback bias

36) When designing a voltage-divider bias circuit, the divider resistors _____.
A) should carry approximately equal current
B) should carry currents that are 10 times the base current
C) determine the base voltage as the drop across base-common resistor
D) All of the above

37) Why is design for a specific bias point desirable for most amplifiers?
A) To meet manufacturer suggested opening point.
B) It allows optimum ac operation of the circuit.
C) It allows optimum dc operation of the circuit.
D) All of the above

38) There are transistors that are called switching transistors because _____.
A) they have a built in switch
B) of the speed at which they can be changed from on to off
C) of the power they can transfer from input to output
D) of the voltage they can transfer from input to output

39) Transistor circuits that are quite stable and relatively insensitive to temperature variations have _____.
A) relative high supply voltages
B) low supply voltages
C) large betas
D) small betas

40) To design a transistor circuit for maximum stability, one must consider _____.
A) the collector leakage current stability factor
B) the base-emitter junction voltage stability factor
C) the transistor's beta stability factor
D) All of the above

41) Variation in h_{fe} is influenced by _____.
A) junction temperature and collector current
B) temperature and base current
C) bias type and device size
D) device size and base current

ANSWER KEY: Chapter 4: DC Biasing-BJTs

1) A
2) D
3) C
4) A
5) D
6) D
7) A
8) C
9) A
10) C
11) C
12) B
13) A
14) C
15) C
16) B
17) C
18) A
19) C
20) D
21) C
22) C
23) A
24) D
25) B
26) D
27) B
28) A
29) D
30) D
31) C
32) D
33) C
34) D
35) A
36) D
37) D
38) B
39) C
40) D
41) A

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 5: BJT ac Analysis

1) The input impedance of a BIT is _____.
A) resistive
B) capacitive
C) inductive
D) a combination of resistive, capacitive, and inductive

2) The output impedance of a BJT is _____.
A) resistive
B) capacitive
C) inductive
D) a combination of resistive, capacitive, and inductive

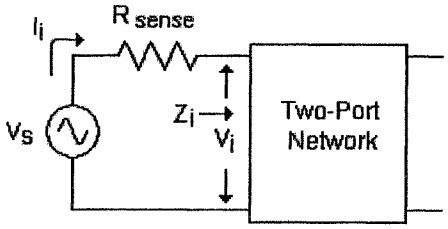
3) To calculate the output impedance the applied signal must be set _____.
A) equal to the smallest value of the input signal
B) equal to the largest value of the input signal
C) equal to zero
D) equal to a value that is half way between the largest and the smallest

4) For a two-port system, like a BIT amplifier, the no-load voltage gain _____.
A) is always greater than the loaded voltage gain
B) is always less than the loaded voltage gain
C) is always equal to the loaded voltage gain
D) can be less than or equal to the loaded voltage gain

5) Depending on the configuration of the amplifier, the magnitude of the no-load voltage gain for a single BJT transistor amplifier typically ranges from _____.
A) 10 to about 10,000
B) a hundred to about a million
C) just a little less than 1 to a few hundred
D) None of the above

6) Depending on the configuration of the amplifier, the magnitude of the no-load current gain for a single BJT transistor amplifier typically ranges from _____.
A) 10 to about 10,000
B) one to about a thousand
C) just a little less than 1 to a level that may exceed one hundred
D) None of the above

7) Determine the input impedance for this two-port network when $V_S = 50$ mV, $I_i = 20$ μA, and $R_{sense} = 500$ Ω.



A) 2000 Ω
B) 20.0 kΩ
C) 200.0 kΩ
D) 2.0 MΩ

8) The r_e transistor model replaces the base-emitter junction with _____.
A) a constant voltage
B) an open circuit
C) the ac resistance of the forward-biased diode at the operating point
D) a diode

9) The h-parameter model uses _____ parameters to describe the equivalent circuit of the BJT transistor.
A) two
B) three
C) four
D) five

10) The h-parameter model and the r_e parameter models are almost identical if the parameter _____ in the h-parameter model is ignored.
A) h_f
B) h_o
C) h_r
D) h_i

11) Determine the equivalent values for β and r_e, given the following h-parameter model values for a common-emitter amplifier: $h_{ie} = 1450$ k Ω, $h_{oe} = 17.5$μS k, $h_{fe} = 125$, and $h_{re}e = 0.4 \times 10^{-3}$.
A) 17.5 μS and 0.4×10^{-3}
B) 125 and 1.450 kΩ
C) 1.450 kΩ and 17.5 μS
D) 0.4×10^{-3} and 125

12) The approximation that allows superposition to be used to isolate the ac analysis and the dc analysis of small-signal amplifiers is that the circuit response is _____.
A) non-linear
B) linear
C) dc linear and ac non-linear
D) dc non-linear and ac linear

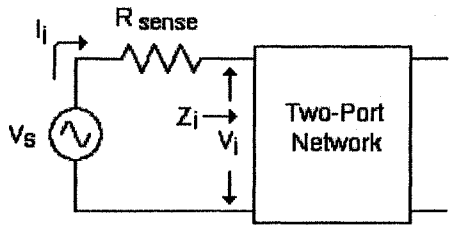
13) A _____ is a combination of circuit elements, properly chosen, that best approximate the actual behavior of a semiconductor device under specific operating conditions.
A) circuit
B) schematic
C) model
D) monolithic IC

14) The input impedance of a BJT is _____.
A) inductive
B) capacitive
C) resistive
D) resistive and capacitive

15) The output impedance of a BJT is _____.
A) capacitive
B) resistive and capacitive
C) resistive
D) resistive and inductive

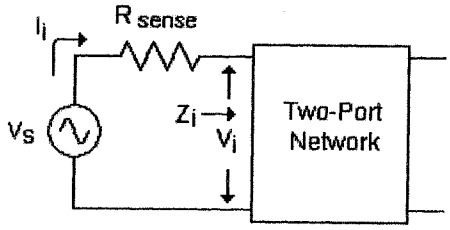
16) For BJT amplifiers, the no-load voltage gain is _____.
A) less than the loaded voltage gain
B) equal to the loaded voltage gain
C) greater than the loaded voltage gain
D) equal to zero

17) Given this configuration, determine the input impedance if $V_S = 40$ mV, $R_{sense} = 0.5$ kΩ, and the input current is 20 μA.



A) 1.5 MΩ
B) 5.822 MΩ
C) 1,500 Ω
D) 582 kΩ

18) Given this configuration, determine the input voltage if $V_S = 40$ mV, $R_{sense} = 0.5$ kΩ, and the input current is 20 μA.



A) 55 m
B) 40 mV
C) 35 mV
D) 30 mV

19) Given the two-port configuration of a BJT amplifier, determine the input voltage if $V_S = 18$ mV, $R_{sense} = 600$ Ω, $V_O = 3.6$ V, and the input current is 20 μA.
A) 12 mV
B) 16 mV
C) 17.994 mV
D) 21.6 m

20) Given the two-port configuration of a BJT amplifier, determine the input impedance if $V_S = 18$ mV, $R_{sense} = 600$Ω, $V_O = 3.6$ V, and the input current is 20 μA.
A) 120 Ω
B) 1,200 Ω
C) 23 kΩ
D) 27 kΩ

21) Given a two-port configuration of a BJT amplifier, determine the no-load voltage gain if $V_S = 18$ mV, $R_{sense} = 600$ Ω, $V_O = 3.6$ V, and the input current is 10 μA.
A) 100
B) 200
C) 300
D) 400

22) Given a two-port BJT amplifier configuration, determine the loaded voltage gain if $V_S = 18$ mV, $R_{sense} = 600$ Ω, $V_O = 3.6$ V, and the input current is 10 μA.
A) 96.66
B) 112.33
C) 133.33
D) 150

23) The r_e transistor model replaces the _____ with the junction diode's ac resistance.
A) collector-base junction
B) collector-emitter junction
C) emitter—base junction
D) All of the above

24) For the common-base configuration, typical values of the output impedance range is _____.
A) 2 Ω to 50 Ω
B) 50 Ω to 1000 Ω
C) 100 Ω to 10,000 Ω
D) l MΩ to 2 MΩ

25) The input impedance of the common-emitter configuration is _____.
A) inversely related to the transistor beta
B) directly related to the transistor beta
C) equal to the transistor beta
D) None of the above

26) The common-emitter configuration has a current gain that is equal to _____.
A) β / 2
B) β
C) 2β
D) 20β

27) The equation that correctly defines one of the hybrid parameters is _____.
A) $V_o = h_{11} \times I_i + h_{21} \times V_i$
B) $V_i = h_{11} \times I_i + h_i \times 2V_o$
C) $I_o = h_{12} \times V_o + h_{22} \times V_o$
D) $I_i = h_{21} \times I_o + h_{22} \times V_o$

28) The h_{12} hybrid parameter is defined as the _____.
A) open-circuit output admittance
B) open-circuit reverse voltage ratio
C) short-circuit forward current ratio
D) short-circuit input impedance

29) The h_{22} hybrid parameter is defined as the _____.
A) open-circuit output admittance
B) open-circuit reverse voltage ratio
C) short-circuit forward current ratio
D) short-circuit input impedance

30) The hybrid parameter that is represented by the name h_f is _____.
A) h_{11}
B) h_{12}
C) h_{21}
D) h_{22}

31) The h-parameter that is the equivalent of the β of a common-emitter circuit is _____.
A) h_{fe}
B) h_{ie}
C) h_{oe}
D) h_{re}

32) The base input impedance of a BJT is listed on its spec sheet as _____.
A) h_{je}
B) h_{re}
C) h_{ie}
D) h_{oe}

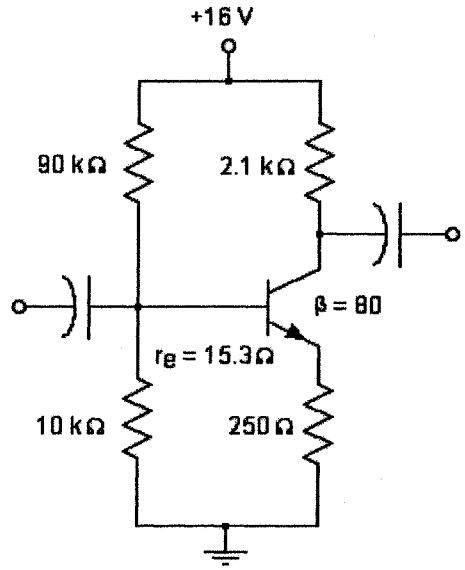
33) A given transistor has the following values: $h_{FE} = 200$, $h_{fe} = 120$, $h_{ie} = 5\ k\Omega$, $h_{re} = 40$, and $h_{oe} = 2500\ \mu S$. What is the value of r_e for the device?
A) 44 Ω
B) 41.7 Ω
C) 400 Ω
D) 25 Ω

34) The hybrid model is used in analysis and design _____.
A) much more than the r_e model
B) more than the r_e model
C) equal to the r_e model
D) less than the r_e model

35) If the resistor in the emitter leg is not bypassed by a capacitor then the input impedance of the small signal amplifier will _____.
A) increase
B) decrease
C) stay the same
D) increase in some cases and decrease in other cases

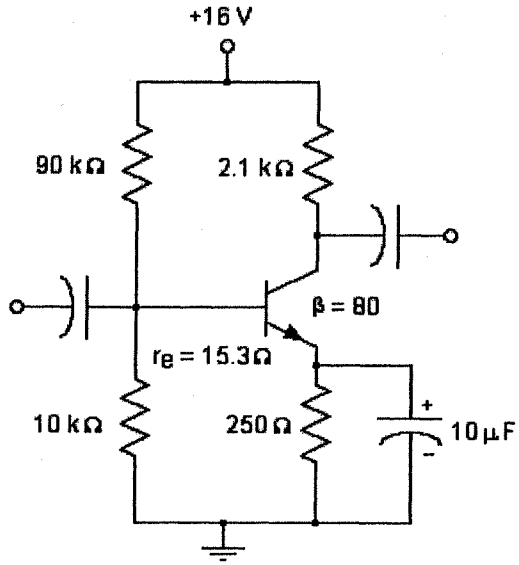
36) If the resistor in the emitter leg is not bypassed by a capacitor then the voltage gain of the small signal amplifier will _____.
A) increase
B) decrease
C) stay the same
D) increase in some cases and decrease in other cases

37) Calculate the voltage gain for this circuit.



A) -137.25
B) -8.4
C) -7.91
D) -16.34

38) Calculate the voltage gain for this circuit.



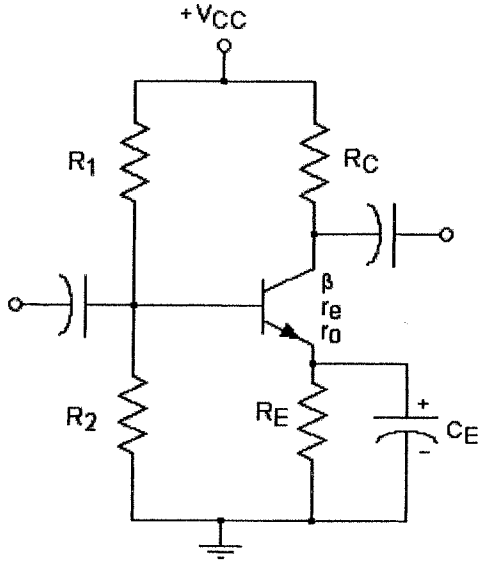
A) -137.25
B) -8.4
C) -7.91
D) -16.34

39) The voltage gain of a very well-designed common collector amplifier configuration, using a *pnp* transistor, is _____.
A) about -0.9
B) about 0.9
C) in the range 0.95 to 0.99
D) in the range -0.95 to -0.99

40) When comparing the common emitter and the common collector amplifiers, the input impedance of the common _____ is much larger and the output impedance of the common _____ is much smaller.
A) collector; emitter
B) collector; collector
C) emitter; collector
D) emitter; emitter

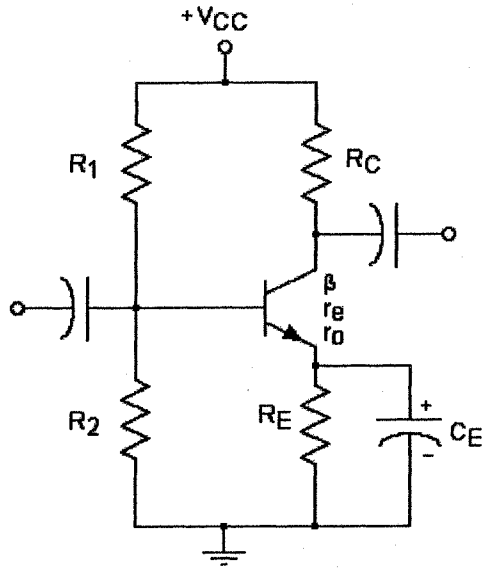
41) The common-base amplifier is characterized as having a relatively _____ input impedance and relatively _____ output impedance.
A) low; high
B) low; low
C) high; low
D) high; high

42) Determine the input impedance for this amplifier circuit.



A) $R_1 \parallel R_2 \parallel (\beta r_e)$
B) $R_1 \parallel R_2$
C) (βr_e)
D) Cannot be determined from the information given

43) Determine the output impedance for this amplifier circuit.



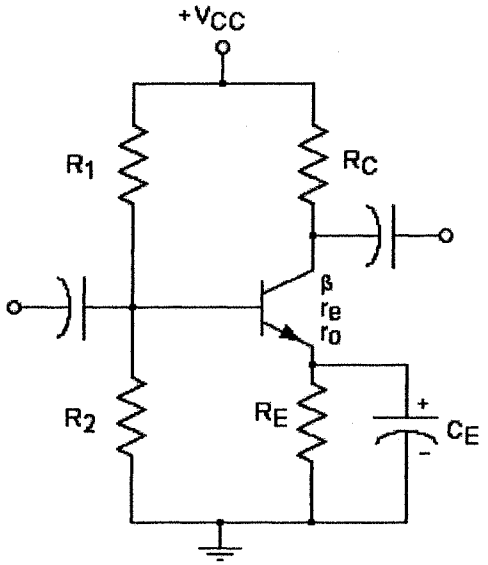
A) $R_C \parallel (\beta r_o)$

B) $R_C \parallel r_o$

C) R_C

D) Cannot be determined from the information given

44) Determine the voltage gain for this amplifier circuit.



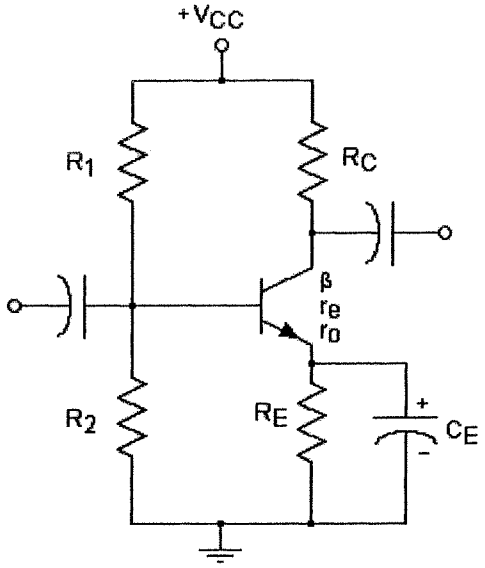
A) $\dfrac{R_C}{(\beta r_e)}$

B) $\dfrac{R_C}{r_e}$

C) $\dfrac{R_C \parallel r_o}{r_e}$

D) Cannot be determined from the information given

45) Determine the current gain for this amplifier circuit.



A) $\frac{-\beta(R_1 \| R_2)}{(R_1 \| R_2) + \beta r_e}$

B) β

C) $\frac{\beta R_C}{R_C + r_o}$

D) Cannot be determined from the information given

46) The common-emitter amplifier has _____.
A) voltage gain, current gain, and power gain
B) voltage gain and power gain, but no current gain
C) current gain and power gain, but no voltage gain
D) current gain and voltage gain, but no power gain

47) A fixed-bias BJT circuit has values of $h_{FE} = 200$ and $h_{fe} = 120$. The ac current gain for the device is _____.
A) 200
B) 120
C) 24,000
D) 320

48) Coupling capacitors are chosen to ensure that the values of X_C are _____ at the amplifier's operating frequency.
A) very small
B) small
C) large
D) very large

49) A common-emitter amplifier has values of $V_E = 1.1$ V, $r_e = 1$ kΩ, and $R_C = 10$ kΩ. What is the value of the voltage gain for the circuit?
A) 10
B) 110
C) 484
D) Cannot be determined with the information given

50) Bypass capacitors are chosen to ensure that the values of X_C are _____ at the amplifier's operating frequency.
A) very small
B) small
C) large
D) very large

51) A common-emitter amplifier with voltage divider bias and a bypassed emitter resistance has values of $R_C = 10$ kΩ, $r_e = 25$ Ω, and $h_{FE} = 150$. What is the value of the voltage gain for the circuit?
A) 3750
B) 60,000
C) 400
D) Cannot be determined with the information given

52) If a bypass capacitor opens, the value of r_e _____.
A) increases
B) decreases
C) remains the same
D) goes to zero

53) Which of the following circuit conditions indicates that a bypass capacitor is open?
A) The presence of a dc voltage at the BJT's emitter terminal.
B) The voltage gain increases significantly.
C) The loss of the ac signal at the base terminal of the BJT.
D) None of the above

54) A fixed-bias common-emitter amplifier has an unbypassed, 1.2 kΩ emitter resistor, and 270 kΩ base resistor. If $r_e = 5$ Ω and $\beta = 200$, what is the voltage gain?
A) 4.64
B) 10.3
C) 24.64
D) 103.3

55) A fixed-bias common-emitter amplifier has an unbypassed, 1.2 kΩ emitter resistor, and 270 kΩ base resistor. If $r_e = 5$ Ω and $\beta = 200$, what is the current gain?
A) 1.05
B) 20.55
C) 105.55
D) 565.5

56) A common-emitter amplifier with emitter bias has values of $r_e = 25\ \Omega$, $h_{fe} = 150$, $h_{FE} = 200$, and $R_E = 2\ k\Omega$. What is the value of Z_o for the circuit?
A) 3750 Ω
B) 303.75 kΩ
C) 5 kΩ
D) 430 kΩ

57) The common—collector amplifier (emitter-follower) has _____.
A) voltage gain, current gain, and power gain
B) voltage gain and power gain, but no current gain
C) current gain and power gain, but no voltage gain
D) current gain and voltage gain, but no power gain

58) The common-base amplifier has _____.
A) voltage gain, current gain, and power gain
B) voltage gain and power gain, but no current gain
C) current gain and power gain, but no voltage gain
D) current gain and voltage gain, but no power gain

59) A transistor amplifier has an input signal applied to its emitter terminal and an output signal taken from its collector terminal. The amplifier is a(n) _____.
A) common-emitter amplifier
B) common-base amplifier
C) common-collector amplifier
D) emitter follower

60) An emitter follower has the following values: $h_{ie} = 3\ k\Omega$, $h_{fe} = 150$, and $h_{RE} = 1.5\ k\Omega$. What is the voltage gain for the circuit?
A) 0.5
B) 0.9925
C) 0.9868
D) Cannot be determined with the information given

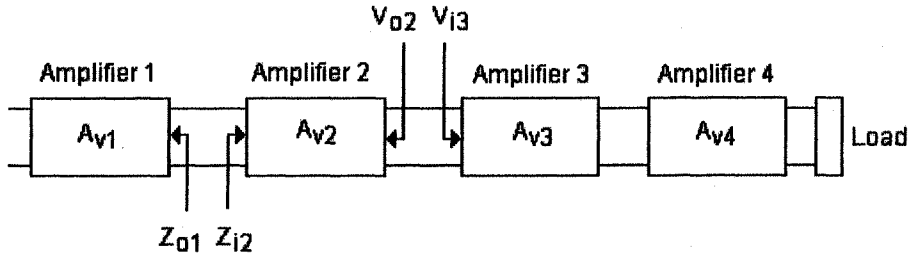
61) Which transistor amplifier configuration has a 180° voltage phase shift from input to output?
A) common-emitter
B) common-collector
C) common-base
D) None of the above

62) Which transistor amplifier configuration has a 180° current phase shift from input to output?
A) common-emitter
B) common-collector
C) common-base
D) None of the above

63) Amplifier ac input and output currents are _____.
A) always 180°s out of phase
B) 180° out of phase in all but one amplifier configuration
C) in phase in all but one amplifier configuration
D) always in phase

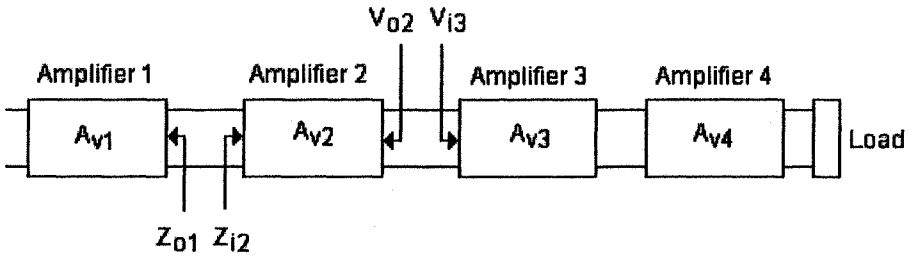
64) Amplifier ac input and output voltages are _____.
A) always 180° out of phase
B) 180° out of phase in all but one amplifier configuration
C) in phase in all but one amplifier configuration
D) always in phase

65) For the cascaded amplifier shown here, input impedance Z_{i2} is _____ the load resistance for Amplifier 1.



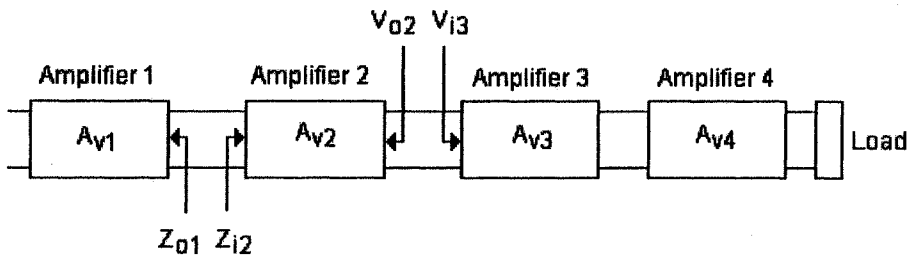
A) less than
B) large than
C) exactly equal to
D) Cannot be determined from the information provided

66) For the cascaded amplifier shown here, the output impedance Z_{o1} is _____ the source resistance seen by Amplifier 2.



A) less than
B) larger than
C) much less than
D) exactly equal to

67) For the cascaded amplifier shown here, the output voltage V_{o2} is _____ the input voltage V_{i3}.



A) less than
B) larger than
C) much larger than
D) exactly equal to

68) The typical β for a Darlington amplifier is _____.
A) a very small value in the range 4 to 40
B) a moderate value in the range of 40 to 400
C) a slightly higher value in the range 400 to 4000
D) a much higher value in the range of 4000 to about 40,000

69) Replacing a standard transistor with a Darlington pair in an emitter follower causes the voltage gain to _____.
A) decrease
B) increase
C) remain the same
D) be exactly equal to 1

70) The feedback pair and the Darlington pair are very similar to each other. One difference between them is _____.
A) the feedback pair uses one *npn* and one *pnp* transistor
B) the Darlington pair uses one *npn* and one *pnp* transistor
C) there is no difference between them
D) there is no similarity between them

71) The feedback pair is similar to the Darlington circuit but it is _____.
A) simpler to analyze
B) more complex
C) more used
D) None of the above

72) Current mirror circuits provide _____ used in _____.
A) constant voltage sources; integrated circuits
B) constant current sources; power circuits
C) constant current sources; integrated circuits
D) constant voltage sources; power circuits

73) Applying an input signal to one of the inputs of differential amplifier and connecting the other input to ground, you create a _____ configuration.
A) single-ended
B) double-ended
C) common-mode
D) None of the above

ANSWER KEY: Chapter 5: BJT ac Analysis

1) A
2) A
3) C
4) A
5) C
6) C
7) A
8) C
9) A
10) C
11) B
12) B
13) C
14) C
15) C
16) C
17) C
18) D
19) A
20) B
21) B
22) C
23) C
24) D
25) B
26) B
27) B
28) B
29) A
30) C
31) A
32) C
33) B
34) D
35) A
36) B
37) C
38) A
39) C
40) B
41) A
42) A
43) B
44) A
45) A
46) A
47) B
48) A
49) D
50) A
51) C
52) C
53) B
54) A
55) C
56) B
57) C
58) B
59) B
60) C
61) A
62) D
63) D
64) C
65) C
66) D
67) D
68) D
69) A
70) A
71) B
72) C
73) A

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 6: Field-Effect Transistors

1) The maximum current in a JFET is defined as I_{DSS} and occurs when V_{GS} is equal to _____.
A) zero Volts
B) pinch-off voltage
C) a small positive voltage
D) a voltage greater than the pinch-off voltage

2) Shockley's equation defines the _____ of the FET and are unaffected by the network in which the device is employed.
A) V_{GS} characteristics
B) drain characteristics
C) input/output characteristics
D) transfer characteristics

3) For an *n*-channel JFFT, I_{DSS} = 8 mA, and V_P = -6 V. If V_{GS} = -2 V. What is the value of the drain current I_D?
A) 2.666 mA
B) 3.5 mA
C) 3.55 mA
D) 5.33 mA

4) For an *n*-channel JFET I_{DSS} = 8 mA and Vp = -6 Volts. If I_D = 6 mA. What is the value of the gate-to-source voltage, V_{GS}?
A) -0.8 V
B) -1.5 V
C) 0.1335 V
D) -4.5 V

5) The drain characteristics for a FET that you see on a curve tracer are drawn for equal step increases in the V_{GS} values, yet they are spaced further apart as V_{GS} gets closer to zero. Why?
A) This is true for only some FET devices, not all.
B) The curve depends on the FET device used.
C) Due to the square relation between I_D and V_{GS}, as V_{GS} gets closer to zero I_D increases faster so the curves are spaced apart further.
D) None of the above

6) The depletion type of MOSFET can operate in the _____.
A) depletion mode only
B) enhancement mode only
C) in the depletion mode and the enhancement mode
D) None of the above

7) For an *n*-channel depletion type of MOSFET, if $V_{GS} > 0$ then I_{DSS} will be _____.
A) less than
B) more than
C) equal to
D) V_{GS} is not allowed to be greater than zero.

8) For an *n*-channel depletion MOSFET, I_{DSS} = 8 mA and V_P = -6 V. If V_{GS} = 0.8 V, what is the value of the drain current, I_D?
A) 8 mA
B) 10.25 μA
C) 10.28 mA
D) 6 mA

9) For an *n*-channel depletion MOSFET I_{DSS} = 8 mA and V_P = -6 V. If I_D = 0.0095 A, what is the value of the gate-to-source voltage, V_{GS}?
A) 0.54 V
B) -0.54 V
C) 0.1335 V
D) 6.54 V

10) For $V_{GS} < V_{TH}$ in an enhancement MOSFET the drain current will be _____.
A) 10.0 μA
B) 1.0 μA
C) zero
D) -1.0 μA

11) Enhancement-type MOSFETs operate in the _____.
A) depletion mode only
B) depletion mode and the enhancement mode
C) enhancement mode only
D) None of the above

12) Many MOSFET devices now contain internal _____ that protect these devices from static electricity.
A) BJT transistors to bypass the static charge
B) back-to-back zener diodes
C) capacitors to collect and store the static charge
D) Nothing can be done to protect these devices from accidental static discharge except very careful handling.

13) The type of FFT that has the best switching speed performance is the _____.
A) CMOS
B) PMOS
C) NMOS
D) VMOS

14) A CMOS inverter is biased with a +10 V V_{SS} supply. The input to the inverter varies between 0 V and +10 V. When the input to the inverter is +10 V, the output from the circuit is _____.
A) +10 V
B) -10 V
C) zero
D) The circuit cannot have an input voltage that is equal to the supply voltage.

15) The primary difference between BJT and FET types of transistors is that _____.
A) BJTs are voltage controlled and FETs are current controlled
B) BJTs are current controlled and FETs are voltage controlled
C) BJTs amplify better than FETs
D) None of the above

16) In the family of FETs, you can expect to find _____.
A) an *n*-channel type
B) a *p*-channel type
C) unipolar structure
D) All of the above

17) FETs usually _____.
A) are less sensitive to temperature change than BJTs
B) have a higher input impudence than BJTs
C) are smaller in construction than BJTs
D) All of the above

18) The level of drain-to-source voltage where the two depletions regions appear to touch is known as _____.
A) the depletion zone
B) channel establishment
C) pinch-off
D) channel saturation

19) The JFET is a _____.
A) voltage-controlled device
B) current-controlled device
C) frequency-controlled device
D) power-controlled device

20) The _____ terminal of the JFFT is the equivalent of the collector terminal of a BJT.
A) gate
B) drain
C) source
D) anode

21) The _____ terminal of the JFET is the equivalent of the base terminal of a BJT.
A) gate
B) drain
C) source
D) anode

22) The _____ terminal of the JFEI' is the equivalent of the emitter terminal of a BJT.
A) gate
B) drain
C) source
D) anode

23) The _____ JFET uses a positive drain supply voltage.
A) *n*-channel
B) *p*-channel
C) MDS
D) CMOS

24) The region of the characteristic curve family for the junction FET that is normally used for linear amplification is _____.
A) the constant-current region
B) the saturation region
C) the linear amplification region
D) All of the above

25) The collector current, I_C, of a BJT flows through two junctions. The drain current of an FET, I_D, flows through _____ junctions.
A) 0
B) 1
C) 2
D) 3

26) As the channel width of a JFET decreases, the source-to-drain resistance _____.
A) increases
B) decreases
C) remains constant
D) is not affected

27) Which of the following is usually used to control the channel width of a given JFET?
A) the source voltage
B) the gate-to-source voltage
C) the operating frequency
D) the drain current

28) The region of the JFET drain curve that lies between pinch-off and breakdown is called _____.
A) the constant-voltage region
B) the ohmic region
C) the saturation region
D) None of the above

29) The value of gate-to-source voltage that causes the drain current to reach its maximum value at a given value of drain voltage is called _____.
A) V_{DMAX}
B) pinch-off voltage
C) V_{DSS}
D) None of the above

30) The FET transfer characteristic curve is defined by Shockley's equation and is _____.
A) unaffected by the network in which it is used
B) directly related to the drain resistor
C) inversely related to the drain resistor
D) inversely related to the sum of the drain and source resistors

31) What two parameters represent the FET transfer characteristic?
A) drain-to-source voltage and gate-to-source voltage
B) drain-to-source voltage and drain current
C) gate-to-source voltage and drain current
D) gate current and drain current

32) The value of drain current is always _____ the value of the short circuit drain current I_{DSS} for a given JFET.
A) less than
B) equal to
C) less than or equal to
D) greater than

33) A JFET has values of I_{DSS} = mA and V_{GSOFF} = -5 V. What is the value of I_D at V_{GS} = -3 V?
A) 1.6 mA
B) 3.6 mA
C) 25.6 mA
D) 4 mA

34) A given JFET has values of V = 10 V and I_{DSS} = 8 mA. What is the value of V_{GSOFF} for the device?
A) +10 V
B) -10 V
C) -5 V
D) Cannot be determined from the information given

35) The enhancement-type and the depletion-type FETs are subclasses of _____.
A) junction FET
B) metal-oxide-semiconductor FETs
C) BJTs
D) bipolar FETs

36) The depletion-type MOSFET' has specifications and many characteristics that are similar to the _____.
A) *pnp* BJT
B) *npn* BJT
C) JFET
D) None of the above

37) Which of the following FETs is the best choice when the gate-source voltage has both positive and negative swings?
A) JFET
B) enhancement MOSFET
C) depletion MOSFET
D) CMOS

38) MOSFETs typically have an input impedance value that is _____.
A) higher than the JFET
B) lower than the JFET
C) equal to theJFET
D) randomly defined relative to the JFET

39) D-MOSFETs can operate in _____.
A) the depletion mode onl
B) the enhancement mode only
C) the depletion mode and the enhancement mode
D) All of the above

40) MOSFETs are also referred to as _____.
A) substrates
B) IGFETs
C) DEFETs
D) SiO-FETs

41) Which of the following is true for an *n*-channel D-MOSFET that is being operated in the depletion mode?
A) $I_D > I_{DSS}$ and V_{GS} is positive.
B) $I_D < I_{DSS}$ and V_{GS} is negative.
C) $I_D > I_{DSS}$ and V_{GS} is negative.
D) $I_D < I_{DSS}$ and V_{GS} is positive.

42) A D-MOSFET has values of D = 15.63 mA and VGS = +1 V. What is the value of I_{DSS}?
A) 0 mA
B) 5 mA
C) 10 mA
D) None of the above

43) For levels of gate-to-source voltage greater than the threshold voltage, the drain current is directly related to the _____.
A) square of the difference between the gate-to-source voltage and the threshold voltage
B) gate-to-drain voltage
C) square of the gate current
D) None of the above

44) For a gate-to-drain voltage less than the threshold level the drain current of an enhancement-type MOSFET is _____.
A) 100 mA
B) 10 mA
C) 1.0 mA
D) 0 mA

45) The EMOSFET can operate in _____.
A) the depletion mode only
B) the enhancement mode only
C) the depletion mode and the enhancement mode
D) All of the above

46) A major disadvantage of MOSFETs is _____.
A) its high input impedance
B) that it is a voltage operated device
C) that it is sensitive to electrostatic discharges
D) None of the above

47) Many MOSFET devices now contain internal _____ that protect them from static electricity.
A) BJTs
B) Zener diodes
C) *p-n* junction diodes
D) capacitors

48) The power-handling levels of a MOSFET _____.
A) is usually less than 1 W
B) is about 10 W
C) is similar to that of a vacuum tube
D) is usually about 100 W

49) When compared with commercially available planar MOSFETs, VMOS FETs have _____.
A) reduced channel resistance
B) higher current capability
C) higher power ratings
D) All of the above

50) The VMOS FET typically has switching times that are _____.
A) very slow
B) half that of the typical BJT
C) twice that of the typical BIT
D) 20 times that of the typical BJT

51) VMOS is a special-purpose type of _____.
A) D-MOSFET
B) E-MOSFET
C) JFET
D) BJT

52) A relatively high input impedance, fast switching speeds, and low operating power describe the characteristics of the _____ family.
A) BJT
B) enhancement-type MOSFET
C) VMOS FET
D) CMOS FET

53) The FET that typically has the best switching speed performance is a(n) _____.
A) CMOS
B) JFET
C) NMOS
D) VMOS

54) CMOS stands for _____.
A) complementary MOS
B) current MOS
C) capacitive MOS
D) conductive MOS

55) A CMOS inverter has a +10 V supply and an input that varies between 0 V and +10 V. When the input to the circuit is +10 V, the output from the circuit is _____.
A) -10 V
B) 0 V
C) +10 V
D) Cannot be determined from the information given

ANSWER KEY: Chapter 6: Field-Effect Transistors

1) A
2) D
3) C
4) A
5) C
6) C
7) A
8) C
9) A
10) C
11) C
12) B
13) A
14) C
15) B
16) D
17) D
18) C
19) A
20) B
21) A
22) C
23) A
24) D
25) A
26) A
27) B
28) C
29) A
30) A
31) C
32) C
33) A
34) A
35) B
36) C
37) C
38) A
39) D
40) B
41) B
42) C
43) A
44) D
45) B
46) C
47) B
48) A
49) D
50) B
51) B
52) D
53) A
54) A
55) B

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 7: FET Biasing

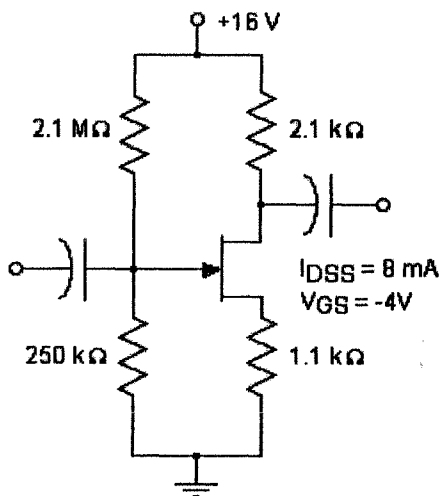
1) A JFET can be biased in several different ways. The common method(s) of biasing an *n*-channel JFET is(are) _____.
A) self-bias configuration
B) voltage-divider bias configuration
C) fixed-bias configuration
D) All of the above

2) In a self-bias circuit for an *n*-channel JFET transistor the self-bias line _____.
A) is straight up and down parallel to the I_D axis
B) is straight left and right parallel to the V_{GS} axis
C) is slanted and passing through the I_D and the V_{GS} axis on the positive side
D) is slanted and passes through origin

3) In a self-bias circuit for an *n*-channel JFET transistor the self-bias line _____.
A) is straight up and down parallel to the I_D axis
B) is straight left and right parallel to the V_{GS} axis
C) is slanted and passing through the I_D and the V_{GS} axis on the positive side
D) is slanted and passes through origin

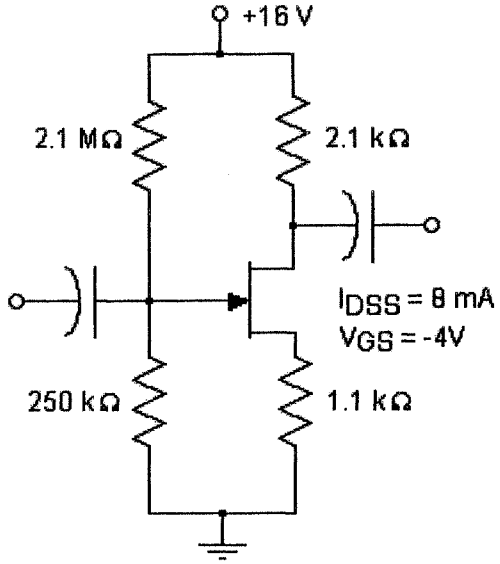
4) In a fixed-bias circuit for an *n*-channel JFET transistor the bias line _____.
A) is straight up and down parallel to the I_D axis
B) is straight left and right parallel to the V_{GS} axis
C) is slanted and passing through the I_D and the V_{GS} axis on the positive side
D) is slanted and passes through origin

5) Calculate the quiescent drain current and the gate-to-source voltage for this voltage-divider bias circuit.



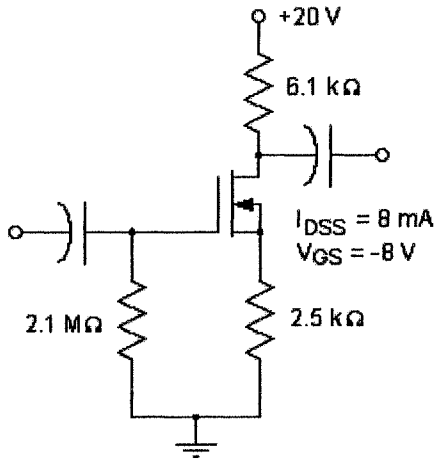
A) I_{DQ} = 2.4 mA and V_{GSQ} = 1.8 V
B) I_{DQ} = 2.4 mA and V_{GSQ} Q = -1.8 V
C) I_{DQ} = 1.2 mA and V_{GSQ} Q = -3.6 V
D) I_{DQ} = 1.2 mA and V_{GSQ} Q = 3.6 V

6) Calculate the drain-gate voltage for this voltage-divider bias circuit.



A) $V_{DG} = 8.42$ V
B) $V_{DG} = 7.42$ V
C) $V_{DG} = 6.42$ V
D) $V_{DG} = 5.42$ V

7) Calculate the quiescent drain current for this self-bias depletion mode MOSFET transistor amplifier.

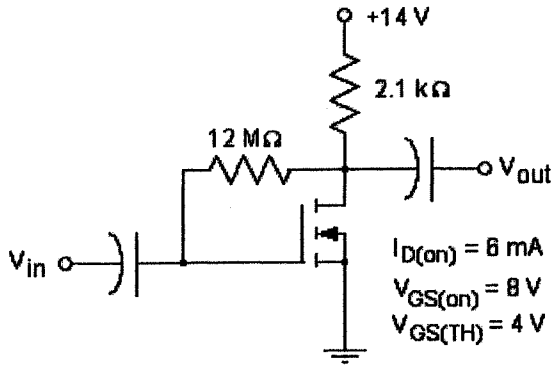


A) $I_{DQ} = 1.9$ mA
B) $I_{DQ} = 1.7$ mA
C) $I_{DQ} = 1.5$ mA
D) $I_{DQ} = 1.3$ mA

8) In the enhancement type of MOSFET the channel is formed when the gate-to-source voltage _____.

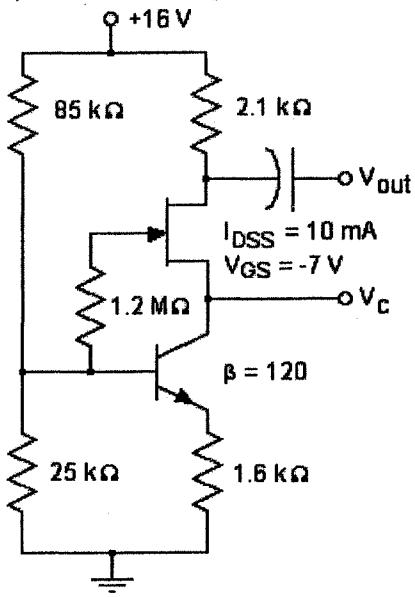
A) exceeds the pinch-off voltage
B) is less than the pinch-off voltage
C) is less than the threshold voltage
D) exceeds the threshold voltage

9) Calculate the quiescent drain current for this circuit.



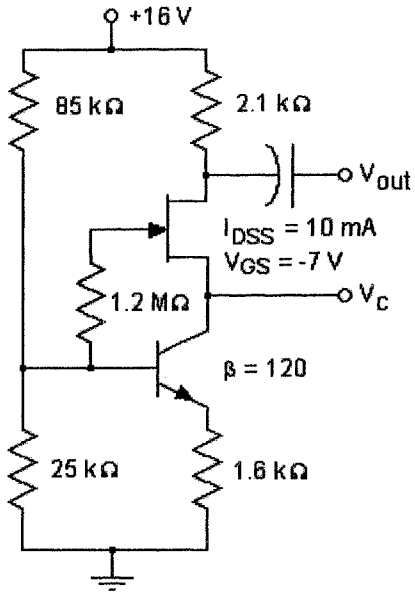
A) $I_{DQ} = 2.5$ mA
B) $I_{DQ} = 2.9$ mA
C) $I_{DQ} = 3.3$ mA
D) $I_{DQ} = 3.7$ mA

10) Calculate the quiescent collector current for this circuit.



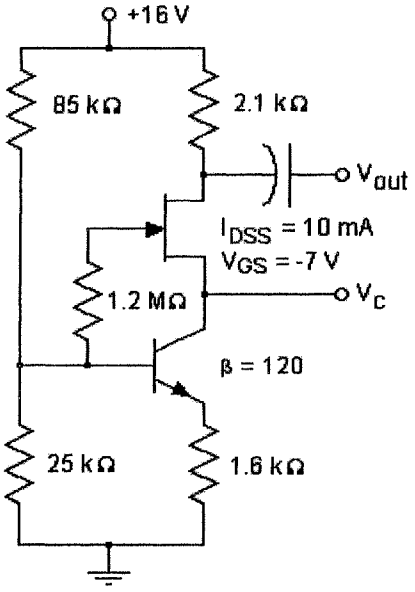
A) $I_{CQ} = 1.7$ mA
B) $I_{CQ} = 1.9$ mA
C) $I_{CQ} = 2.1$ mA
D) $I_{CQ} = 2.3$ mA

11) Calculate the quiescent collector-to-emitter voltage for the BJT in this circuit.



A) $V_{CE} = 3.63$ V
B) $V_{CE} = 7.78$ V
C) $V_{CE} = -4.14$ V
D) $V_{CE} = 5.11$ V

12) Calculate the voltage at the drain of the JFET in this combination network.



A) $V_D = 8.22$ V
B) $V_D = 4.14$ V
C) $V_D = 12.5$ V
D) $V_D = 3.5$ V

13) Generally, it is a good design practice for linear amplifiers to choose the operating point that is approximately _____.
A) near the saturation region
B) near the cut-off region
C) in the center of the active region
D) near the origin

14) The analysis that we mostly work with is that of the *n*-channel device. For *p*-channel devices the transfer curve employed is the _____ image and the defined current directions are _____.
A) identical; the same
B) mirror; the same
C) mirror; reversed
D) identical; reversed

15) It is important to remember that when the JFET is used as a voltage variable resistor, which is one of its practical applications, the voltage V_{DS} is _____ $V_{DS(max)}$ and $|V_{GS}|$ is _____ $|V_P|$.
A) very much greater than; very much greater than
B) very much less than; very much greater than
C) very much greater than; very much less than
D) very much less than; very much less than

16) The simplest biasing arrangement for the *n*-channel JFET is _____.
A) voltage-divider bias
B) variable bias
C) drain-feedback bias
D) fixed bias

17) The fixed-bias technique requires _____ power supplies.
A) 1
B) 2
C) 3
D) 4

18) A JFET has the following ratings: VP = -2 V to -5 V and an I_{DSS} = 4 mA. The device is being used in a fixed-bias circuit with a gate supply voltage of V_{GG} = 1 V. What is the difference between the minimum and maximum values of I_D values for the circuit?
A) 7.6 mA
B) 9.6 mA
C) 6.68 mA
D) 8.6 mA

19) The self-bias configuration develops the controlling gate-to-source voltage across a resistor introduced in the _____.
A) drain leg
B) gate leg
C) source leg
D) None of the above

20) A characteristic of voltage divider-bias in FET circuits is _____.
A) the current in both R_1 and R_2 is the same
B) the voltage drop across R_2 is V_{GS}
C) the gate current is zero
D) All of the above

21) When using voltage divider-bias in FET amplifiers, increasing the size of the source resistor results in _____.
A) lower quiescent values
B) more positive of V_{GS}
C) a larger value of drain current
D) All of the above

22) The primary difference between JFETs and depletion-type MOSFETs is _____.
A) JFETs can have positive values of V_{GS} and levels of drain current that exceed I_{DSS}
B) depletion-type MOSFETs can have positive values of V_{GS} and levels of I_D that exceed I_{DSS}
C) depletion-type MOSFETs can have only positive of V_{GS}
D) JFETs can have only positive values of V_{GS}

23) _____ biasing may be used with D-MOSFETs but not with JFETs.
A) Gate-drain
B) Zero
C) Gate-cutoff
D) Current-source

24) A popular arrangement for enhancement type MOSFET biasing is _____.
A) drain-feedback biasing
B) fixed bias
C) source-resistor bias
D) All of the above

25) An E-MOSFET has values of $V_{GSth} = 2$ V and $I_{DON} = 8$ mA when $V_{FS} = 10$ V. What is the value of k for the device?
A) 0.0001
B) 0.000125
C) 80
D) Cannot be determined from the information given

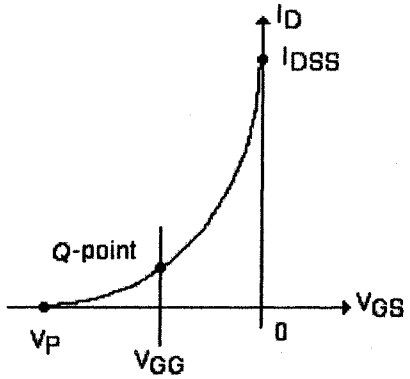
26) An E-MOSFET has values of $V_{GSth} = 4$ V and $I_{DON} = 12$ mA when $V_{GS} = 10$ V. The device is being used in a circuit that has a value of $V_{GS} = 6$ V. What is the value of I_D for the circuit?
A) 13.33 mA
B) 1 mA
C) 1.33 m
D) 0 mA

27) Which of the following biasing circuits can be used with E-MOSFETs?
A) self bias
B) zero bias
C) drain-feedback bias
D) current-source bias

28) Generally, it is good design practice for linear amplifiers to have operating points that close to _____.

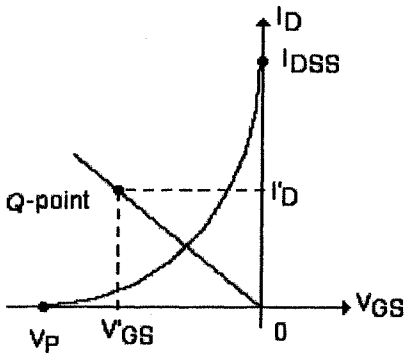
A) are close to saturation level
B) the cut-off region
C) the midpoint of the load line
D) None of the above

29) This graphical solution represents _____.



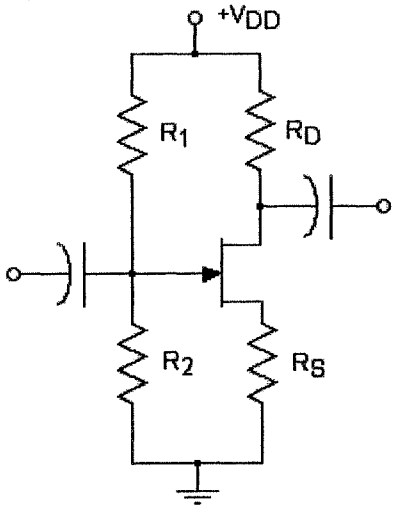
A) voltage-divider bias for an *n*-channel JFET
B) self bias for an *n*-channel JFET
C) fixed-bias configuration for an *n*-channel JFET.
D) None of the above

30) This graphical solution represents _____.



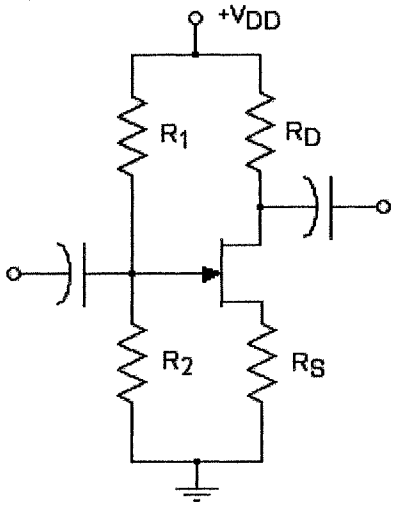
A) fixed bias for an *n*-channel JFET
B) voltage-divider bias for an *n*-channel JFET
C) self bias for an *n*-channel JFET
D) None of the above

31) Which of the following is true for this circuit?



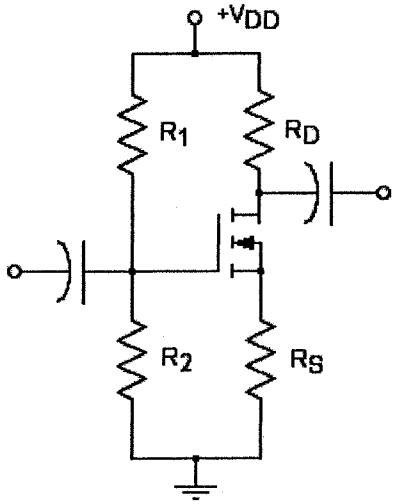
A) V_G is measured between the gate and common.
B) V_G is measured between the gate and source terminals.
C) V_G is equal to the voltage across R_S.
D) V_G is always close to +0.7 V.

32) Which one of the following statements about this circuit is true?



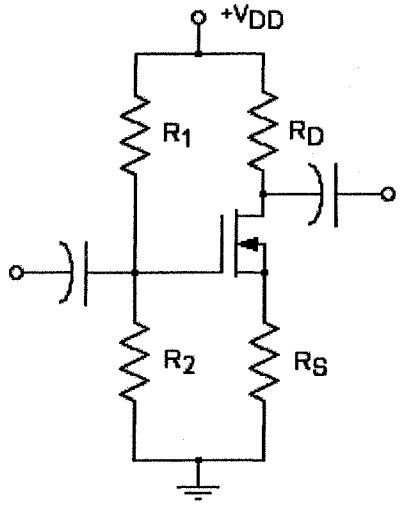
A) V_{GS} is measured across R_2.
B) V_{GS} is measured between the gate and source terminals.
C) V_{GS} is equal to the voltage across R_S.
D) V_{GS} is always close to +0.7 V.

33) Which of the following equations properly characterize the value of V_{DS} for this circuit?



A) $V_{DS} = V_D - V_S$
B) $V_{DS} = V_{DD} - I_D(R_D + R_S)$
C) $V_{DS} = V_{R1} + V_{R2} - I_D(R_D + R_S)$
D) All of the above

34) Which of the following expressions is correct for this circuit?



A) $V_{GS} = V_G - I_D R_S$
B) $V_{GS} = V_G - I_S R_S$
C) $V_{GS} = V_G - V_S$
D) All of the above

ANSWER KEY: Chapter 7: FET Biasing - Answer Key

1) D
2) D
3) D
4) A
5) B
6) A
7) B
8) D
9) C
10) A
11) D
12) C
13) C
14) C
15) D
16) D
17) B
18) C
19) C
20) D
21) A
22) B
23) B
24) A
25) B
26) C
27) C
28) C
29) C
30) B
31) A
32) B
33) D
34) D

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 8: FET Amplifiers

1) A FET amplifier is considered a voltage-controlled device, whereas a BJT is a current-controlled device.

2) The input impedance of a FET amplifier tends to be much greater than a comparable BJT amplifier.

3) The voltage gain of a FET amplifier is generally larger than that of a comparable BJT amplifier.

4) The output impedance of a BJT amplifier tends to be much less than that of a comparable FET amplifier.

5) The input current for a FET amplifier is generally assumed to be zero.

6) For the typical transistor amplifier, the phase shift due to reactive components of input and output impedances _____.
A) depends upon the values of external reactive components
B) depends upon the values of stray reactance
C) is negligible in most instances
D) is unpredictable

7) The input impedance of a common-base BJT configuration is typically _____.
A) between 100 kΩ and 10 MΩ
B) between 100 Ω and 100 kΩ
C) less than 50 Ω
D) so large that the input current can be considered to be zero

8) The input impedance of a common-emitter configuration is typically _____.
A) between 100 kΩ and 10 MΩ
B) between 300 Ω and 10 kΩ
C) less than 50 Ω
D) so large that the input current can be considered to be zero

9) The expression, $r_e = \frac{26mV}{I_E}$, shows that _____.
A) r_e is a constant value that is determined by the internal structure of the transistor
B) the emitter-base voltage for the configuration is a constant 26 mV (as opposed to 700 mV for a forward-biased, common-emitter BJT configuration)
C) it is rather easy to determine the correct value for the emitter resistor
D) the ac input resistance of the configuration is determined by the amount of dc emitter current

10) The input impedance for a common-emitter configuration can be expressed as _____.
A) $Z_i = \beta r_e$
B) $Z_i = r_e$
C) $Z_i = r_e (\alpha - 1)$
D) $Z_i = r_e \left(\frac{1}{\alpha - 1} \right)$

11) The unit of measure for FET transconductance (g_m) is _____.
A) ohms per volt
B) volts per ampere
C) Siemens
D) The term is unitless

12) A source-follower FET configuration is also know as _____ circuit.
A) common-source
B) common-gate
C) common-drain
D) current-controlled

13) In the hybrid equivalent model, h_{ie} is the expression for _____.
A) input impedance
B) voltage feedback ratio
C) small-signal current gain
D) output admittance

14) In the hybrid equivalent model, h_{re} is the expression for _____.
A) input impedance
B) voltage feedback ratio
C) small-signal current gain
D) output admittance

15) In the hybrid equivalent model, h_{fe} is the expression for _____.
A) input impedance
B) voltage feedback ratio
C) small-signal current gain
D) output admittance

16) In the hybrid equivalent model, h_{oe} is the expression for _____.
A) input impedance
B) voltage feedback ratio
C) small-signal current gain
D) output admittance

17) The FET version of the BJT's common-emitter configuration is the _____ circuit.
A) common-source
B) common-gate
C) common-drain
D) common-current

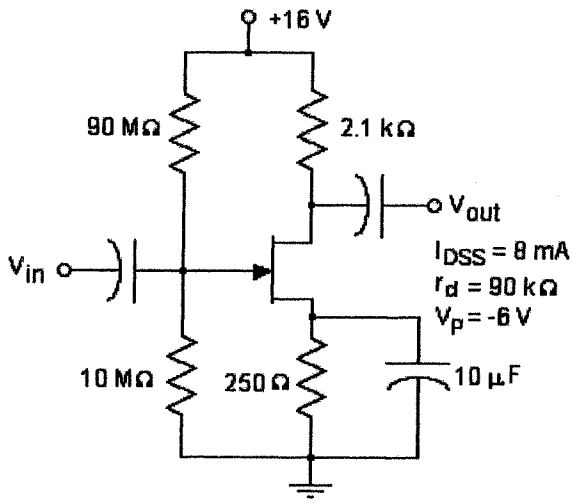
18) The current gain $\left(\frac{i_o}{i_i}\right)$ of an FET amplifier is _____.

A) generally assumed to be zero
B) undefined
C) depends upon the value of g_m for the device
D) less than or equal to 1

19) If the resistor bypass capacitor in the source leg is removed, the voltage gain of the small signal FET amplifier _____.
A) will increase
B) will decrease
C) will stay the same
D) may increase in some cases and decrease in other cases

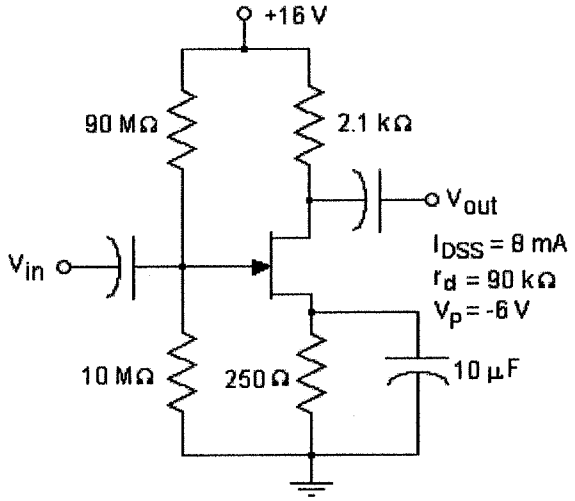
20) The input impedance (Z_i) of an FET is hundreds of times greater than the Z_i of a BJT. This _____ effect on the overall gain of the system.
A) has a very negative
B) has a very positive
C) has no
D) may be positive or negative and depends on the rest of the circuit

21) Calculate the input impedance for this FET amplifier.



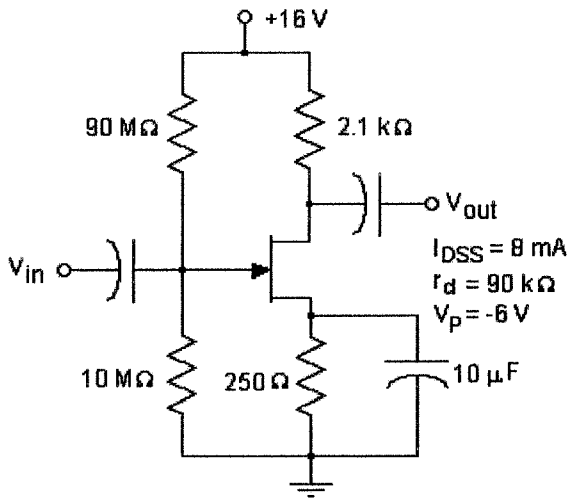
A) $Z_i = R_1 \parallel R_2 = 9\ M\Omega$
B) $Z_i = R_2 = 10\ M\Omega$
C) $Z_i = R_1 = 90\ M\Omega$
D) Z_i = would depend on the drain current I_D.

22) Calculate the output impedance for this FET amplifier.



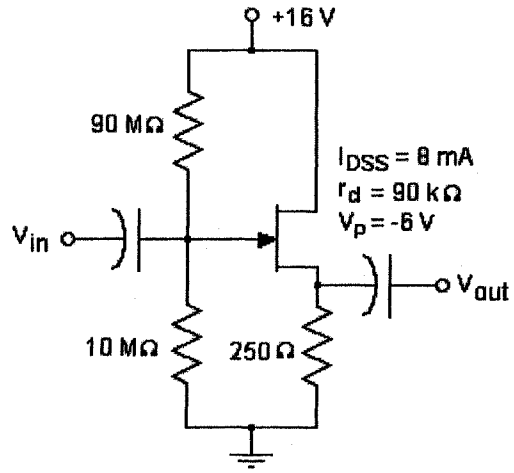
A) $Z_O = R_D = 2.1\ k\Omega$
B) $Z_O = r_d = 90\ k\Omega$
C) $Z_O = R_D \parallel r_d = 2052\ \Omega$
D) Z_O depends on the drain current I_D.

23) Calculate the voltage gain for this FET amplifier. Assume that the transconductance (g_m) for this circuit is 1.85 mS



A) $A_V = -g_m R_D = -3.885$
B) $A_V = -g_m (R_D \parallel rd) = -3.7962$
C) $A_V = -g_m r_d = -166.5$
D) A_V depends on the drain current, I_D.

24) Calculate the voltage gain for this FET amplifier. Assume that $g_m = 1.85$ mS.



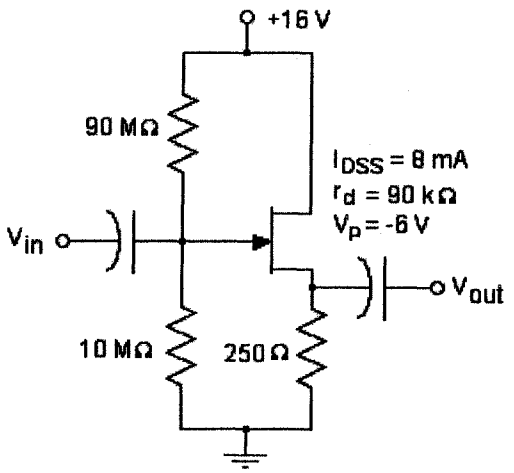
A) $A_V = \dfrac{-g_m(r_d \| R_1 \| R_2)}{1+g_m(r_d \| R_1 \| R_2)} = 0.9939$

B) $A_V = g_m r_d = 166.5$

C) $A_V = g_m R_S = 0.462$

D) A_V would depend on the drain current, I_D.

25) Calculate the input impedance for this amplifier. Assume that $g_m = 1.85$ mS.



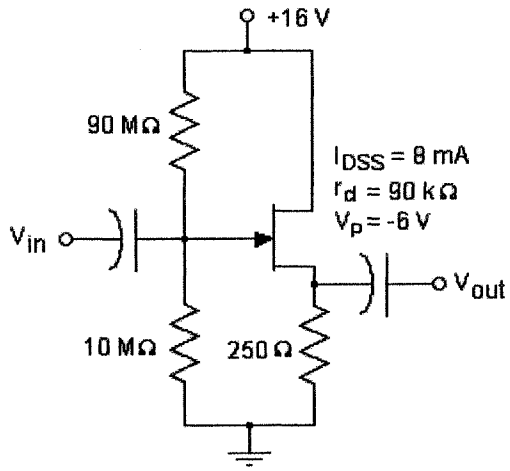
A) $Z_i = R_1 \| R_2 = 9$ MΩ

B) $Z_i = R_2 = 10$ MΩ

C) $Z_i = R_1 = 90$ MΩ

D) Z_i depends on the value of I_D.

26) Calculate the output impedance for this FET amplifier. Assume that $g_m = 1.85$ mS.



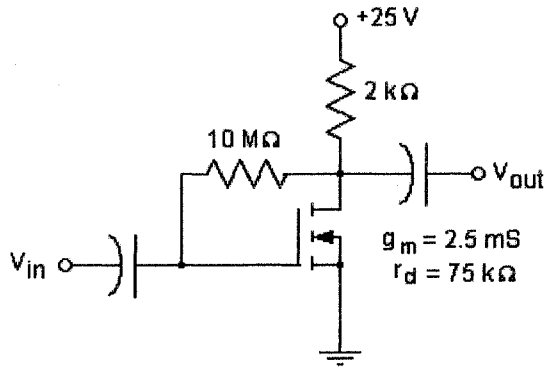
A) $Z_O = R_S = 250\ \Omega$

B) $Z_O = r_d = 90\ k\Omega$

C) $Z_O = R_S \parallel r_d \parallel \dfrac{1}{g_m} = 170.6\ \Omega$

D) Z_O depends on the value of I_D.

27) Calculate the input impedance for this E-MOSFET circuit.



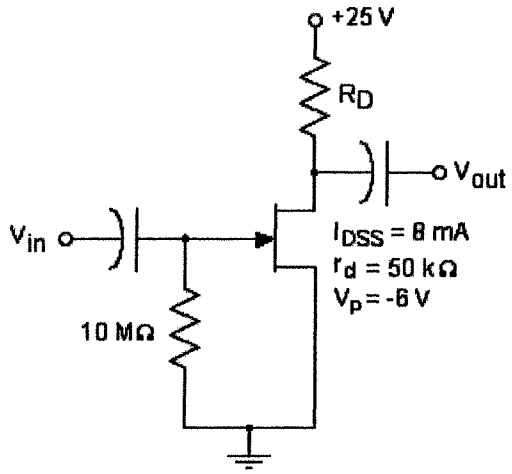
A) $Z_i = \dfrac{R_F}{1 + (r_d \parallel R_D)} = 5.130\ k\Omega$

B) $Z_i = \dfrac{R_F}{1 + g_m r_d} = 53.05\ k\Omega$

C) $Z_i = R_F = 10.0\ M\Omega$

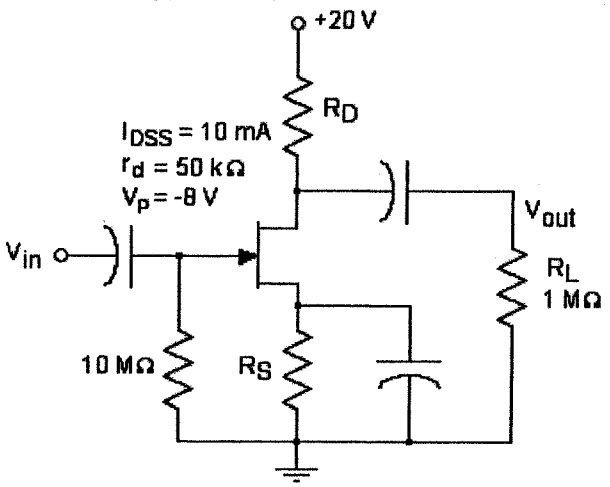
D) $Z_i = \dfrac{R_F}{1 + (r_d \parallel R_D)} = 1.0735\ M\Omega$

28) Design this amplifier for a voltage gain of 8, and indicate the required value of resistor R_D.



A) $R_D = 9.0\ k\Omega$
B) $R_D = 10.0\ k\Omega$
C) $R_D = 3103.44\ \Omega$
D) $R_D = 3.0\ k\Omega$

29) Design this circuit for a voltage gain of 10. You have to calculate the value of resistor R_D and R_S. It is desired that the transistor operate with a relatively high value of g_m. For this device, a high value of g_m is defined as $V_{GS} = 0.2\ V_P$.



A) $R_D = 9.0\ k\Omega$, $R_S = 1.0\ k\Omega$
B) $R_D = 5.555\ k\Omega$, $R_S = 250\ \Omega$
C) $R_D = 5.555\ k\Omega$, $R_S = 1.0\ k\Omega$
D) $R_D = 9.0\ k\Omega$, $R_S = 250\ \Omega$

30) The operating value of g_m is always _____ the value of g_{mo} for a given JFET.
A) less than
B) equal to
C) less than or equal to
D) greater than

31) A given JFET has values of g_{mo} = 1200 μS and V_{GSOFF} = -4 V. What is the value of g_m for the device at V_{GS} = -2 V
A) 500 μ.S
B) 1200 μ.S
C) 300 μ.S
D) Cannot be determined from the information given

32) A JFET has values of = 1200 μS and V_{GSOFF} = -4 V. What is the approximate value of I_{DSS}?
A) 4.8 mA
B) 9.6 mA
C) 2.4 mA
D) Cannot be determined from the information given

33) The _____ amplifier has high input impedance, low output impedance, and low voltage gain.
A) common-gate
B) common-drain
C) common-source
D) None of the above

34) The _____ FET amplifier has low input impedance, high output impedance, and high voltage gain.
A) common-gate
B) common-drain
C) common-source
D) None of the above

ANSWER KEY: Chapter 8: FET Amplifiers – Answer Key

1) TRUE

2) TRUE

3) FALSE

4) FALSE

5) TRUE

6) C

7) C

8) B

9) D

10) A

11) C

12) C

13) A

14) B

15) C

16) D

17) A

18) B

19) B

20) B

21) A

22) C

23) B

24) A

25) A

26) C

27) D

28) C

29) B

30) C

31) A

32) C

33) B

34) A

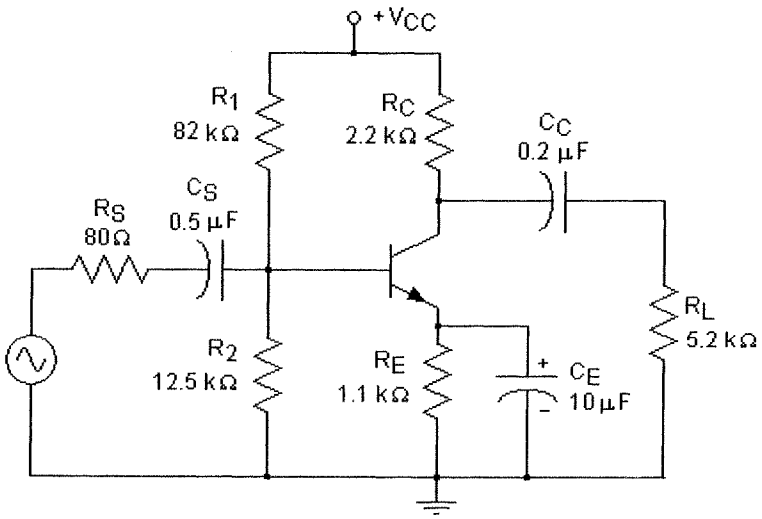
Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 9: BJT and JFET Frequency Response

1) The input power to a device is 5000 W at 400 V. The output power of the device is 750 W, and the output impedance is 25 Ω. Calculate the power gain in decibels.
A) -8.239 dB
B) 8.239 dB
C) -16.478 dB
D) 16.478 dB

2) The input power to a device is 5000 W at 400 V. The output power of the device is 750 W, and the output impedance is 25 Ω. Calculate the voltage gain in decibels.
A) -4.6556 dB
B) 4.6556 dB
C) -9.311 dB
D) 9.311 dB

3) To fix the frequency boundaries of relatively high gain, 0.7 A_{mid} was chosen to be at the cutoff levels. The corresponding frequencies, f_1 and f_2, are generally called _____.
A) corner frequencies
B) cutoff frequencies
C) half-power frequencies
D) All of the above

4) Calculate the low frequency break point due to the capacitor C_S for this BJT amplifier.



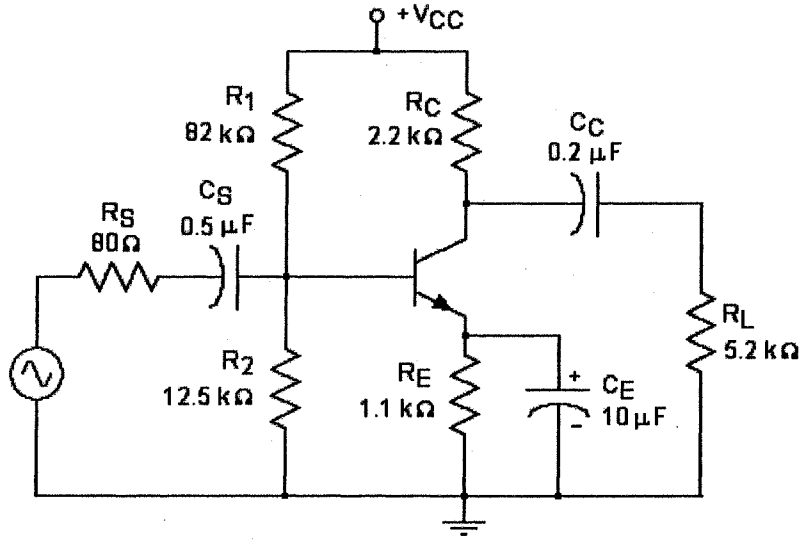
A) $f_{Ls} = \dfrac{1}{2\pi(R_S + R_i)C_S} = 280.25$ Hz

B) $f_{Ls} = \dfrac{1}{2\pi(R_S + R_1 \parallel R_2)C_S} = 30.6$ Hz

C) $f_{Ls} = \dfrac{1}{2\pi(R_S \parallel R_i)C_S} = 5585.4$ Hz.

D) $f_{Ls} = \dfrac{1}{2\pi(R_S + R_1 + R_2)C_S} = 4.269$ Hz.

5) Calculate the low frequency break point due to the capacitor C_C for this BJT amplifier.



A) $f_{LC} = \dfrac{1}{2\pi R_C C_C} = 361.71$ Hz.

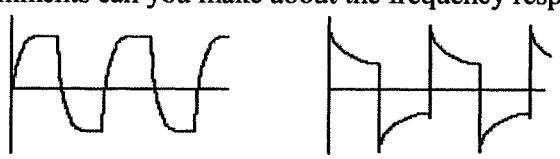
B) $f_{LC} = \dfrac{1}{2\pi (R_C \parallel R_L) C_C} = 514.75$ Hz

C) $f_{LC} = \dfrac{1}{2\pi (R_C + R_L) C_C} = 107.53$ Hz.

D) $f_{LC} = \dfrac{1}{2\pi R_L C_C} = 153.03$ Hz.

6) If several identical stages of amplifiers, each having the exact same upper and lower cutoff frequencies, are connected in cascade, then the bandwidth of the resulting amplifier will _____.
A) increase
B) remain unchanged
C) decrease
D) be equal to the sum of all the individual bandwidths

7) When using square-wave testing on two different amplifiers, you see the waveforms shown here. What comments can you make about the frequency response of the two amplifiers?



Response from amplifier A Response from amplifier B

A) Amplifier A has poor low frequency response, while amplifier B has poor high frequency response.
B) Amplifier A has poor low frequency response, while amplifier B has poor low frequency response.
C) Amplifier A has poor high frequency response, while amplifier B has poor high frequency response.
D) Amplifier A has poor high frequency response, while amplifier B has poor low frequency response.

8) The base-10 logarithm of 100,000 is _____.
A) 4
B) 5
C) 6
D) None of the above

9) The base-10 logarithm of 1780.331 _____.
A) 0.335
B) 3.2505
C) 33.5
D) None of the above

10) Which of the following are properties of logarithms?
A) $\log_n 0 = 1$
B) $\log_n(a/b) = \log_n a + \log_n b$
C) $\log_n(1/b) = -\log_n b$
D) None of the above

11) Which of the following are properties of logarithms?
A) $\log_n 1 = 0$
B) $\log_n(a/b) = \log_n a + \log_n b$
C) $\log_n(1/b) = \log_n b$
D) None of the above

12) Which of the following are properties of logarithms?
A) $\log_n 1 = 0$
B) $\log_n(a/b) = \log_n a - \log_n b$
C) $\log_n(1/b) = -\log_n b$
D) All of the above

13) The term *semi-log* refers to a graphical scale that has _____.
A) a linear axis and a log axis
B) a log-log structure
C) a linear vertical axis and a log horizontal axis
D) All of the above

14) The common log of the ratio of two power levels is called a _____.
A) decibel
B) bel
C) big bel
D) None of the above

15) For the gain in decibels to be completely correct it should be referred to as voltage or current gain in decibels to differentiate it from the normal power level consideration. This occurs when _____.
A) $R_1 < R_L$
B) $R_1 > R_L$
C) $R_1 \neq R_L$
D) All of the above

16) The gain in decibels of a power gain of 10,000,000 is _____.
A) 5 dB
B) 6 dB
C) 70 dB
D) 80 dB

17) An amplifier has a midband power gain of 24,500. What is the value of the power gain in dB for the circuit?
A) 87.78 dB
B) 43.9 dB
C) 4.39 dB
D) None of the above

18) An amplifier has values of P_{in} = 20 mW and P_{out} = 60 W. What is the value of the power gain in dB for the circuit?
A) 3000 dB
B) 69.5 dB
C) 34.8 dB
D) None of the above

19) An amplifier normally has a power gain of 12,000. If the power gain of the circuit drops by 3 dB, the value of the new power-gain will be approximately _____.
A) 6,000
B) 4,000
C) 9,000
D) zero

20) Negative dB values represent _____.
A) power gain
B) power losses
C) power values that do not change
D) None of the above

21) Which of the following is an advantage of using dB representations of gain values?
A) Positive and negative dB values represent gain and loss values that are reciprocals of each other.
B) In multistage amplifiers, gain calculations are simplified by the use of dB values.
C) Using db values, we can represent large gain values with relatively small numbers.
D) All of the above

22) An amplifier has an output power of 500 W. What is the value of the power gain in dB for the circuit?
A) 26.99 dB
B) 53.98 dB
C) 56.99 dB
D) Cannot be determined from the information given

23) An amplifier delivers 500 W to a 600 Ω load. What is the gain in dBm?
A) 26.99 dBm
B) 53.98 dBm
C) 56.99 dBm
D) Cannot be determined from the information given

24) An amplifier has values of power gain 49 dB and voltage gain dB of 30 dB. The operating frequency of the circuit is increased until the power gain drops to 42 dB. What is the dB voltage gain at this frequency?
A) 23 dB
B) 42 dB
C) 25.7 dB
D) Cannot be determined from the information given

25) What frequency is two decades above 5 kHz?
A) 105 kHz
B) 25 kHz
C) 500 kHz
D) Cannot be determined from the information given

26) What frequency lies four octaves above 1 kHz?
A) 9 kHz
B) 16 kHz
C) 8 kHz
D) None of the above

27) The roll-off rate for a BJT amplifier is approximately _____.
A) 20 dB per octave
B) 6 dB per decade
C) 6 dB per octave
D) None of the above

28) The low-frequency response of a BJT amplifier is affected by _____.
A) the BJT internal capacitances
B) the supply voltage
C) the coupling and bypass capacitor values
D) All of the above

29) The input value of f_1 for a FET amplifier normally is _____ the input value of f_1 for a comparable BJT amplifier.
A) similar to
B) much lower than
C) greater than
D) less than or equal to

30) A common-emitter amplifier has values of C_{bc} = 5 pF and $A_{V(dB)}$ = 23.5218 dB. What is the Miller input capacitance for the circuit?
A) 80 pF
B) 7 pF
C) 163.1 pF
D) None of the above

31) The BJT gain-bandwidth product is approximately _____.
A) $\alpha_{mid}\beta$
B) $\alpha_{mid}\alpha$
C) $\beta_{mid}\,\alpha\,[1-\alpha]$
D) None of the above

32) The BJT gain-bandwidth product (f_T) is _____.
A) inversely related to r_e
B) inversely related to ($C_{be} + C_{bc}$)
C) not related to DC bias conditions
D) All of the above

33) A common-emitter amplifier uses a transistor where $\beta = 50$, $r_e = 10\ \Omega$, $C_{be} = 30$ pF, $C_{bc} = 3.5$ pF, $C_{ce} = 1$ pF, $C_W = 5$ pF. The value of f_β is _____.
A) 19.04 MHz
B) 1.95 MHz
C) 9.502 MHz
D) None of the above

34) The high-frequency response of a BJT amplifier is affected by _____.
A) the BJT internal capacitances
B) the supply voltage
C) the coupling and bypass capacitor values
D) All of the above

35) Determine the high end cut-off frequency for a FET amplifier that has devices capacitances $C_{wo} = 5$ pF, $R_{sig} = 12\ k\Omega$, $R_G = 0.5\ M\Omega$, $R_D = 5.6\ k\Omega$, and $A_v = -5$.
A) 1.617 MHz
B) 16.17 MHz
C) 0.1617 MHz
D) None of the above

36) Two identical amplifiers are cascaded. The overall bandwidth of the multistage amplifier is _____ the bandwidth of each individual stage.
A) equal to
B) less than
C) greater than
D) less than or equal to

37) The Miller effect tends to _____ of an inverting amplifier.
A) increase both the input and output capacitances
B) increase the input capacitance and decrease the output capacitance
C) decrease the input capacitance and increase the output capacitance
D) decrease both the input and output capacitances

38) Which one of the following equations provides the Miller effect input capacitance for a BJT amplifier?
A) $\dfrac{1}{(1-A_v)\,C_f}$
B) $\left(1-\dfrac{1}{A_v}\right)C_f$
C) $(1 - A_v)C_f$
D) $\dfrac{1}{2\pi R_{Th_i} C_i}$

39) Which one of the following equations provides the Miller effect output capacitance for a BJT amplifier?

A) $\frac{1}{(1-A_v)\,C_f}$

B) $\left(1-\frac{1}{A_v}\right)C_f$

C) $(1 - A_v)C_f$

D) $\frac{1}{2\pi R_{Th_i} C_i}$

40) A 3-dB drop in β occurs at _____.

A) G_{db}

B) F_{Ls} and f_{Lo}

C) f_β

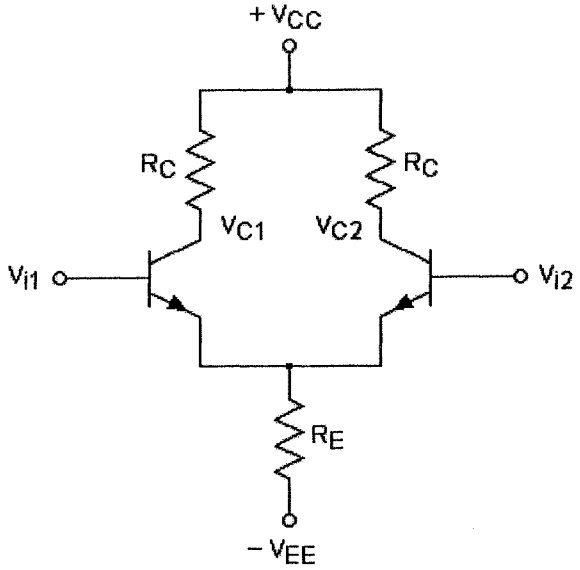
D) f_{Hi}

ANSWER KEY: Chapter 9: BJT and JFET Frequency Response – Answer Key

1) A
2) C
3) D
4) A
5) C
6) C
7) D
8) B
9) B
10) C
11) A
12) D
13) A
14) B
15) D
16) C
17) B
18) C
19) A
20) B
21) D
22) D
23) C
24) A
25) C
26) B
27) C
28) C
29) A
30) A
31) C
32) D
33) C
34) A
35) A
36) B
37) A
38) C
39) B
40) C

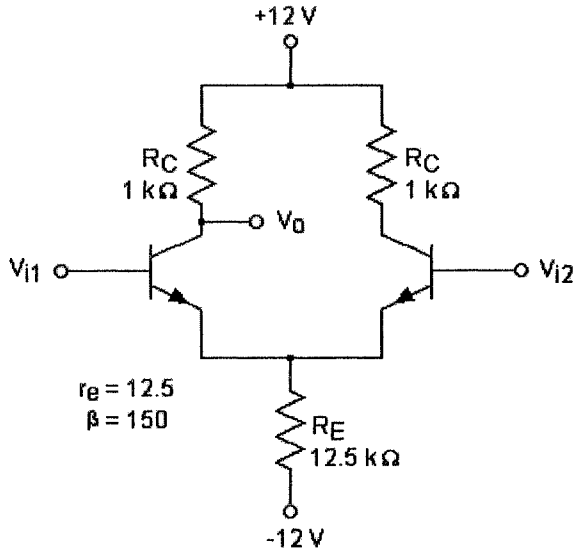
Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 10: Operational Amplifiers

1) The main features of a difference pair, such as the circuit shown here, are _____.



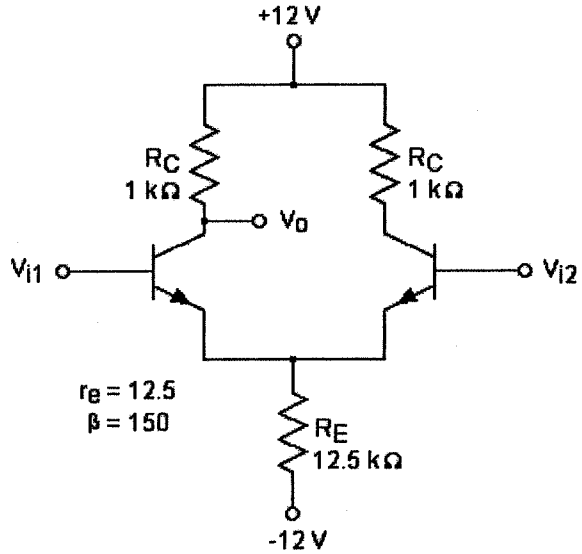
A) very high input impedance and very high voltage gain
B) very high input impedance and moderate voltage gain
C) moderate input impedance and very high voltage gain
D) moderate input impedance and moderate voltage gain

2) Under single-ended operation, the voltage gain for this is _____.



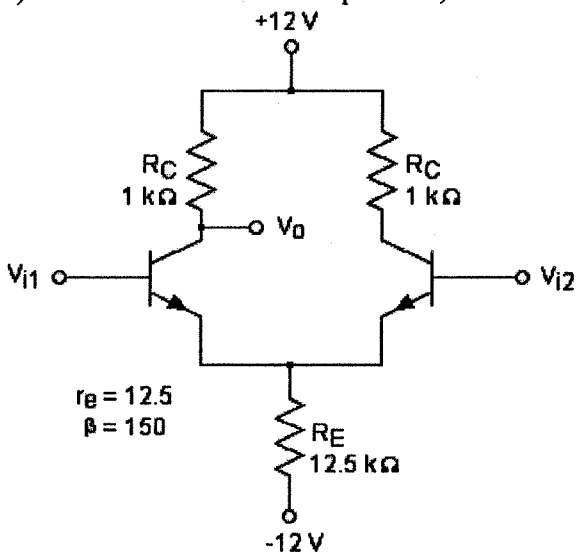
A) 20
B) 40
C) 80
D) 160

3) Under common-mode operation, the common-mode voltage gain for this circuit is _____.



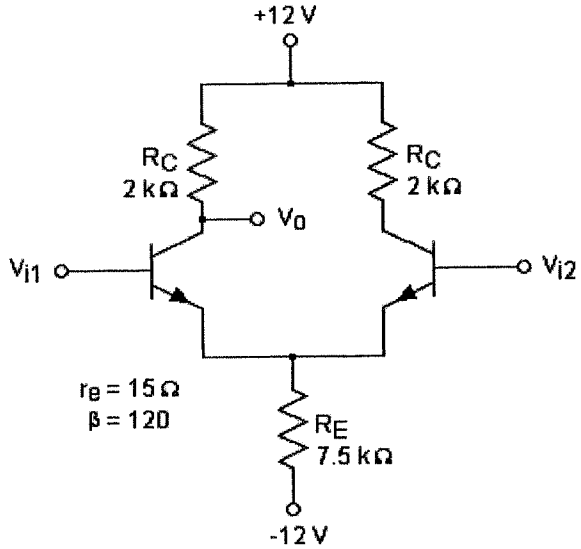
A) 0.0397
B) 80
C) 40
D) 0.08

4) Under difference-mode operation, the difference-mode voltage gain for this circuit is _____.



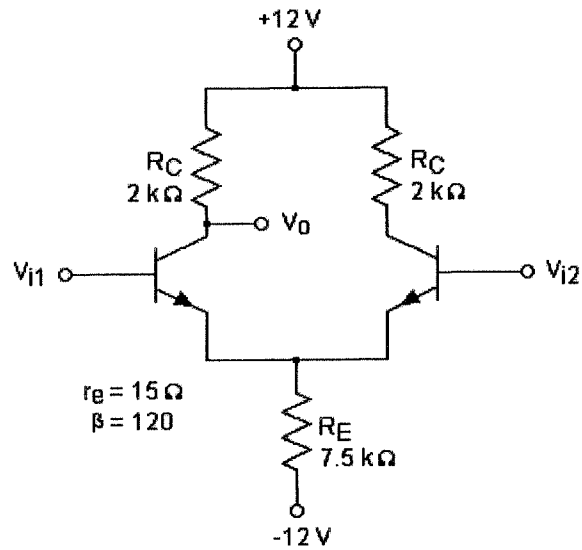
A) 0.0397
B) 80
C) 40
D) 0.08

5) Under single-ended operation the voltage gain for this circuit is _____.



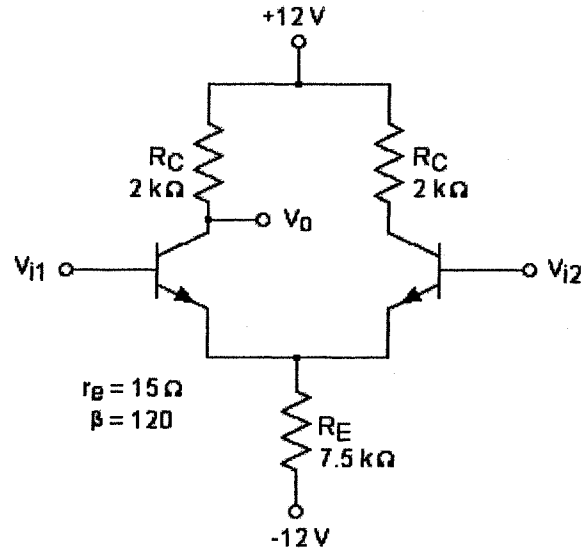
A) 33.33
B) 66.66
C) 133.33
D) 266.66

6) Under common-mode operation, the common-mode voltage gain is _____.



A) 0.133
B) 33.33
C) 66.66
D) 0.266

7) Under difference-mode operation, the difference-mode voltage gain for this circuit is _____.



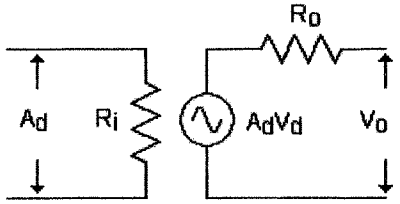
A) 0.133
B) 33.33
C) 66.66
D) 0.266

8) The operational amplifier will only slightly amplify signals _____.
A) when the supply voltages are more then ±25 V
B) when the supply voltages are less then ±5 V
C) that are common on both the inputs
D) that are different on both the inputs

9) Which of the following statements is true about operational amplifiers?
A) Op-amps are very high-gain DC amplifiers.
B) Op-amps have very low output impedance.
C) Op-amps have very high input impedance.
D) All of the above

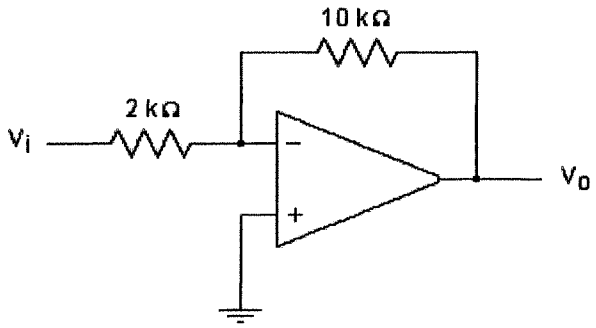
10) The common mode rejection ratio (CMRR) is the ratio of _____.
A) the difference mode gain to the common mode gain
B) the common mode gain to the difference mode gain
C) noninverting gain to inverting gain
D) inverting gain to noninverting gain

11) For this AC equivalent circuit of an op-amp, the input signal is applied between the input terminals and sees an input impedance R_i. This impedance _____.



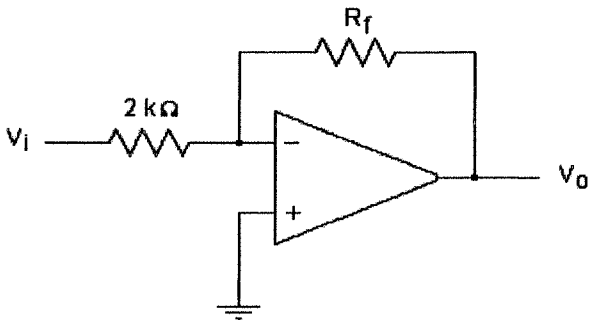
A) is a few hundred ohms
B) is a few kilohms
C) is a few hundred kilohms
D) depends on the op-amp used. It could be a few hundred ohms to a few hundred kilohms

12) One basic circuit connection using an op-amp is shown here. If the input voltage is 0.25 V, the output voltage is _____.



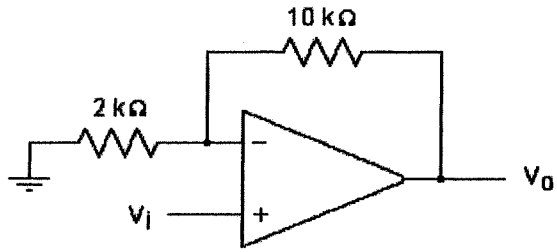
A) -10.0 V
B) -5.0 V
C) -2.5 V
D) -1.25 V

13) If the input voltage is 0.25 V and the output is -2.0 V, the value of R_f must be _____.



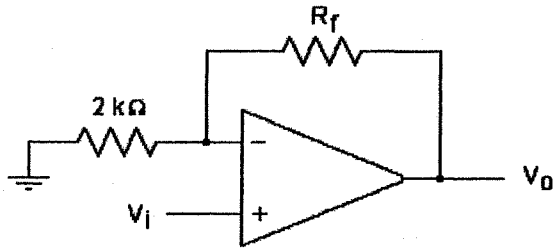
A) 40.0 kΩ
B) 20.0 kΩ
C) 10.0 kΩ
D) 5.0 kΩ

14) When the input voltage to this circuit is 0.25 V, the output voltage is _____.



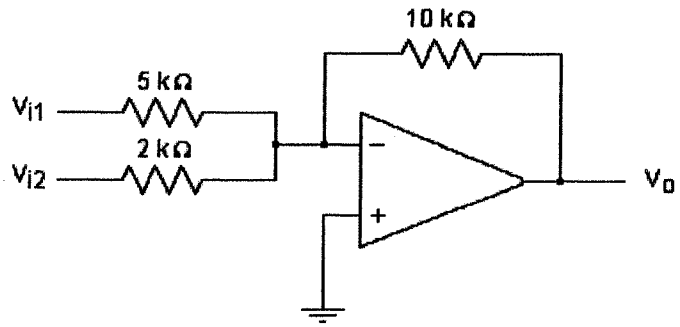
A) 10.0 V
B) 5.0 V
C) 3.0 V
D) 1.5 V

15) If the input voltage is 0.25 V and the required output voltage is 2.75 V, the value for R_f must be _____.



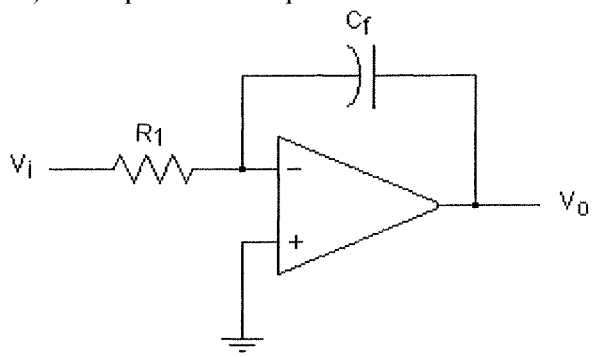
A) 40.0 kΩ
B) 20.0 kΩ
C) 10.0 kΩ
D) 5.0 kΩ

16) When the input voltages to this circuit are $V_{i1} = 0.25$ V and $V_{i2} = 0.5$ V, the output is _____.



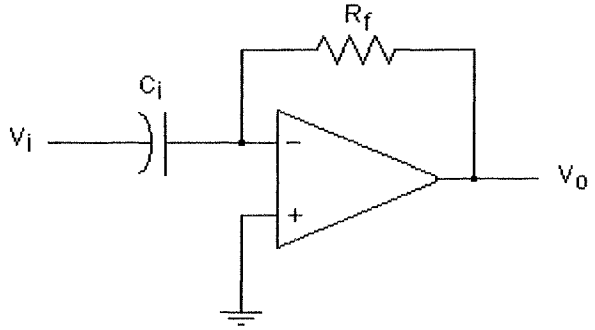
A) 1.125 V
B) 2.25 V
C) 4.5 V
D) 9.0 V

17) The operational amplifier circuit shown here is a(n) _____.



A) level comparator
B) differentiator
C) integrator
D) difference amplifier

18) The operational amplifier circuit shown here is a(n) _____.



A) level comparator
B) differentiator
C) integrator
D) difference amplifier

19) The inverting and noninverting inputs to an op-amp are used to drive a(n) _____ amplifier.
A) inverting
B) noninverting
C) differential
D) open-loop

20) An op-amp amplifies only slightly when its _____.
A) supply voltages are less than +5 V
B) input offset voltage is less than 100 mV
C) inverting or noninverting inputs have a common input
D) input offset current is less than 1 mA

21) When a given op-amp has a common-mode input of 10 V, the output of the device is 10 V. When the device has a differential input of 2 mV, the output of the device is 10 V. What is the CMPR of the device?
A) 5 : 1
B) 5000 : 1
C) 1000 : 1
D) 5,000,000 : 1

22) Which of the following statements is true?
A) Op-amps are high-gain dc amplifiers
B) Op-amps have extremely high input impedance.
C) Op-amps have extremely low output impedance.
D) All of the above

23) The CMRR of an inverting amplifier always lower than that of its op-amp because _____.
A) the common-mode gain of an op-amp increases when it is used in an inverting amplifier
B) the value of differential gain for an inverting amplifier is lower than that of its op-amp
C) slew-rate limiting decreases the common-mode gain of the op-amp
D) of the lower input impedance of the inverting amplifier

24) The op-amp circuit that add each input and multiplies the sum by a fixed amount is called a(n) _____.
A) unity follower
B) integrator
C) differentiator
D) summing amplifier

25) The op-amp circuit that has a capacitor as the feedback component is called a(n) _____.
A) unity follower
B) integrator
C) differentiator
D) summing amplifier

26) An op-amp integrator circuit has a 2 MΩ input resistor and a 5 μF feedback loop capacitor. If the inverting input voltage is 2 VDC, the final value of the output voltage is _____.
A) -20 V
B) -2 V
C) -0.2 V
D) 0.02 V

27) A summing integrator is an op-amp integrator that has _____.
A) multiple feedback capacitors
B) multiple input resistors
C) multiple input resistors and feedback capacitors
D) None of the above

28) Op-amp differentiator circuits differ from the integrators in that the differentiators _____.
A) are not as useful
B) have a scale factor of -RC
C) have a resistor in the feedback loop
D) All of the above

29) An inverting amplifier and a noninverting amplifier are built using the same values of R_f and R_1. Assuming that the op-amps being used in the two circuits have identical common-mode gain values, _____.
A) the inverting amplifier has the higher CMRR
B) the noninverting amplifier has the higher CMRR
C) the CMRR is the same for the inverting and noninverting amplifiers
D) None of the above

30) An inverting amplifier with +11 V supply voltages normally has a sinusoidal output of 10 V_{PP}. When checking the circuit with an oscilloscope, you find that the output is 0 V. Which of the following could account for this problem?
A) R_1 is open.
B) $V_1 = 0$
C) R is shorted by a solder bridge.
D) All of the above

31) The voltage follower typically has a voltage gain value of _____.
A) 1000
B) 100
C) 10
D) 1

32) Op-amp output offset voltage can be totally explained by _____.
A) the input offset voltage
B) input offset current
C) the external circuit connection components
D) None of the above

33) A standard inverting op-amp circuit has an R_1 of 10 kΩ and an R_f of 220 kΩ. If the offset voltage is 0.003 V, the $V_{o(offset)}$ is _____.
A) 3 mV
B) 6.6 mV
C) 69 mV
D) 200 mV

34) A standard inverting op-amp circuit has an R_1 of 10 kΩ and an R_f of 220 kΩ. If the offset current is 100 nA the output offset voltage due to this current is _____.
A) 10 mV
B) 22 mV
C) 32 mV
D) 8 mV

35) When calculating the total offset voltage the absolute values are used to accommodate the fact that _____.
A) offset current can be negative or positive
B) offset voltage can be negative or positive
C) both the offset current and voltage can be negative or positive
D) None of the above

36) Op-amp roll-off characteristics are caused by the fact that _____.
A) they are designed to have high-gain and wide-bandwidth
B) the uncompensated circuit would be unstable
C) an effective compensation circuit is used
D) All of the above

37) Determine the op-amp cutoff-frequency for a device whose unity-gain bandwidth is 1.5 MHz and the differential-gain is 300 V/mV.
A) 5 Hz
B) 10 Hz
C) 50 Hz
D) 150 Hz

38) Slew rate is the _____.
A) ratio of the change in time to the change in output voltage
B) maximum rate at which the op-amp output voltage can change
C) maximum rate at which the amplifier input voltage can change
D) None of the above

39) Exceeding the op-amp slew rate results in _____.
A) improved gain and reduced distortion
B) increased power and reduced distortion
C) clipping and increased distortion
D) None of the above

40) A certain op-amp has a slew rate of 4 V/μsec. What is the maximum closed-loop voltage gain if the input voltage rises at a rate of 5 V/l00 μsec?
A) 20
B) 40
C) 80
D) 200

41) A certain inverting op-amp has a closed-loop gain of 50 and a slew rate of 5 V/μsec. What is the maximum useful frequency of a 2 V input signal applied to this circuit?
A) 5×10^3 rad/s
B) 10×10^3 rad/s
C) 50×10^3 rad/s
D) 100×10^3 rad/s

42) The bandwidth of an amplifier is _____.
A) the range of frequencies over which gain remains relatively constant
B) the range of frequencies between the lower and upper 3 dB frequencies
C) the range of frequencies found using $f_2 - f_1$
D) All of the above

43) Op-amps are available in a number of packages. Among the most common forms is the _____.
A) dual in-line package (DIP)
B) surface-mount package
C) TO-5 metal can
D) TO-8 metal can

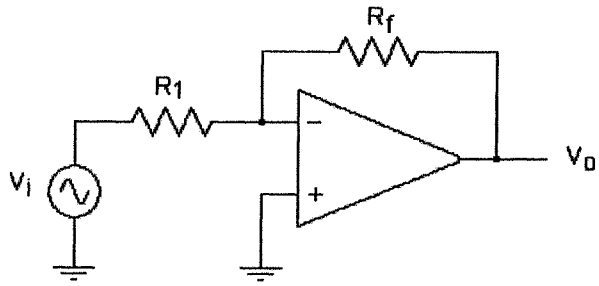
44) An op-amp circuit has ±15 V supply voltages and a voltage gain of 20. The noninverting voltage (v+) is 0.3 V and the inverting voltage (v-) is 0.35 V. What is the output voltage from the device?
A) +1 V
B) +6 V
C) -0.8 V
D) -7 V

ANSWER KEY: Chapter 10: Operational Amplifiers

1) A
2) A
3) A
4) C
5) B
6) D
7) C
8) C
9) D
10) A
11) C
12) D
13) B
14) D
15) B
16) B
17) C
18) B
19) C
20) C
21) D
22) D
23) B
24) D
25) B
26) C
27) B
28) D
29) B
30) D
31) D
32) D
33) C
34) B
35) C
36) D
37) A
38) B
39) C
40) C
41) C
42) D
43) A
44) C

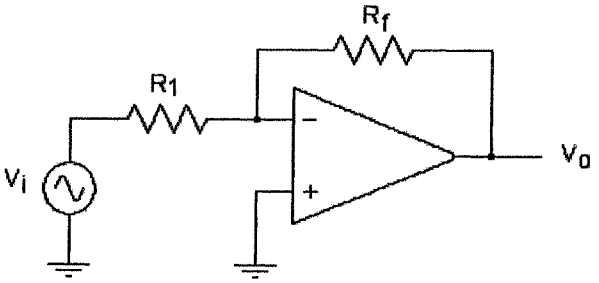
Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 11: Op-Amp Applications

1) This op-amp is connected as a(n) _____.



A) inverting constant-gain multiplier
B) noninverting constant-gain multiplier
C) voltage buffer
D) None of the above

2) The voltage gain for this circuit is give as _____.



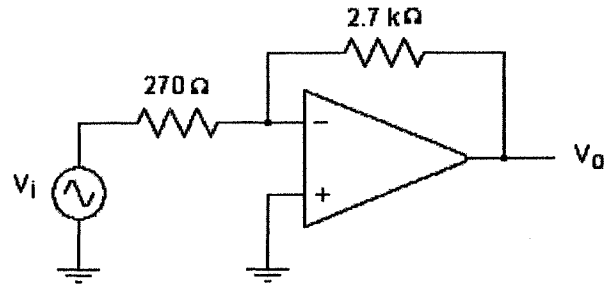
A) $A = 1 + \frac{R_f}{R_1}$

B) $A = -\frac{R_f}{R_1}$

C) $A = \frac{R_1}{R_f}$

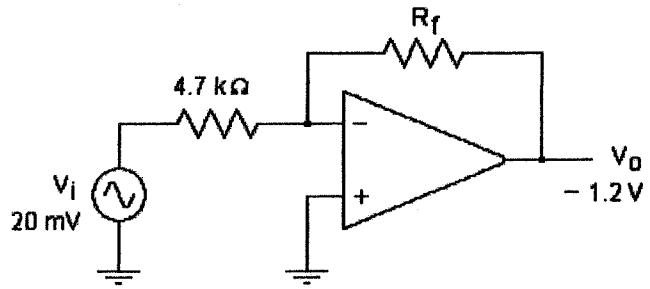
D) $A = -\frac{R_1}{R_f}$

3) Determine the gain of this circuit.



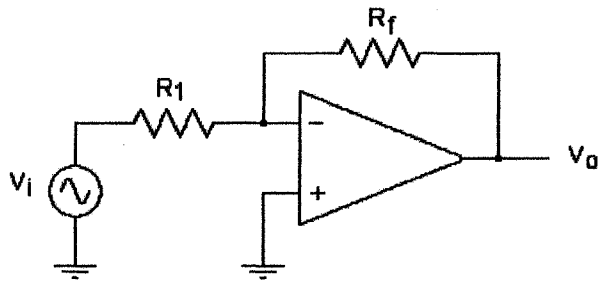
A) 10
B) 0.1
C) -0.1
D) -10

4) What is the design value for R_f?



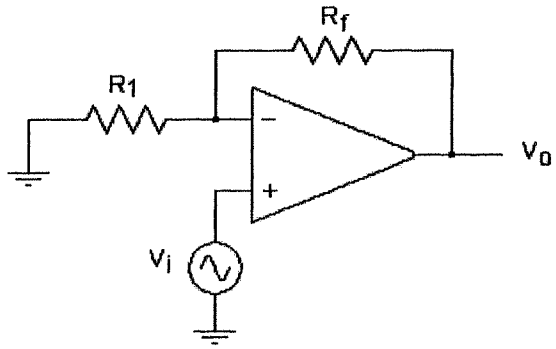
A) 60 Ω
B) 282 kΩ
C) 78.3 Ω
D) 180 kΩ

5) If this circuit is to have an inverting voltage gain of 1, the ratio of R_f : R_1 should be _____.



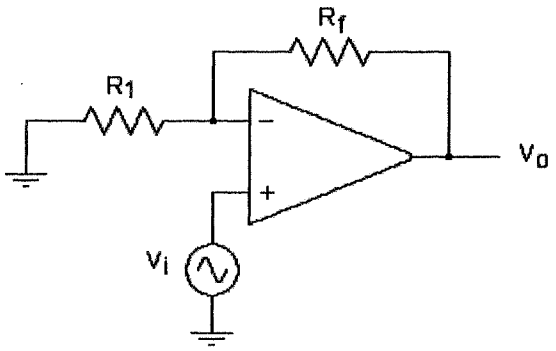
A) 1 : 1
B) 2 : 1
C) 1: 2
D) Cannot be done in theory. It can be done in practice by making R_1 much larger than R_f.

6) This op-amp is connected as a(n) _____.



A) inverting constant-gain multiplier
B) noninverting constant-gain multiplier
C) voltage buffer
D) None of the above

7) The voltage gain for this circuit is given as _____.



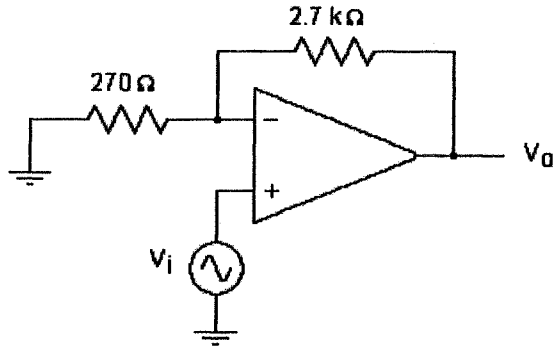
A) $A = 1 + \frac{R_f}{R_1}$

B) $A = - \frac{R_f}{R_1}$

C) $A = 1 + \frac{R_1}{R_f}$

D) $A = - \frac{R_1}{R_f}$

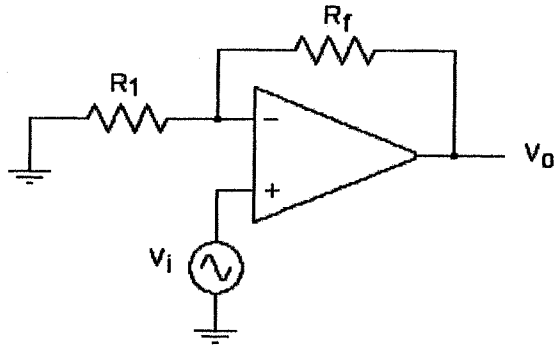
8) Determine the gain of this circuit.



A) 10
B) -10
C) 11
D) 1.1

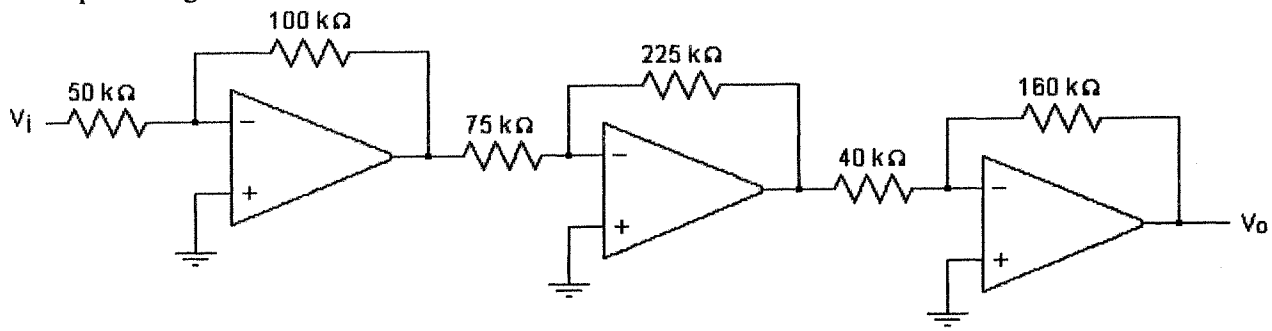
9) A gain of 7 is required from a noninverting constant-multiplier op-amp circuit. If the input resistor is 5 k Ω, what should be the value of the feedback resistor ?
A) 30 kΩ
B) 15 kΩ
C) 7.5 kΩ
D) 5 kΩ

10) If this circuit is to have a voltage gain of 1, the ratio of R_f : R_1 should be _____.



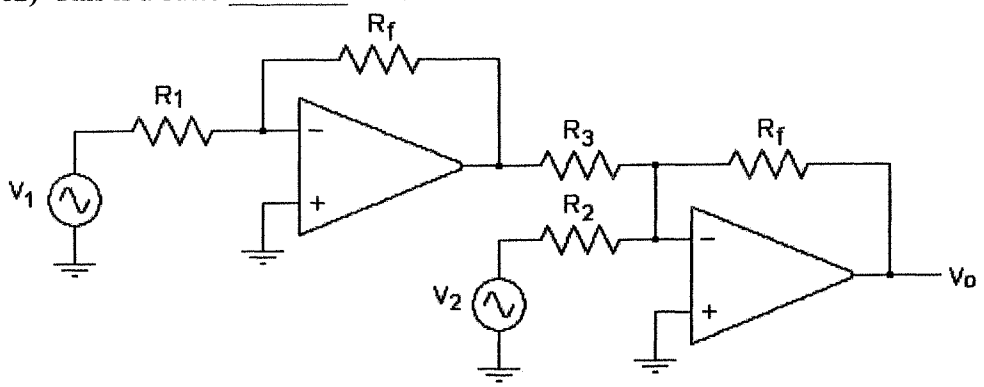
A) 1 : 1
B) 2 : 1
C) 1: 2
D) Cannot be done in theory. It can be done in practice by making R_1 much larger than R_f.

11) For this cascade amplifier, the input voltage swing is 0.2 V_{PP}. Calculate the peak-to-peak swing on the output voltage.



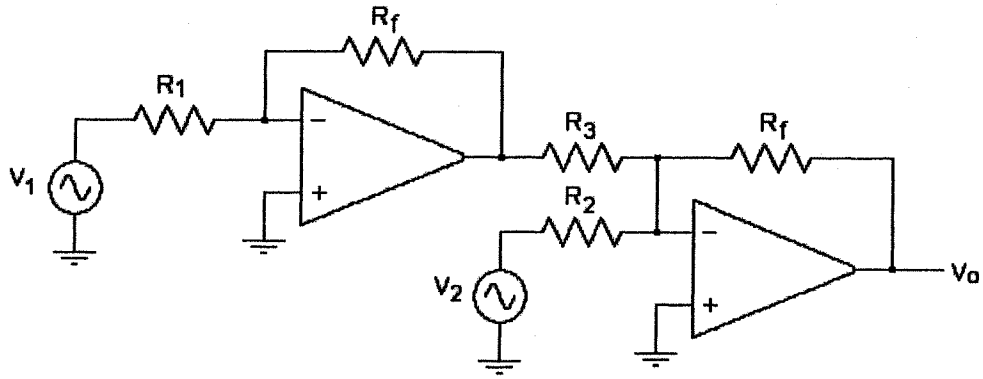
A) 9.6 V_{PP}
B) 4.8 V_{PP}
C) 2.4 V_{PP}
D) 1.2 V_{PP}

12) This is a basic _____ circuit.



A) differentiator
B) integrator
C) two-input multiplier
D) subtractor

13) The output voltage, V_O, is given by _____.



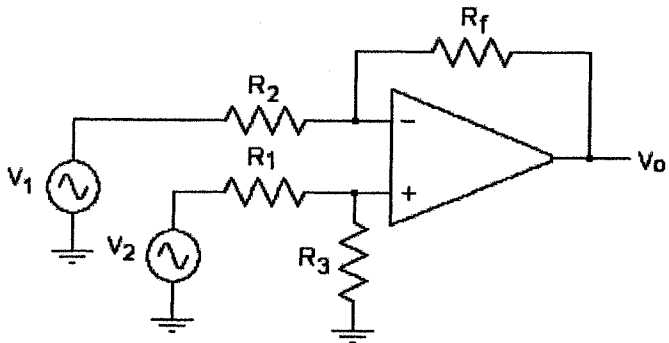
A) $V_O = -\left(\frac{R_f}{R_2}V_2 - \frac{R_f^2}{R_1R_3}V_1\right)$

B) $V_O = \frac{R_3}{R_1+R_3}\frac{R_2+R_f}{R_2}V_1 - \frac{R_f}{R_2}V_2$

C) $V_O = -V_1\frac{R_f}{R_1} + V_2\frac{R_f}{R_2\|R_3}$

D) None of the above

14) The output voltage, V_O, is given by _____.



A) $V_O = -\left(\frac{R_f}{R_2}V_2 - \frac{R_f^2}{R_1R_3}V_1\right)$

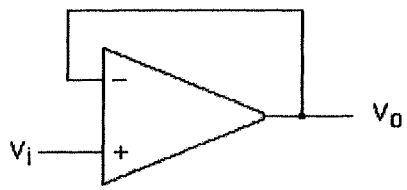
B) $V_O = \frac{R_3}{R_1+R_3}\frac{R_2+R_f}{R_2}V_1 - \frac{R_f}{R_2}V_2$

C) $V_O = -V_1\frac{R_f}{R_1} + V_2\frac{R_f}{R_2\|R_3}$

D) None of the above

15) How many feedback resistors are found in a 3-input voltage summing circuit that is constructed around an op-amp?
A) 0
B) 1
C) 2
D) 3

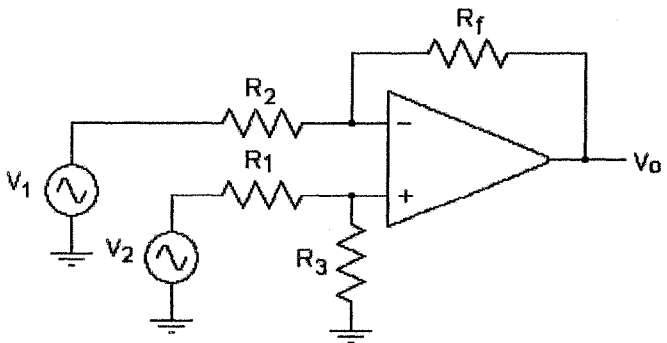
16) This circuit is known as a _____.



A) noninverting amplifier
B) voltage buffer
C) low-pass filter
D) high-pass filter

17) The output of an op-amp voltage buffer is characterized by _____.
A) $V_O = 1$
B) $V_O = -\frac{R_f}{R_1} V_i$, where $R_f = R_1$
C) $V_O = \left(1 + \frac{R_f}{R_1}\right) V_i$
D) $V_O = V_i$

18) If all resistances in this circuit are equal, the output is _____.



A) $V_1 - V_2$
B) $V_2 - V_1$
C) $V_1 + V_2$
D) $V_1 \times V_2$

19) By carefully configuring the op-amp external circuit components the op-amp can be made to function as _____.
A) voltage-controlled voltage source
B) voltage-controlled current source
C) current-controlled voltage source
D) All of the above

20) Which one of the following expressions model the function of a voltage-controlled current source?
A) $V_O = kV_i$
B) $V_O = kI_i$
C) $I_O = kV_i$
D) $I_O = kI_i$

21) Which one of the following expressions model the function of a voltage-controlled voltage source?
A) $V_O = kV_i$
B) $V_O = kI_i$
C) $I_O = kV_i$
D) $I_O = kI_i$

22) Which one of the following expressions model the function of a current-controlled current source?
A) $V_O = kV_i$
B) $V_O = kI_i$
C) $I_O = kV_i$
D) $I_O = kI_i$

23) Which one of the following expressions model the function of a current-controlled voltage source?
A) $V_O = kV_i$
B) $V_O = kI_i$
C) $I_O = kV_i$
D) $I_O = kI_i$

24) An op-amp low-pass active filter provides a constant output _____.
A) from dc to f_{oh}
B) for all frequencies higher than f_{ol}
C) from f_{ol} to f_{oh}
D) from dc to infinite frequency

25) An op-amp high-pass active filter provides a constant output _____.
A) from dc to f_{oh}
B) for all frequencies higher then f_{ol}
C) from f_{ol} to f_{oh}
D) from dc to infinite frequency

26) An op-amp bandpass active filter provides a constant output _____.
A) from dc to f_{oh}
B) for all frequencies higher then f_{ol}
C) from f_{ol} to f_{oh}
D) from dc to infinite frequency

27) The roll-off rate of a second order filter is _____.
A) 60 dB/decade or 18 dB/octave
B) 40 dB/decade or 12 dB/octave
C) 20 dB / decade or 6 dB/octave
D) 0 dB/decade or 0 dB/octave

28) A constant-gain multiplier has three stages and a total gain of 22,200. For all three feedback resistors, $R_f = 470$ kΩ. Two of the R_1 resistors have a value of 33 kΩ. What is the gain of the stage where R_1 is unknown?
A) 14.2
B) -14.2
C) 110.3
D) -110.3

29) A constant-gain multiplier has three stages and a total gain of 22,200. For all three feedback resistors, $R_f = 470$ kΩ. Two of the R_1 resistors have a value of 33 kΩ. What is the value of the third R_1, and what is the nature of its stage?
A) 4.26 kΩ, ohms, inverting
B) 4.3 kΩ, ohms, noninverting
C) 4.3 kΩ, inverting
D) 4.6 kΩ, noninverting

30) A constant-gain multiplier has three stages and a total gain of 22,200. For all three feedback resistors, $R_f = 470$ kΩ. Two of the R_1 resistors have a value of 33 kΩ. What is the individual gain of the two identical stages where R_1 is known?
A) 14.2
B) -14.2
C) 110.3
D) -110.3

31) Which type of op-amp circuit has unity gain, no phase inversion, high input impedance, and low output impedance?
A) voltage buffer
B) subtractor
C) summing amplifier
D) differentiator

32) By careful configuring the external circuit components, an op-amp can be made to function as a _____.
A) voltage-controlled current source
B) current-controlled voltage source
C) current-controlled current source
D) All of the above

33) An active filter that provides a constant output from dc to f_{oh}, and then passes no signal above f_{oh}, is called an ideal _____.
A) low-pass filter
B) high-pass filter
C) bandpass filter
D) None of the above

34) An active filter that provides a constant output for input signals above f_{ol} is called an ideal _____.
A) low-pass filter
B) high-pass filter
C) bandpass filter
D) None of the above

35) An active filter that provides a constant output for input signals from f_{ol} to f_{oh} is called an ideal _____.
A) low-pass filter
B) high-pass filter
C) bandpass filter
D) None of the above

36) A second order high-pass filter has a low-end roll-off of _____.
A) 60 dB/octave
B) 40 dB/decade
C) 20 dB/octave
D) 6 dB/decade

37) A second order low-pass filter has a high-end roll-off of _____.
A) 60 dB/octave
B) 40 dB/decade
C) 20 dB/octave
D) 6 dB/decade

ANSWER KEY: Chapter 11: Op-Amp Applications

1) A
2) B
3) D
4) B
5) D
6) B
7) A
8) C
9) A
10) D
11) A
12) D
13) A
14) B
15) B
16) B
17) D
18) A
19) D
20) C
21) A
22) D
23) B
24) A
25) B
26) C
27) B
28) C
29) B
30) B
31) A
32) D
33) A
34) B
35) C
36) B
37) B

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 12: Power Amplifiers

1) A class A amplifier _____.
A) conducts through 360° of the input waveform
B) conducts through 180° of the input waveform
C) conducts between 180° and 360° degrees of the input waveform, depending on the amount of dc bias
D) conducts through less than 180° of the input waveform

2) The maximum possible efficiency of a class A amplifier is equal to _____.
A) 30%
B) 25%
C) 20%
D) 15%

3) Transformer action with a class A power amplifier allows the engineer to design for _____.
A) voltage transformation
B) current transformation
C) impedance transformation
D) All of the above

4) In a transformer-operated power amplifier, the amount of secondary load resistance that is reflected back to the primary is _____.
A) proportional to the transformer turns ratio
B) inversely proportional to the transformer turns ratio
C) proportional to the square of the turns ratio
D) equal to the load resistance

5) A class B amplifier (not push-pull) _____.
A) conducts through 360° of the input waveform
B) conducts through 180° of the input waveform
C) conducts between 180° and 360° degrees of the input waveform, depending on the amount of dc bias
D) conducts through less than 180° of the input waveform

6) A class B push-pull amplifier _____.
A) conducts through 180° of the input waveform
B) conducts between 180° and 360° degrees of the input waveform, depending on the amount of dc bias
C) conducts through less than 180° of the input waveform
D) works only with digital and pulsed waveforms

7) Crossover distortion occurs _____.
A) only in class C amplifiers
B) in class B push-pull amplifiers that use complementary transistors
C) only in class D digital amplifiers
D) only in transformerless amplifiers

8) A class AB amplifier _____.
A) conducts through 180° of the input waveform
B) conducts between 180° and 360° degrees of the input waveform, depending on the amount of dc bias
C) conducts through less than 180° of the input waveform
D) works only with digital and pulsed waveforms

9) A class C amplifier _____.
A) conducts through 180° of the input waveform
B) conducts between 180° and 360° degrees of the input waveform, depending on the amount of dc bias
C) conducts through less than 180° of the input waveform
D) works only with digital and pulsed waveforms

10) Class C amplifiers are used mostly in _____ circuits.
A) audio
B) power amplifier
C) digital
D) tuned

11) A class D amplifier _____.
A) conducts through 180° of the input waveform
B) conducts between 180° and 360° degrees of the input waveform, depending on the amount of dc bias
C) conducts through less than 180° of the input waveform
D) works only with digital and pulsed waveforms

12) With transformer coupling the maximum theoretical efficiency of a class A amplifier can be increased up to _____.
A) 60%
B) 55%
C) 50%
D) 45%

13) A transformer coupled class A amplifier has a transformer turns ratio of 4.5 : 1 and a load resistance of 30 Ω. The peak-to-peak value of V_{CE} is 12 V. What is the approximate power that is delivered to the load?
A) 47 W
B) 71 W
C) 95 W
D) 119 W

14) A transformer coupled class A amplifier has a transformer turns ratio of 4.5 : 1 and a load resistance of 30 Ω. What is approximately the value of the effective AC load resistance seen by the collector of the transistor?
A) 400 Ω
B) 500 Ω
C) 600 Ω
D) 700 Ω

15) The crossover distortion in a class B amplifier is prevented by _____.
A) biasing the individual transistors deeply into cutoff
B) biasing the transistors just slightly above cutoff
C) biasing the transistors just slightly into cutoff
D) adjusting the load resistance so that the transistor will turn on and off faster

16) A class AB transistor amplifier using complementary transistors is biased by power supply that is ±18Vdc. The value of $V_{CE(OFF)}$ for either transistor is approximately equal to _____.
A) 8 V
B) 10 V
C) 18 V
D) 36 V

17) For class B or class AB amplifiers, closely matched complementary transistors are used because they _____.
A) can carry more current than push-pull transistors
B) require individual power supplies
C) do not need an output transformer
D) are complementary so use complementary transistors

18) Calculate the total harmonic distortion for a signal that has the following amplitude components:
Fundamental = 2.5 V
Second harmonic = 0.25 V
Third harmonic = 0.1 V
Fourth harmonic = 0.025 V
A) THD% = 10.81%
B) THD% = 10.95%
C) THD% = 10.74%
D) THD% = 10.68%

19) Determine the maximum dissipation that should be allowed for a 75 W silicon transistor rated at 22° C. Derating is required above 22° C by a derating factor of 0.35 W/° C at a case temp of 142° C.
A) 30 W
B) 27 W
C) 25 W
D) 22 W

20) A given transistor has a power derating factor of 0.25 mW/°C. This transistor has a power dissipation rating of 0.5 W at 27°C. What is the max temp the device can be allowed to operate as if it has to dissipate 450 mW?
A) 220° C
B) 210° C
C) 200° C
D) 190° C

21) Which of the following is true?
A) Efficiency is the ratio of power output to power input.
B) The power that an amplifier delivers to a load is equal to the difference between the power that the circuit draws from the power supply and the power that the circuit dissipates.
C) Power amplifiers are typically used to drive low impedance loads.
D) All of the above

22) Class B amplifiers _____.
A) provide an output signal for half the input signal cycle
B) usually contain an LC tank circuit in the BJT collector circuit
C) usually contain a single BJT that conducts through 360° of the ac input cycle
D) usually contain a single BJT that conducts through 270° of the ac input signal

23) Class C amplifiers usually contain _____.
A) two transistors
B) an LC tank circuit in the BJT collector circuit
C) a single BJT that conducts through 360° of the ac input cycle
D) a signal BJT that conducts through 270° of the ac input cycle

24) Class D amplifiers have a maximum theoretical efficiency of _____.
A) 25%
B) 78.5%
C) 50%
D) over 90%

25) A class A amplifier has values of V_{CC}= 10 V, I_B = 450 μA, and I_{CQ} = 10.55 mA. What is the total power that the circuit is drawing from the dc power supply?
A) 4.5 mW
B) 1.21 mW
C) 110 mW
D) Cannot be determined from the information given

26) A class A amplifier has an 8 V_{PP} output that is being applied to a 200 Ω load. What is the total ac load power?
A) 320 mW
B) 640 mW
C) 40 mW
D) 80 mW

27) 22. The maximum theoretical efficiency of an RC-coupled class A amplifier is _____.
A) 25%
B) 50%
C) 78.5%
D) 99%

28) An RC-coupled class A amplifier has values of I_B = 1 ma, I_{CQ} = 50 ma, V_{CC} = 15 V, V_{PP} = o.3 V, and R_C = 68 Ω. What is the maximum efficiency of the amplifier?
A) 16.6%
B) 15.9%
C) 21.2%
D) 25%

29) Impedance matching is important for _____.
A) maximum voltage transfer from source to load
B) maximum power transfer from source to load
C) maximum impedance transfer from source to load
D) maximum current transfer from source

30) The transformer-coupled class A amplifier _____.
A) is 60% efficient
B) is able to provide for impedance transformation
C) usually contains two transistors
D) All of the above

31) The dc load line of a transformer-coupled class A amplifier is _____.
A) identical to that of an RC-coupled class A amplifier
B) a near-horizontal line
C) a near-vertical line
D) None of the above

32) The maximum value of $V_{CE(MAX)}$ in a transformer-coupled class A amplifier will be greater than V_{CC}. This is caused by _____.
A) the input biasing network
B) the efficiency characteristics of the amplifier
C) the counter emf produced by the transformer primary
D) the natural relationship between V_{CE}, V, and $V_{p\text{-}p}$

33) A transformer-coupled class A amplifier has the following values:
turns ratio = 3:1
RL = 200 Ω
V_{CEQ} = 6 V
I_{CQ} = 12 mA
What is the maximum possible change in V_{CE} for the circuit?
A) 3 V
B) 6 V
C) 12 V
D) Cannot be determined from the information given

34) A transformer-coupled class A amplifier has a transformer turns ratio of 4 : 1 and a load resistance of 25 Ω. The peak-to-peak value of V_{CE} is 12 V. What is the approximate load power for the circuit?
A) 45 mW
B) 160 mW
C) 90 mW
D) 60 mW

35) A transformer-coupled class A amplifier has a transformer turns ratio of 4 : 1 and a R_L of 25 Ω. The peak-to-peak value of V_E is 12 V and it draws 220 mW from the dc power supply. What is the efficiency of the circuit?
A) 27.3%
B) 40.9%
C) 73%
D) 20.5%

36) Complementary-symmetry amplifiers are generally preferred over standard push-pull amplifiers because _____.
A) they use complementary transistors
B) they do not require the use of an output transformer
C) they have high efficiency ratings
D) they can drive lower impedance loads

37) Crossover distortion in class B amplifiers is prevented by _____.
A) biasing the transistors deeply into cutoff
B) biasing the transistors slightly above cutoff
C) using complementary-symmetry transistors
D) increasing the load resistance

38) A class AB amplifier has a supply voltage that is equal to +15 VDC. The value of $V_{CE(OFF)}$ for either transistor is approximately equal to _____.
A) 15 V
B) 5 V
C) 7.55 V
D) 0.7 V

39) A given transistor has a power derating factor of 1.8 mW/° C and a power dissipation rating of 400 mW at 25° C. How much power can the device dissipate at 120° C?
A) 216 mW
B) 81.9 mW
C) 184 mW
D) 355 mW

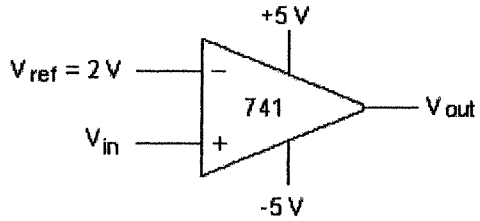
40) The unit of measurement for thermal resistance is _____.
A) °C / W
B) Ω / °C
C) W / V
D) Ω · W / °C

ANSWER KEY: Chapter 12: Power Amplifiers

1) A
2) B
3) D
4) C
5) B
6) B
7) B
8) B
9) C
10) D
11) D
12) C
13) C
14) C
15) B
16) C
17) C
18) A
19) B
20) C
21) D
22) A
23) B
24) D
25) A
26) C
27) A
28) C
29) B
30) D
31) C
32) D
33) B
34) A
35) D
36) B
37) B
38) C
39) B
40) A

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 13: Linear-Digital ICs

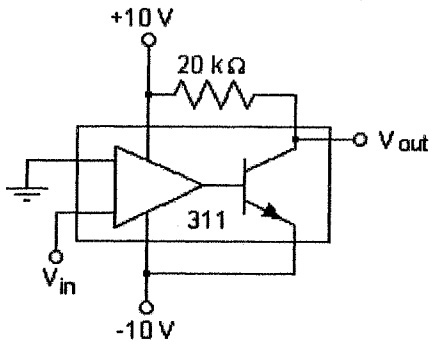
1) What is the output of this 741 op-amp comparator when the input is 1.5 V?



A) + 5 V
B) + 2 V
C) - 2 V
D) - 5 V

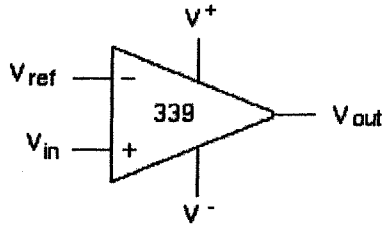
2) The output of a 311 comparator is taken from _____ so that it can be used to drive a variety of loads such as a lamp or a relay.
A) the strobe pins of the op-amp
B) the output pin of the op-amp
C) a bipolar transistor
D) the noninverting input of the op-amp

3) The output of this 311 comparator changes state when the input voltage _____.



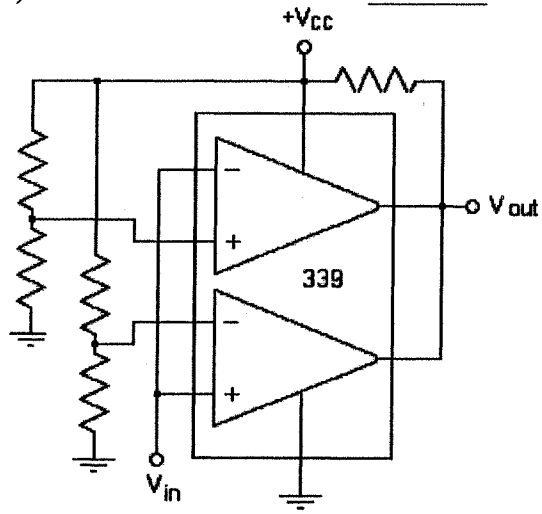
A) crosses 0 V and going from negative to positive only
B) crosses 0 V in any direction
C) crosses 0 V and going from positive to negative only
D) None of the above

4) The input voltage at the noninverting terminal of this comparator is greater than the reference voltage at the inverting terminal. What is the status of the output?



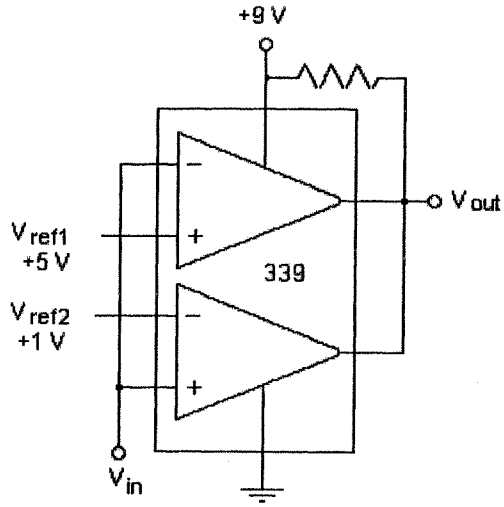
A) open circuit
B) equal to V^+
C) equal to V^-
D) short circuit

5) The circuit shown here is a _____.



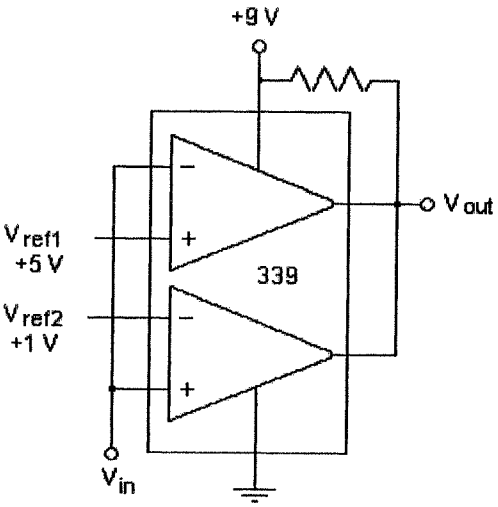
A) level detector
B) window detector
C) door detector
D) zero crossing detector

6) When the input to this circuit is less than 1 V, the output is _____.



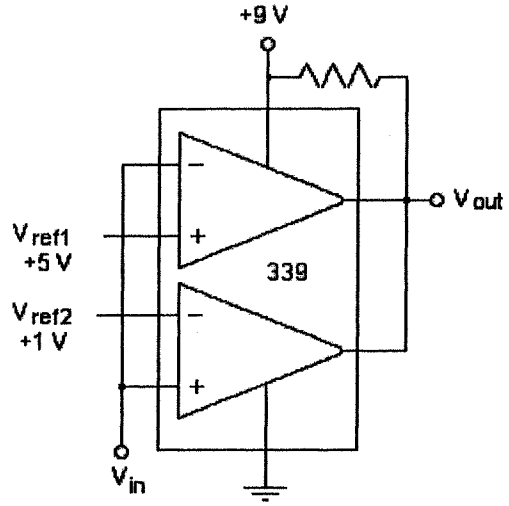
A) 0 V
B) 9 V
C) open circuit
D) The input is not allowed to be less than 1 V.

7) When the input voltage to this circuit is more than 1 V, but less than 5 V, the output is _____.



A) 0 V
B) 9 V
C) open circuit
D) The input is not allowed to be in the specified voltage range.

8) When the input to this circuit is more than 5 V, the output is _____.



A) 0 V
B) 9 V
C) open circuit
D) The input voltage is not allowed to be more than 5 V.

9) The ladder network for digital-to-analog conversion uses resistors _____.
A) that are required to be very precise
B) that are wire-wound so they can carry a large current
C) that have a ratio that is 2 : 1, but the actual value is not very important
D) that have a ratio that is 1 : 2 : 4 : 8 : . . ., depending on how many legs you need for the ladder

10) Which one of the following is most closely associated with a R-2R ladder network?
A) comparator
B) 555 timer
C) D/A converter
D) PLL

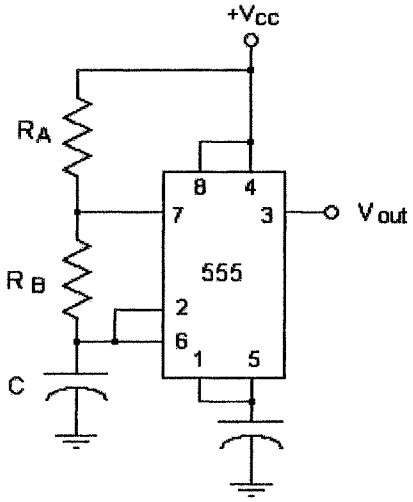
11) An 8-stage D/A converter has a reference voltage of +12 V. What is the resolution of the converter?
A) 1.5 V
B) 666.7 mV
C) 46.875 mV
D) comparator
E) 47.0 mV

12) In the dual-slope A/D converter, the first slope (the one that charges the capacitor) _____.
A) has a constant slope
B) is at a constant voltage level
C) has a slope until the capacitor charges to a predetermined level
D) has a constant slope over a fixed time interval

13) In the dual-slope A/D converter, the second slope (the one that discharges the capacitor) _____.
A) has a constant slope
B) is at a constant voltage level
C) has a constant slope until the capacitor discharges to a predetermined level
D) has a constant slope over a fixed time interval

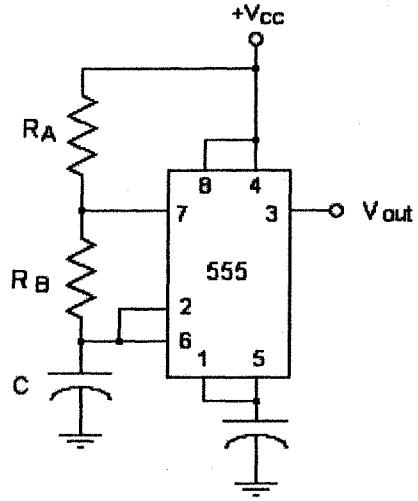
14) The 555 timer can be used as a(n) _____.
A) monostable multivibrator
B) astable multivibrator
C) pulse width modulator
D) All of the above

15) This 555 IC circuit is a(n) _____.



A) monostable flip-flop
B) bistable flip-flop
C) free-running oscillator
D) phase-locked loop

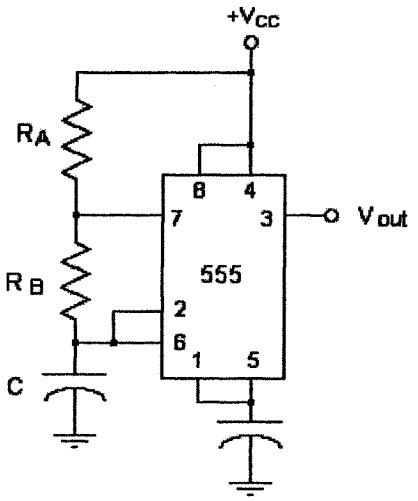
16) Resistor R_B and capacitor C in this circuit determine _____.



A) high time of the output waveform
B) low time of the output waveform
C) output dc voltage level
D) output phase

17) Decreasing the $+V_{cc}$ supply voltage for a 555 free-running oscillator causes _____.
A) the output frequency to increase
B) the output frequency to decrease
C) no significant change in the output frequency
D) causes a positive shift in the output phase

18) Capacitor C in this circuit both charges and discharges through _____.



A) R_A
B) R_B
C) R_A and R_B
D) terminal 7

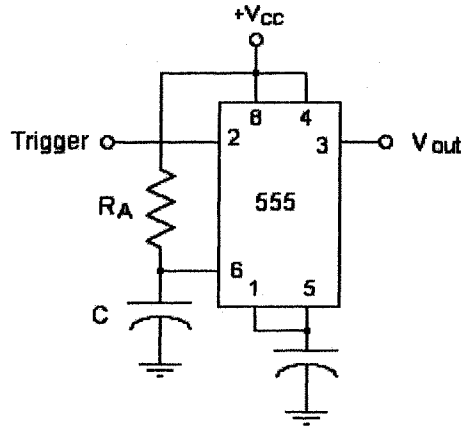
19) Which of the following expressions indicates the time-high for the waveform from a 555 astable multivibrator?
A) $0.7(R_A + R_B)C$
B) $0.7R_BC$
C) $\dfrac{1.44}{(R_A + 2R_B)C}$
D) $1.1R_AC$

20) Which of the following expressions indicates the time-low for the waveform from a 555 astable multivibrator?
A) $0.7(R_A + R_B)C$
B) $0.7R_BC$
C) $\dfrac{1.44}{(R_A + 2R_B)C}$
D) $1.1R_AC$

21) Which of the following expressions indicates the frequency of the waveform from a 555 astable multivibrator?
A) $0.7(R_A + R_B)C$
B) $0.7R_BC$
C) $\dfrac{1.44}{(R_A + 2R_B)C}$
D) $1.1R_AC$

22) Which of the following expressions indicates the duration of the output pulse from a 555 monostable multivibrator?
A) $0.7(R_A + R_B)C$
B) $0.7R_BC$
C) $\dfrac{1.44}{(R_A + 2R_B)C}$
D) $1.1R_AC$

23) This 555 IC is being used as a(n) _____.

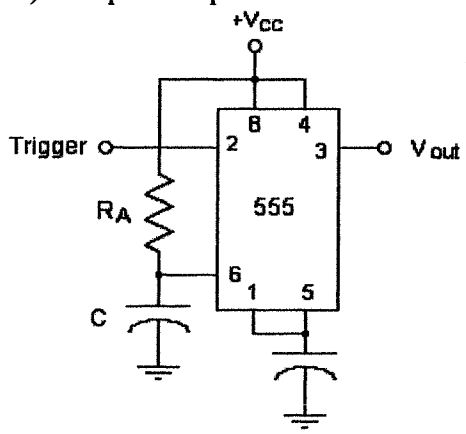


A) one-shot multivibrator
B) astable flip-flop
C) VCO
D) PLL

24) In the 566 VCO, the output frequency is controlled by the external _____.
A) capacitor
B) resistor
C) input voltage
D) All of the above

25) A 555 monostable multivibrator begins it timing cycle _____.
A) at intervals determined by the values of R and C
B) in the lock range
C) negative-going edge of the input trigger waveform
D) positive-going edge of the input trigger waveform

26) The positive pulse width for this circuit is mainly determine by _____.

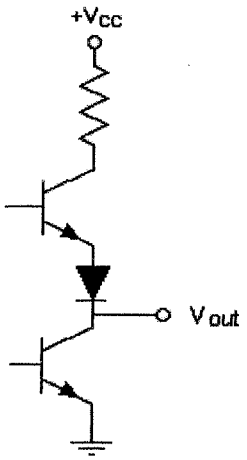


A) the value of $+V_{cc}$
B) the amplitude of the trigger waveform
C) the values of R_A and C
D) None of the above

27) When a sinusoidal waveform is applied to the modulating input a basic 566 VCO, the output frequency increases _____.
A) as the input goes more positive
B) as the input goes more negative
C) whenever the input crosses through zero in either direction
D) The answer cannot be determined without knowing the values of R and C for the circuit.

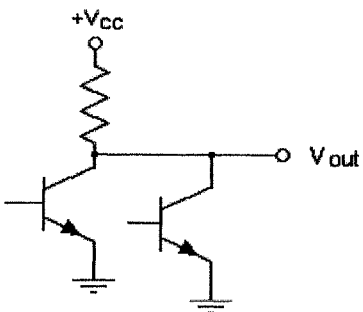
28) In a simple 556 VCO application, increasing time constant of the external capacitor and resistor causes the output frequency to _____. (Assume all other variables remain unchanged.)
A) increase
B) decreases
C) There will be no significant change.
D) The answer cannot be determined without knowing the ratio of input-to-supply voltage.

29) This circuit represents a(n) _____.



A) open-collector output
B) open-drain output
C) regular output of an op-amp
D) totem pole, or tri-state, output

30) This circuit represents a(n) _____.



A) open-collector output
B) open-drain output
C) regular output of an op-amp
D) totem-pole, or tri-state, output

31) A comparator has a _____.
A) digital input and a linear output
B) digital input and a digital output
C) linear input and a linear output
D) linear input and a digital output

32) Typical gain of a comparator is _____.
A) 100
B) 1000
C) 10,000
D) 100,000

33) Which of the following represents an improvement in comparators due to IC technology?
A) faster switching
B) built-in noise immunity
C) variety of output drive capability
D) All of the above

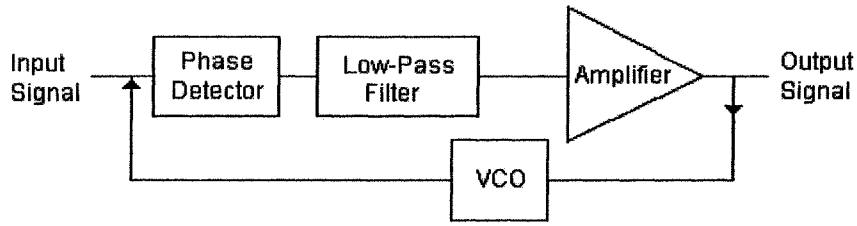
34) Which of these is an application of the 555 IC?
A) astable multivibrator
B) A/D converter
C) D/A converter
D) comparator

35) Which of the following is an application of the 566 IC?
A) monostable multivibrator
B) VCO
C) Comparator
D) astable multivibrator

36) A circuit that contains a phase detector, a low-pass filter, and a VCO is a(n) _____.
A) comparator
B) astable multivibrator
C) phase-locked loop
D) None of the above

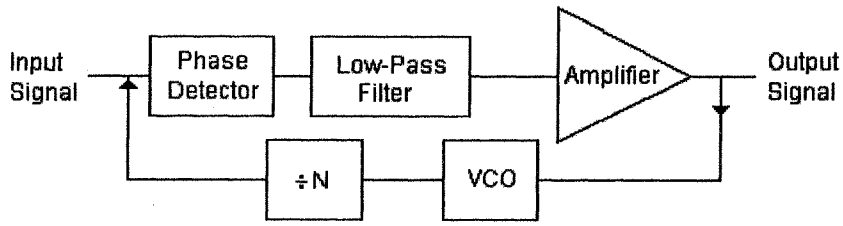
37) A common application of the PLL is _____.
A) frequency synthesis
B) FM demodulation
C) demodulation of two carrier frequencies
D) All of the above

38) This figure is a block diagram of a(n) _____.



A) frequency synthesizer
B) A/D converter
C) PLL
D) 555 timer

39) This figure is a block diagram of a(n) _____.



A) frequency synthesizer
B) A/D converter
C) PLL
D) 555 timer

40) A circuit that provides an output signal at the voltage and/or current levels suitable for operating a particular load is called a(n) _____.
A) RS-232C
B) driver
C) polarizer
D) None of the above

ANSWER KEY: Chapter 13: Linear-Digital ICs

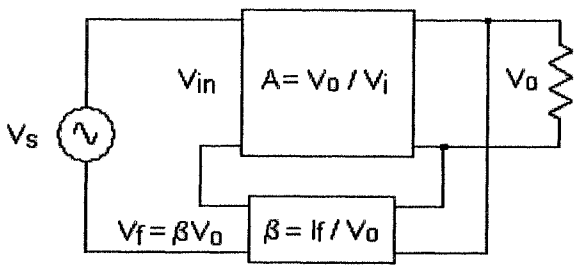
1) D
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6) A
7) B
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9) C
10) C
11) C
12) C
13) C
14) D
15) C
16) A
17) C
18) B
19) A
20) B
21) C
22) D
23) A
24) D
25) C
26) C
27) B
28) B
29) D
30) A
31) D
32) D
33) D
34) A
35) B
36) C
37) A
38) C
39) A
40) B

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 14: Feedback and Oscillator Circuits

1) The advantage of negative feedback is _____.
A) higher input impedance
B) voltage gain that is more stable
C) lower output impedance
D) All of the above

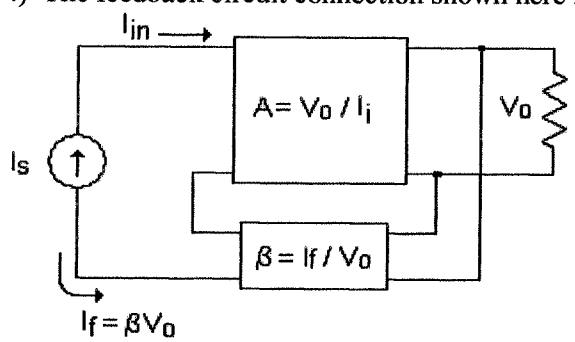
2) Calculate the gain of a negative feedback amplifier having A = -2000 and β = - 1/16.
A) -2000
B) -1/16
C) -125
D) 125

3) The feedback circuit connection shown here is _____.



A) voltage series feedback
B) voltage shunt feedback
C) current series feedback
D) current shunt feedback

4) The feedback circuit connection shown here is _____.



A) voltage series feedback
B) voltage shunt feedback
C) current series feedback
D) current shunt feedback

5) Calculate the voltage gain for voltage series feedback when the amplifier gain is A = -2000 and β = -1/50.
A) -2000
B) -1/50
C) 50
D) -48.75

6) Calculate the input impedance for voltage series feedback when the amplifier has R_i = 10 kΩ, amplifier gain is A = -2000, and $\beta = -\frac{1}{50}$.
A) input impedance with feedback 10 kΩ
B) input impedance with feedback 410 kΩ
C) input impedance with feedback 244 Ω
D) input impedance with feedback 50 Ω

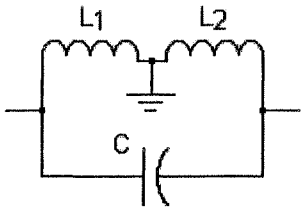
7) The feedback amplifier is unstable if the *Nyquist curve* plotted _____ the (-1 + j0) point and it is otherwise stable.
A) passes to the left of
B) encloses (encircles)
C) passes right over
D) does not come close to

8) The _____ margin is defined as the negative of the value of |βA| decibels at the frequency at which the phase angle is 180°.
A) error
B) phase
C) gain
D) feedback

9) The _____ margin is defined as the angle of 180° minus the magnitude of the angle at which the value |βA| is unity (0 dB).
A) error
B) phase
C) gain
D) feedback

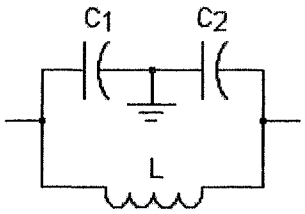
10) For oscillations to exist and the voltage to sustain the loop operations, the *Barkhausen Criterion* tells us that the loop gain 3A must be exactly equal to _____.
A) unity (1)
B) minus one
C) equal to the reciprocal of β
D) more than one

11) The tank circuit shown here is for a _____ oscillator.



A) Hartley
B) Colpitts
C) phase shift
D) crystal

12) The tank circuit shown here is for a _____ oscillator.



A) Hartley
B) Colpitts
C) phase shift
D) crystal

13) Adding a negative voltage-feedback network to an amplifier has no effect on the value of _____ for the circuit.
A) input impedance
B) frequency response
C) signal distortion
D) None of the above

14) Voltage-series feedback _____ the input impedance of an op-amp.
A) increases
B) decreases
C) reduces by half
D) has no effect on

15) Voltage-series feedback _____ the output impedance of an op-amp.
A) increases
B) decreases
C) reduces by half
D) has no effect on

16) Voltage-series feedback _____ the bandwidth of an op-amp.
A) increases
B) decreases
C) reduces by half
D) has no effect on

17) The input impedance of a voltage-shunt feedback amplifier _____ the input impedance of its op-amp.
A) is increased when compared to
B) is decreased when compared to
C) is reduced by half when compared to
D) has no effect on

18) The output impedance of a voltage-shunt feedback amplifier _____ the output impedance of its op-amp.
A) is increased when compared to
B) is decreased when compared to
C) is reduced by half when compared to
D) has no effect on

19) Current-series feedback _____ the input impedance of an op-amp.
A) increases
B) decreases
C) reduces by half
D) has no effect on

20) Current-series feedback _____ the output impedance of an op-amp.
A) increases
B) decreases
C) reduces by half
D) has no effect on

21) Current-series feedback _____ the bandwidth of an op-amp.
A) increases
B) decreases
C) reduces by half
D) has no effect on

22) The input impedance of current-shunt feedback amplifier _____ the input impedance of its op-amp.
A) is increased when compared to
B) is decreased when compared to
C) is reduced by half when compared to
D) has no effect on

23) An amplifier has a gain-bandwidth product of 200 MHz. A feedback network is added that has a feedback factor ($1 + \beta A$) of 18.88. What is the gain-bandwidth product for the circuit with the added feedback network?
A) 10.59 MHz
B) 18.88 MHz
C) 200 MHz
D) None of the above

24) Negative voltage feedback _____.
A) increases A_v
B) decreases bandwidth
C) decreases A_v
D) increases A_i

25) Negative current feedback _____.
A) increases A_i
B) decreases A_i
C) decreases bandwidth
D) increases A_v

26) Positive feedback is used to produce a special type of circuit called a(n) _____.
A) inverting amplifier
B) noninverting amplifier
C) oscillator
D) feedback regulator

27) Positive feedback is also called _____.
A) degenerative feedback
B) additive feedback
C) Barkhausen oscillation
D) regressive feedback

28) An oscillator has the following values: $A_v = 188$ and $\beta = 0.00488$. Which of the following statements is true?
A) The circuit has a constant amplitude output.
B) The output from the circuit fades out after several cycles.
C) The output from the circuit clips after several cycles.
D) None of the above

29) The Barkhausen criterion states that _____ in an oscillator.
A) $\beta = 10$
B) $\beta = -1$
C) $\beta A_v = 1$
D) None of the above

30) Which of the following is *not* a requirement for oscillator operation?
A) The circuit must fulfill the Barkhausen criterion.
B) The circuit must initially be triggered into operation.
C) The feedback network must contain an RC circuit.
D) The circuit must provide positive feedback.

31) The total phase shift around a negative feedback loop of a common-emitter circuit is _____.
A) 360° or 0°
B) 180°
C) 90°
D) 45°

32) The total phase shift around a positive feedback loop for a common-emitter circuit is _____.
A) 360° or 0°
B) 180°
C) 90°
D) 45°

33) In a positive feedback system, the feedback signal and the amplifier input signal are _____.
A) in phase
B) 45° out of phase
C) 90° out of phase
D) 180° out of phase

34) In a practical phase-shift oscillator, each RC circuit section produces a _____.
A) 30° phase shift
B) 60° phase shift
C) 90° phase shift
D) 180° phase shift

35) The negative feedback circuit in an op-amp Wien-bridge oscillator is used to _____.
A) determine the frequency of operation
B) control the gain of the circuit
C) bias the positive feedback network
D) prevent unwanted oscillations

36) The positive feedback circuit in a Wien-bridge oscillator is used to _____.
A) determine the frequency of operation
B) control the gain of the circuit
C) bias the negative feedback network
D) prevent unwanted oscillations

37) The circuit recognition feature of the Colpitts oscillator is _____.
A) a pair of tapped capacitors in parallel with an inductor
B) a pair of tapped inductors in parallel with a capacitor
C) a feedback transformer with a capacitor in parallel with its primary winding
D) a pair of tapped capacitors in parallel with an inductor and a third small-value capacitor

38) The circuit recognition feature of the Hartley oscillator is a _____.
A) pair of tapped capacitors in parallel with an inductor
B) pair of tapped inductors in parallel with a capacitor
C) feedback transformer with a capacitor in parallel with its primary winding
D) pair of tapped capacitors in parallel with an inductor and a third small-value capacitor

39) A Colpitts oscillator has values of $C_1 = 1$ μF, $C_2 = 33$ μF, and $L = 100$ μH. What is the frequency of oscillation?
A) 12.154 kHz
B) 16.154 kHz
C) 20.3 kHz
D) 100.5 kHz

40) A Colpitts oscillator has C = 1 μF, C2 = 33 μF and L = 4.7 mH. What is the approximate operating frequency of the circuit?
A) 74.5 kHz
B) 25.6 kHz
C) 12.8 kHz
D) None of the above

41) A Hartley oscillator has the following values: RFC = 1 mH, L_1 = 100 μH, L_2 = 22 μH, and C = 0.001 μF. Assuming that the mutual inductance in the circuit is too small to be considered, what is the approximate output frequency of the oscillator?
A) 456 kHz
B) 152 kHz
C) 503 kHz
D) None of the above

42) The biggest advantage that LC oscillators have over RC oscillators is the fact that LC oscillators generally _____.
A) can be operated at a much higher frequency
B) can be constructed more economically
C) require less physical space
D) None of the above

43) At the _____ a crystal acts as a short circuit.
A) series-resonant frequency
B) parallel-resonant frequency
C) non-resonant frequency
D) None of the above

44) At the _____ a crystal acts as an open circuit.
A) series-resonant frequency
B) parallel-resonant frequency
C) non-resonant frequency
D) None of the above

45) Crystal-controlled oscillators have _____.
A) very low operating frequencies
B) very stable output frequencies
C) extremely simple biasing circuits
D) All of the above

46) The effect that is responsible for the resonant quality of a crystal is known as the _____ effect.
A) Miller
B) Barkhausen
C) hysteresis
D) piezoelectric

47) The electrical equivalent circuit for a crystal actually models _____ effects.
A) electromechanical
B) optoelectrical
C) electrical noise
D) thermal

48) When the reactances in the RLC leg of a crystal are equal and opposite, the resulting condition is known as _____.
A) cutoff
B) Barkhausen criterion
C) parallel resonance
D) series resonance

49) The Q of a crystal is typically _____.
A) 5000
B) 20,000
C) 500,000
D) 1,000,000

ANSWER KEY: Chapter 14: Feedback and Oscillator Circuits

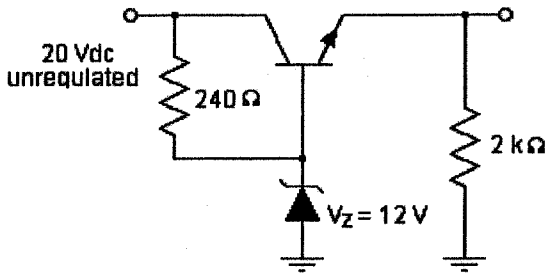
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16) A
17) B
18) C
19) A
20) A
21) A
22) B
23) C
24) C
25) B
26) C
27) C
28) B
29) C
30) C
31) A
32) B
33) A
34) B
35) B
36) A
37) D
38) B
39) A
40) A
41) A
42) A
43) A
44) A
45) B
46) D
47) A
48) D
49) B

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 15: Power Supplies (Voltage Regulators)

1) The typical parts of a power supply are _____.
A) a transformer
B) diode rectifier and filter
C) voltage drop element
D) All of the above

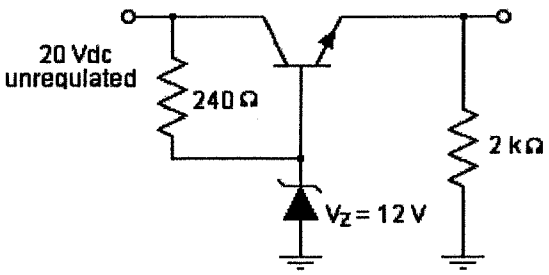
2) A dc voltage supply provides an output of 25 V under no-load condition. This output voltage drops to 22.5 V under full-load condition. Calculate the percent voltage regulation.
A) 10%
B) 90%
C) 11.1%
D) 2.5%

3) Calculate the output voltage for the voltage regulator shown here.



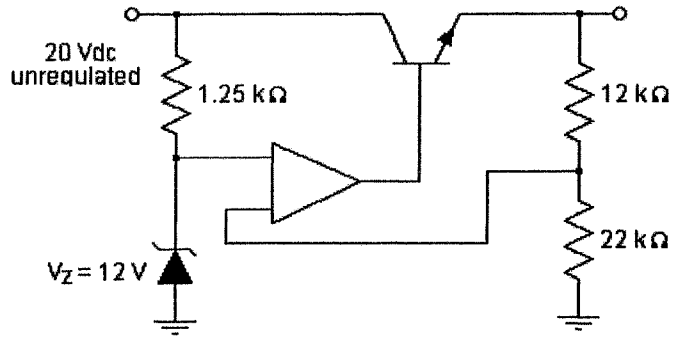
A) 20.0 V
B) 12.0 V
C) 11.3 V
D) 8.0 V

4) Calculate the current through the Zener diode for this voltage regulator.



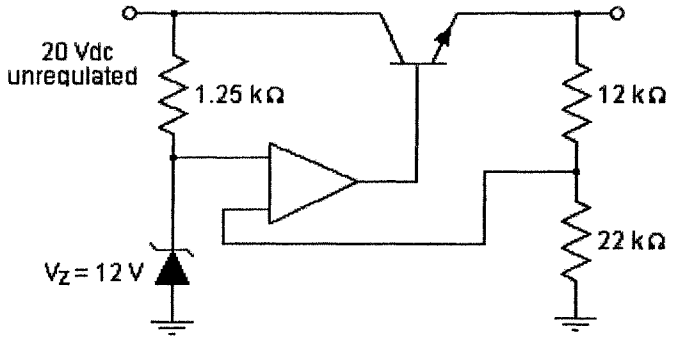
A) 50 mA
B) 33 mA
C) 83.33 mA
D) None of the above

5) Calculate the output voltage for this improved series voltage regulator.



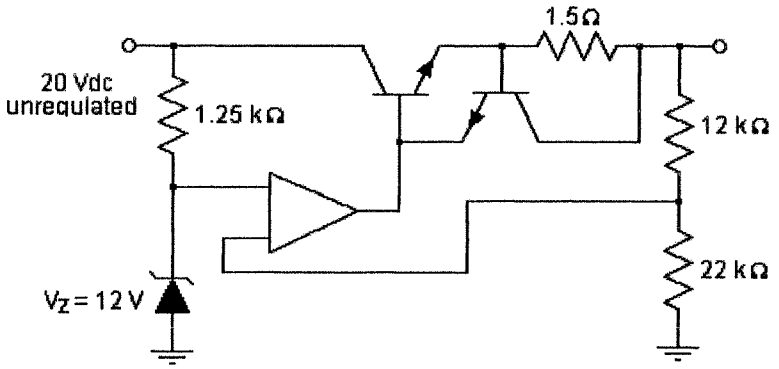
A) 20.0 V
B) 12.0 V
C) 18.5 V
D) 8.0 V

6) Calculate the current through the Zener diode for this series voltage regulator.



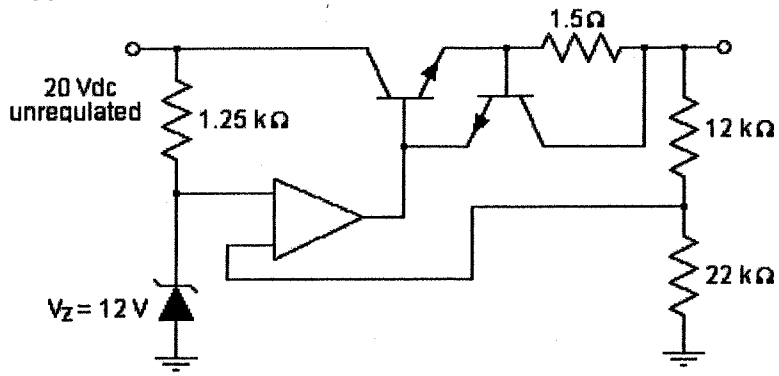
A) 9.6 mA
B) 16 mA
C) 6.4 mA
D) 14.8 mA

7) Calculate the approximate output current when the circuit triggers the short circuit protection.



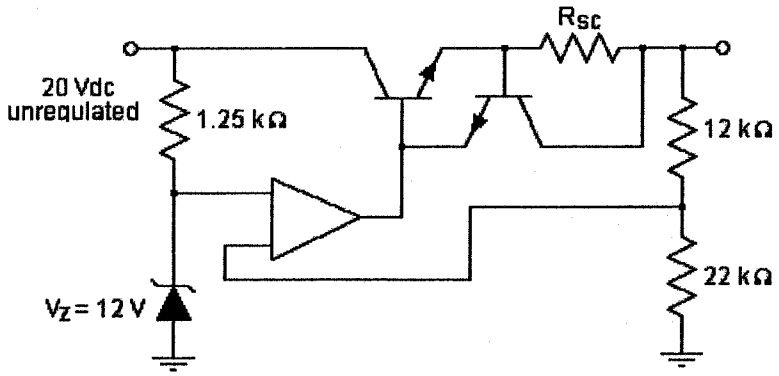
A) 0.5 A
B) 8 A
C) 5.33 A
D) 13.33 A

8) Calculate the approximate load resistance that the voltage regulator can drive before the circuit triggers the current limit circuit protection.



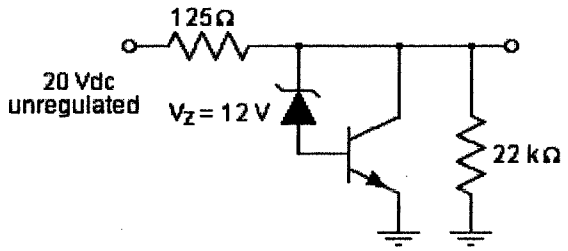
A) 40 kΩ and more
B) 400 Ω and more
C) 40 Ω and more
D) 37 Ω and more

9) Calculate the approximate resistance Rsc required to provide current limit circuit protection at the 0.75 A level.



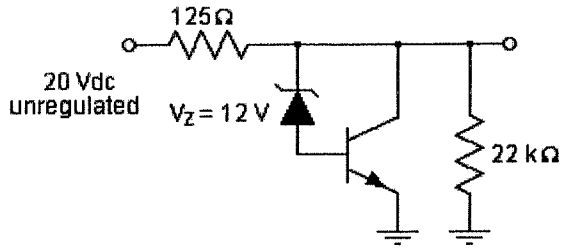
A) 0.5 Ω
B) 0.75 Ω
C) 1.0 Ω
D) 0.93 Ω

10) Referring to this shunt voltage regulator, calculate the output voltage that the load sees.



A) 10 V
B) 8.2 V
C) 20 V
D) 8.9 V

11) For this shunt voltage regulator, determine the minimum load resistance that the voltage regulator can drive without dropping out of regulation.



A) 10000 Ω
B) 1000 Ω
C) 100 Ω
D) 10 Ω

12) A three-terminal voltage regulator affords several different types of _____ protection.
A) short-circuit
B) thermal shut-down
C) Both A and B
D) Neither A nor B

13) A three-terminal voltage regulator drops out of regulation if the difference between the input voltage and the expected regulated output voltage is less than _____.
A) 10% expected regulated output voltage
B) 90% expected regulated output voltage
C) 2.0 V
D) The input and the expected output voltage can be exactly same.

14) A rectifier is used to _____.
A) convert ac to pulsating dc
B) reduce the variations in a pulsating dc signal
C) maintain a constant power supply dc output voltage
D) convert one dc level to another

15) The basic power supply is made up of _____.
A) a regulator, a follower, and a rectifier
B) a filter, a follower, and a regulator
C) a rectifier, a filter, and a regulator
D) None of the above

16) A voltage regulator _____.
A) maintains a constant power supply dc output voltage
B) limits the primary voltage of a power supply transformer
C) reduces the power supply ripple output voltage
D) None of the above

17) The ideal voltage regulator maintains a constant dc output voltage regardless of changes in _____.
A) its input voltage
B) its output voltage demand
C) its load current demand
D) either its load current demand or its input voltage

18) The term *full load* means _____.
A) load resistance is at a maximum value
B) load resistance is at a minimum value
C) no load resistance is present
D) load current is at a minimum value

19) A rectified dc voltage was measured with both an ac and a dc voltmeter. It was found that $V_{dc} = 50$ V and $V_{ac} = 2.16\ V_{rms}$. What was the percent ripple?
A) 6%
B) 4.32%
C) 0.432%
D) 0.86%

20) A voltage regulator is rated for an output current range of $I_L = 0$ to 40 mA. Under no-load conditions the output voltage from the circuit is 4 V_{dc}. Under full-load conditions, the output voltage from the circuit is 3.984 V_{dc}. What is the percent load regulation of the circuit?
A) 400%
B) 2.5
C) 0.4%
D) None of the above

21) The ideal line percent regulation rating is _____.
A) 100%
B) 75%
C) 50%
D) zero

22) The _____ the percent load regulation rating of a voltage regulator, the higher the quality of the circuit.
A) lower
B) higher
C) larger the change in
D) None of the above

23) A voltage regulator has a dc supply voltage of 50 V when the output is unloaded. When a load is connected the output voltage is 46 V. What is the percent regulation?
A) 4%
B) 8.7%
C) 92%
D) None of the above

24) A capacitive filter is added to a half-wave rectifier. The initial value of capacitance is 22 μF. If this value is increased to 100 μF, the ripple output from the circuit will _____.
A) increase
B) decrease
C) remain the same
D) Cannot be predicted

25) A filtered rectifier has a 15 V_{dc} output with 100 mV_{p-p} of ripple. The peak output voltage for the circuit is _____.
A) 15.1 V_{peak}
B) 14.9 V_{peak}
C) 15.05 V_{peak}
D) 47.2 V_{peak}

26) A capacitive filter is added to a full-wave rectifier. The value of capacitance is 22 μF. If the circuit R_L = 1000 Ω, what is the circuits ripple factor r?
A) 15.2%
B) 12.7%
C) 10.9%
D) Cannot be determined from the information given

27) Calculate the ripple of a capacitor filter for a peak rectified voltage of 40 V, a capacitor of 80 μF and a 100 mA load current.
A) 2.84%
B) 4.3%
C) 8.33%
D) 10.24%

28) Calculate the percent ripple for a RC filter if the inputs are V_{dc} = 150 V, $V_{r(rms)}$ = 15 V. The filter components are $C_1 = C_2$ = 50 μF, R = 500Ω and R_L = 5 KΩ.
A) 0.572%
B) 0.884%
C) 1.27%
D) 4.39%

29) The reduction of the ac component of the output voltage is due to the _____ action of the RC filter sections.
A) voltage divider
B) current gain
C) current divider
D) none of the above

30) Shunt voltage regulators require _____.
A) shorted-load protection
B) load voltage sampling
C) high-frequency protection
D) open-load protection

31) Which of the following is not a type of voltage regulator?
A) fixed negative
B) adjustable
C) variable polarity
D) fixed positive

32) The connections on a three-terminal, IC voltage regulator are _____.
A) R, C, gnd
B) in/out, filter, gnd
C) in, out, gnd
D) A, B, gnd

33) The ability of an IC voltage regulator to attenuate any input ripple voltage is called its _____ rating.
A) ripple attenuation
B) ripple reduction
C) ripple rejection ratio in dB
D) ripple elimination

ANSWER KEY: Chapter 15: Power Supplies (Voltage Regulators)

1) D
2) C
3) C
4) B
5) C
6) C
7) A
8) D
9) D
10) D
11) D
12) C
13) C
14) A
15) C
16) D
17) D
18) B
19) B
20) C
21) D
22) A
23) B
24) B
25) C
26) C
27) C
28) A
29) C
30) B
31) C
32) C
33) C

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 16: Other Two-Terminal Devices

1) This is the symbol for a(n) _____.



A) varactor diode
B) tunnel diode
C) Schottky diode
D) thermistor

2) This is the symbol for a(n) _____.



A) tunnel diode
B) varactor diode
C) Schottky diode
D) thermistor

3) This is the symbol for a(n) _____.



A) varactor diode
B) Schottky diode
C) tunnel diode
D) photodiode

4) This is the symbol for a(n) _____.



A) varicap diode
B) thermistor
C) Schottky diode
D) tunnel diode

5) This is the symbol for a(n) _____.



A) photodiode
B) light-emitting diode
C) thermistor
D) varicap

6) This is the symbol for a(n) _____.



A) thermistor
B) photoresistor
C) optoresistor
D) variac

7) Varactor diodes are also known as _____ diodes.
A) varigen
B) reactance
C) XCvar
D) varicap

8) *Hot-carrier* refers to _____ diodes.
A) photoconductive
B) pin
C) tunnel
D) Schottky

9) The capacitance of a varactor diode depends on _____
A) area of the depletion region
B) width of the depletion region
C) the permittivity of the semiconductor material
D) All of the above

10) As the reverse bias potential across a varactor diode increases, the transition capacitance _____.
A) decreases
B) increases linearly
C) remains constant
D) increases exponentially

11) Power diodes are most often used for _____.
A) rectification
B) high-frequency control
C) high-voltage operation
D) low-voltage operation

12) The most unique feature of the characteristics of the tunnel diode is _____.
A) very low positive resistance region
B) very high positive resistance region
C) negative resistance region
D) almost zero breakdown voltage implying almost non-existent depletion region

13) Optoelectronic devices are generally classified as _____.
A) optical couplers or optical isolators
B) optically discrete or optically integrated
C) optical emitters or optical detectors
D) optical diodes or optical transistors

14) The photodiode characteristics show equal spacing between the diode current curves for an equal increase in photon energy. So a graph of radiant flux versus diode current is a(n) _____ relationship.
A) exponential
B) parabolic
C) linear
D) square

15) The resistance of the photoconductive device varies with the intensity of the incident light. The relationship between the resistance and the light intensity is a(n) _____ relationship.
A) exponential
B) parabolic
C) linear
D) square

16) An LCD has power dissipation on the order of _____.
A) watts
B) milli-watts
C) micro-watts
D) None of the above

17) Field effect, or twisted nematic, LCDs require a(n) _____ to be applied in order for the light to be seen by the viewer.
A) constant current
B) constant voltage
C) alternating current
D) alternating voltage

18) A typical solar cell has an efficiency of approximately _____.
A) 1 - 3%
B) 10 - 12%
C) 25 - 30%
D) 75 - 80%

19) A thermistor is a temperature sensitive semiconductor resistor that has _____.
A) negative temperature coefficient
B) positive temperature coefficient
C) negative temperature coefficient at low temperature, positive temperature coefficient at high temperature
D) a temperature coefficient that may be either positive or negative

20) The Schottky diode is used for _____.
A) very high frequency applications
B) low noise applications
C) low voltage/high current power supplies
D) All of the above

21) In the Schottky diode, the injected carriers have a very high kinetic energy level compared to the electrons of the metal; as a result the device is called a(n) _____ diode.
A) junction
B) energy
C) hot-carrier
D) None of the above

22) The varactor acts as a _____.
A) current-controlled capacitance when forward biased
B) voltage-controlled capacitance when forward biased
C) current-controlled capacitance when reverse biased
D) voltage-controlled capacitance when reverse biased

23) The capacitance of a varactor is _____.
A) inversely proportional to the permittivity of the semiconductor material
B) directly proportional to the width of the depletion layer
C) inversely proportional to the amount of diode reverse voltage
D) None of the above

24) A varactor with a high capacitance ratio (CR) rating is well-suited for _____.
A) fine-tuning applications
B) coarse-tuning applications
C) crystal-controlled oscillator applications
D) extremely high-Q applications

25) The tank circuit in a tuned amplifier consists of a 2.2 mH inductor and a varactor with the following specs: $CT_i = 80$ pf when $V_R = 3$ Vdc, and $CR = 3$ for $V_R = 3$ Vdc to 6 Vdc. Calculate the resonant frequency when $V_R = 3$ Vdc.
A) 904.3 kHz
B) 758.7 kHz
C) 379.4 kHz
D) 189.9 kHz

26) The tank circuit in a tuned amplifier consists of a 2.2 mH inductor and a varactor with the following specs: $CT_i = 80$ pf when $V_R = 3$ Vdc, and CR = 3 for $V_R = 3$ Vdc to 6 Vdc. Calculate the resonant frequency when $V_R = 6$ Vdc.
A) 599.8 kHz
B) 479.9 kHz
C) 219.0 kHz
D) 1.20 MHz

27) Power diodes are constructed of silicon because of its _____.
A) higher current
B) higher temperature capacity
C) higher PIV
D) All of the above

28) The tunnel diode is a _____.
A) lightly doped diode that greatly reduces the depletion region
B) lightly doped diode that greatly increases the depletion region
C) heavily doped diode that greatly reduces the depletion region
D) heavily doped diode that greatly increases the depletion region

29) The tunnel diode region of operation between V_P and V_V is called the _____.
A) constant resistance region
B) constant current region
C) negative resistance region
D) negative current region

30) Tunnel diodes are often used as the active devices in _____.
A) constant resistance amplifier
B) negative resistance oscillators
C) negative current rectifiers
D) negative resistance linear amplifiers

31) Optoelectronic devices are generally classified as being either _____.
A) couplers or isolators
B) discrete or integrated
C) emitters or detectors
D) input or output devices

32) The typical infrared-emitting diode has a radiant flux versus dc current curve that is _____.
A) an exponential relationship
B) a square relationship
C) an almost linear relationship
D) None of the above

33) LCDs have power dissipation on the order of _____.
A) Watts
B) milli-Watts
C) micro-Watts
D) None of the above

34) LCDs require a(n) _____.
A) constant current
B) constant voltage
C) large heat sink
D) internal light source

35) LCDs are limited to a temperature range of _____.
A) 0 to 10 degrees C
B) 0 to 60 degrees C
C) 0 to 100 degrees C
D) 30 to 100 degrees C

36) LCDs degrade _____.
A) chemically
B) quickly
C) in a dry place
D) All of the above

37) Typical solar cell efficiency is _____.
A) 1%
B) 10%
C) 30%
D) 70%

38) The most widely used material for solar cells are _____.
A) selenium and silicon
B) indium arsenide and gold
C) gallium arsenide and cadmium sulfide
D) None of the above

39) A typical four solar cell array can deliver _____ of power.
A) 20 mW
B) 10.7 mW
C) 5.83 mW
D) 4.16 mW

40) A thermistor is a temperature-sensitive semiconductor resistor that has _____ *p-n* junctions.
A) no
B) one
C) two
D) three

ANSWER KEY: Chapter 16: Other Two-Terminal Devices

1) C
2) B
3) C
4) D
5) A
6) A
7) D
8) D
9) D
10) A
11) A
12) C
13) C
14) C
15) C
16) C
17) B
18) B
19) B
20) C
21) C
22) D
23) C
24) B
25) C
26) C
27) D
28) C
29) C
30) B
31) C
32) C
33) C
34) C
35) B
36) A
37) B
38) A
39) D
40) A

Electronic Devices and Circuit Theory, 9e (Boylestad)
Chapter 17: *pnpn* and Other Devices

1) Some of the popular applications of SCRs include _____.
A) relay control, time delay circuits, and regulated power supplies
B) choppers, inverters, and battery chargers
C) static switches, heater controls, and phase controls
D) All of the above

2) Thyristors are electronic devices that act as a _____.
A) silicon controlled rectifier
B) unijunction transistor
C) gate turn-off switch
D) All of the above

3) One region of the SCR curve when it is forward biased represents the non-conducting region of operation. What is this region called?
A) forward turn-off region
B) forward dropout region
C) forward blocking region
D) forward non-conduction region

4) A silicon controlled switch (SCS) can be turned off by applying a(n) _____.
A) opposite-level pulse to the anode gate
B) a low-level pulse on the cathode gate terminal
C) essential short-circuits from anode to cathode
D) All of the above

5) One of the very important characteristics of the GTO is _____.
A) improved current handling characteristics
B) improved noise characteristics
C) improved blocking region characteristics
D) improved turn-off time

6) The LASCR is a device that can be turned on by _____.
A) noise intensity of dB level
B) harmonic content of the signal present at the gate
C) light intensity
D) wavelength of the light input

7) Determine the frequency of oscillation for a UJT relaxation oscillator where $R_1 = 50\ k\Omega$, $C = 0.1\ pF$, and $\eta = 0.6$.
A) 18 Hz
B) 180 Hz
C) 218 Hz
D) 82 Hz

8) Thyristors are electronic devices that act as a _____.
A) silicon-controlled switch
B) silicon-controlled rectifier
C) unijuction transistor
D) all of the above

9) The *pnpn* device that is of greatest interest today is the _____.
A) triac
B) SCR
C) PUT
D) diac

10) Silicon was chosen for the construction of the SCR because of its _____.
A) high frequency characteristic
B) switching speed characteristics
C) temperature and power capabilities
D) All of the above

11) Anode-current interruption and forced commutation are the two methods that are used to _____ an SCR.
A) turn off
B) turn on
C) hold on
D) None of the above

12) An SCR is forced into forward conduction when V_F exceeds the _____ rating of the device.
A) forward conducting voltage
B) forward breakover voltage
C) forward trigger voltage
D) forward breakdown voltage

13) Once an SCR is forced into forward conduction, it continues to conduct until I_F drops below the _____ rating of the device.
A) minimum forward current
B) forward breakover current
C) holding current
D) dropout current

14) What are the two methods that are commonly used to return an SCR to its conconducting state?
A) Anode current interruption and forced commutation
B) Current holding and forced commutation
C) Anode current interruption and current holding
D) Forced commutation and current dropout

15) The region of the SCR forward operating curve that represents the nonconducting region of operation is called the _____.
A) forward off-state region
B) forward blocking region
C) forward dropout region
D) None of the above

16) Which of the following distinguishes the SCR from the SCS?
A) The SCS has a fourth terminal, called the anode gate.
B) The SCS is driven into cutoff using entirely different methods.
C) The SCS has less means of being forced into its forward conducting state.
D) All of the above

17) The forward operating curve of the SCR is identical to that of a _____.
A) SCS
B) GTO
C) diac
D) triac

18) The silicon controlled switch has two _____.
A) anodes
B) cathodes
C) gates
D) None of the above

19) Which of the following devices can be driven into its conducting or nonconducting state by applying the proper pulse to its gate terminal?
A) Shockley diode
B) SBS
C) GTO device
D) triac

20) The LASCR is a device whose state is controlled by _____.
A) light intensity
B) spectral wavelength
C) light amplitude
D) the light-area product

21) Which of the following devices acts as an SCR with $I_G = 0$?
A) Shockley diode
B) SBS
C) GTO device
D) triac

22) The primary difference between the diac and the SCS is the fact that the _____.
A) diac has a higher maximum power dissipation rating
B) SCS is capable of conducting in only one direction
C) diac is no longer used in any practical application
D) SCS requires the use of a snubber

23) A triac is a _____ switching device.
A) bilateral
B) unilateral
C) multilateral
D) trilateral

24) The _____ is commonly used to control SCR triggering.
A) UJT
B) diac
C) SCR
D) JFET

25) A UJT has the following values: $\eta = 0.72$ (maximum) and $V_{EBB} = 12$ V. What is the maximum value of V_{EB1} required to trigger the device into conduction?
A) 8.64 V
B) 17.4 V
C) 12 V
D) 9.34 V

26) UJTs are commonly used as _____.
A) breakover devices
B) amplifiers
C) thyristor triggering devices
D) tuned oscillators

27) A PUT has a value of $V_{GK} = +8$ V. What value of V_{AK} is needed to trigger the device into conduction?
A) -8.7 V
B) +16.7 V
C) +8.7 V
D) Cannot be determined from the information given

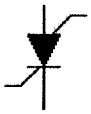
28) Which of the following devices is actually an integrated circuit rather than a single discrete component?
A) SIDAC
B) optoisolator
C) GTO device
D) triac

29) This is the symbol for a(n) _____.



A) SCS
B) diac
C) SCR
D) triac

30) This is the symbol for a(n) _____.



A) SCR
B) SCS
C) diac
D) triac

31) This is the symbol for a(n) _____.



A) GTO
B) Darlington transistor
C) LASCR
D) triac

32) This is the symbol for a(n) _____.



A) LASCR
B) optoisolator
C) photocell
D) phototransistor

33) This is the symbol for a(n) _____.



A) UJT
B) SCS
C) GTO
D) Shockley diode

34) This is the symbol for a(n) _____.



A) PUT
B) SCS
C) GTO
D) Shockley diode

35) This is the symbol for a(n) _____.



A) SCR
B) triac
C) diac
D) GTO

36) This is the symbol for a(n) _____.



A) diac
B) triac
C) UJT
D) GTO

37) This is the symbol for a(n) _____.



A) UJT
B) BJT
C) SCS
D) FET

ANSWER KEY: Chapter 17: pnpn and Other Devices

1) D
2) D
3) C
4) D
5) D
6) C
7) D
8) D
9) B
10) B
11) A
12) B
13) C
14) A
15) B
16) D
17) A
18) C
19) C
20) A
21) A
22) B
23) A
24) A
25) D
26) C
27) C
28) B
29) C
30) C
31) A
32) A
33) D
34) A
35) A
36) A
37) A